The Price of "Free" Advice: Measuring Agency Problems in the Insurance Broker Market*

Elaine Shen† Margaret Kallus

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Abstract

Consumers often rely on intermediaries to navigate complex markets, but the very information frictions that make expert advice valuable can leave consumers unable to evaluate the experts themselves. We study how commissions and competition shape agency problems in the Medicare market, where nearly 70 million seniors and people with disabilities choose between complex health insurance plans. To measure the costs and causes of agency problems, we conduct a large-scale audit study in which senior "secret shoppers" make 637 calls to randomly assigned brokers. Brokers steer seniors toward a higher-commission plan in 52% of calls. Dominated recommendations carry a 29% premium markup, costing seniors $377 per year and over $12,000 in present-value lifetime wealth. Recommendations improve with broker tenure, consistent with learning. A natural experiment that eliminates broker commissions reduces steering to the higher-commission plan by 21 p.p. but does not eliminate steering. A randomly assigned treatment designed to increase perceived competition–by signaling that seniors will seek multiple recommendations–causes brokers to recommend lower-priced plans. This shift increases consumer surplus when plans are homogeneous, but decreases it when plans differ in coverage. A structural model shows that supply-side reforms, such as equalizing commissions and capping premium markups, generate larger consumer surplus gains than consumer-side policies such as ratings feedback and increased competitive pressure. The findings show that when consumers cannot evaluate expert advice, competitive pressure can shift recommendations without necessarily aligning them with consumer interests, highlighting the importance of well-designed intermediary incentives.

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†Shen: UC Berkeley (elaineshen@berkeley.edu); Kallus: UC Berkeley (margaret_kallus@berkeley.edu).

1 Introduction

Governments rely on private intermediaries to help consumers navigate complex public programs in healthcare, housing, and welfare. These intermediaries can help make programs more accessible, but the same information frictions that make advice valuable can prevent consumers from evaluating the quality of the advice itself. Governments respond to these agency problems by regulating intermediary compensation, requiring licensing and training, monitoring sales practices, and providing free information and resources to consumers as an alternative to expert advice. How effective these policies are at constraining agency problems remains an open question.

This paper studies how financial incentives and competition shape intermediary behavior in an important market: the market for Medicare advice. Over half of the nearly 70 million seniors and individuals with disabilities enrolled in Medicare rely on insurance brokers to help them choose between complex health insurance plans that differ in premiums, provider networks, and long-term restrictions on care. Medicare plan choice also has important financial implications; despite near-universal Medicare coverage, roughly one in four seniors report carrying medical debt, one in three report forgoing medical care because of costs, and healthcare costs make up roughly 14% of household budgets.[1] The consequential nature of the decision makes outsourcing to licensed experts appealing. Broker advice is ostensibly "free" since brokers are paid by insurance companies for each enrollment. Commissions are capped by regulators and sales behavior is closely monitored to protect the vulnerable population; misconduct is punished with sanctions and fines.

Data limitations and difficulty measuring consumer preferences have made it difficult to measure broker quality and agency problems in insurance markets. To overcome these limitations, we conduct a large-scale audit study where we hire 109 senior secret shoppers to make 637 phone calls to 554 Medicare insurance brokers. This allows us to generate a dataset of broker advice for the two main decisions most beneficiaries face: the choice between Medicare Advantage (MA) and Traditional Medicare plus Medigap (Decision 1), and which Medigap plan to choose (Decision 2). Two experiments within the audit study allow us to decompose the drivers of broker steering.

A key innovation is our ability to measure broker quality: Decision 2 gives us a clear monetary measure of lost consumer surplus. Because Medigap plans are homogeneous and have identical coverage within a plan letter, we are able to measure lost consumer surplus using the Medigap markup, or the amount the senior overpays for identical coverage due to a dominated plan recommendation. Since brokers directly enroll beneficiaries in the plans they recommend, our audit study outcomes are reasonable proxies for the enrollment outcomes seniors face. To detect agency problems in Decision 1, we assign the secret shoppers all the preferences that Medicare.gov states as reasons not to enroll in Medicare Advantage, so that the plan that matches these preferences conflicts with the financial incentives of brokers.

[1] Jacobson and Blumenthal (2022); Gaffney et al. (2024); Gunja et al. (2024); Ochieng et al. (2024).

We find that brokers recommend a higher-commission but less suitable plan on 52% of calls. In Decision 1, brokers recommend a Medicare Advantage plan that pays them roughly $400 more, but does not match consumer preferences on 32% of calls. In Decision 2, where brokers are paid a percent of the Medigap premium but higher-premium plans have identical coverage and are strictly dominated, the average Medigap markup is $377 per year, a 29% markup. We estimate that dominated Medigap recommendations cost seniors at least $12,000 in present-value lifetime wealth given average premium growth rates. Using a back-of-the-envelope calculation, we estimate that the recommendations we observe would cost seniors $1.3 billion a year in lost consumer surplus, or $894 per advised senior if scaled to the 1.41 million seniors who turn 65 each year and who use a broker to choose a plan in the individual market. Strictly dominated Medigap plan recommendations alone would cost these seniors $211 million a year.

Two experiments allow us to decompose contributors to broker advice. We first take advantage of an unexpected commission change during the audit study to test how broker advice responds to commissions. In this natural experiment, Experiment A, all the insurance companies in one audit region stopped paying call center brokers commissions for Medicare Advantage (MA) enrollments. These plans originally paid higher commissions, but did not match consumer preferences. We collect audit study data before and after this commission change. For the exact same consumer profile, the rate of unsuitable MA recommendations falls by 21 p.p. However, the rate does not fall to zero. Call center brokers continue to recommend MA on 24% of calls when the plan both does not match consumer preferences and does not pay a commission. Experiment A provides causal evidence that broker recommendations respond to commissions, but also provides evidence consistent with broker information frictions.

In Experiment B, we experimentally randomize a treatment designed to test whether perceived competition disciplines advice. Within the audit study, we randomly assign secret shoppers to tell brokers that they plan to talk to multiple brokers. The treatment is motivated by the fact that 66% of seniors only talk to one broker before enrolling in Medicare. We find that advice quality is elastic to the perceived competition treatment. Brokers respond to the treatment by shifting their recommendations to the lower-price plan for both the Medicare Advantage (MA) and the Medigap recommendations. When plans are homogeneous and vary in price alone, the treatment response increases consumer surplus. When plans vary by both price and coverage, however, the broker response to perceived competition also lowers plan coverage. According to estimates calibrated from brokers' own beliefs, the net impact is a loss in consumer surplus.

To evaluate which broker attributes consumers value, we also ask the secret shoppers to rate how likely they are to recommend the broker to a friend or family member enrolling in Medicare after each call. This allows us to test whether broker ratings are correlated with our two measures of broker recommendation quality. Broker ratings are uncorrelated with our main measures of

broker quality, but are correlated with call length. This suggests that seniors' ratings do not distinguish good from bad advice by objective quality and instead reflect other heuristics. The results highlight how consumer information frictions can limit feedback mechanisms and allow low-quality advice to persist.

Finally, we estimate a structural model by combining the audit study data with a broker survey. We then use the model to compare consumer surplus gains under supply-side policies that can be implemented by the government to changes in how seniors search for advice. We estimate lost consumer surplus from broker advice to be about $2,154 a year for a senior with our audit-study profile. Equalizing first-year Medicare Advantage and Medigap commissions recovers 24% of lost consumer surplus. Capping the Medigap markup at 25% recovers 6% of lost consumer surplus. In contrast, policies that rely on consumer feedback to discipline advice, such as competitive pressure (signaling plans to solicit multiple recommendations) and limiting search to highly-rated brokers, are far less effective and may even reduce consumer surplus.

This paper studies how misaligned incentives shape advice in a new, important financial market serving an older population. First, we contribute both to the literature on expert intermediation and to a growing literature on aging. Existing research has documented agency problems and broker steering in financial advice (Malmendier and Shanthikumar, 2007; Bhattacharya et al., 2012; Inderst and Ottaviani, 2012b; Gennaioli et al., 2015; Egan, 2019; Robles-Garcia, 2019; Linnainmaa et al., 2021; Bucher-Koenen et al., 2025). A related literature shows that older adults face difficulty making financial decisions (Agarwal et al., 2009; Kézdi and Willis, 2014; Lusardi et al., 2014; Kim et al., 2021; Ameriks et al., 2023; Mazzonna and Peracchi, 2024). We provide some of the first direct evidence of agency problems in this large market serving a vulnerable population.

Second, we build on a literature using audit studies as a measurement tool to translate broker advice into monetary loss. Furthermore, by experimentally randomizing a treatment increasing competitive pressure within the audit study, we are able to directly detect strategic broker behavior. The paper builds on existing research using audit studies to identify discrimination and misconduct difficult to detect in observational data (Bertrand and Mullainathan, 2004; Bertrand and Morse, 2011; Mullainathan et al., 2012; Anagol et al., 2017; Alsan et al., 2019; Kessler et al., 2019; Kline et al., 2022). This allows us to measure the cost of agency problems in a setting where data is scarce and quality is difficult to measure.

Third, we contribute to the literature on complexity and mistakes in health insurance choice by showing that demand-side frictions can create an economic environment in which supply-side distortions can generate additional losses. Prior work has documented several demand-side frictions: consumers overweight premiums (Abaluck and Gruber, 2011), choose dominated plans (Afendulis et al., 2015; Bhargava et al., 2017), misunderstand coverage (Brot-Goldberg et al., 2017), and remain in unsuitable plans due to inattention (Ericson, 2014; Ketcham et al., 2015; Heiss et al., 2021;

Brot-Goldberg et al., 2023). This paper builds on the literature studying the supply-side response to consumer information frictions by highlighting the importance of well-designed regulations in the face of these frictions (Aizawa and Kim, 2018; Cabral et al., 2018; Gruber et al., 2020).

Finally, this paper speaks to the theoretical and experimental literature on expert advice in credence-good markets and physician-induced demand by testing the model predictions in a field setting (Pitchik and Schotter, 1987; McGuire and Pauly, 1991; Wolinsky, 1993; Inderst and Ottaviani, 2009, 2012a; Dulleck et al., 2011; Carroll, 2019; Balafoutas and Kerschbamer, 2020; Karakoç et al., 2024). In these models competition can either raise or lower the probability of conflicted recommendations; we test these model predictions empirically within the audit study.

2 Background on Medicare insurance plan choice

This section provides institutional context on Medicare and the Medicare broker market. In a nationally representative survey, we find that 51% of current Medicare beneficiaries enrolled through a broker. This number may be an underestimate since many seniors think they are talking to "Medicare" when they enroll and do not realize they are enrolling using a broker.[2]

When seniors first join Medicare Part B (typically, when they turn 65 or retire), they qualify for a 7-month Initial Enrollment Period when they can choose any Medicare insurance plan.[3] This initial choice can be difficult to reverse; depending on the plan chosen, medical underwriting can leave seniors stuck in their initial plan type and have long-term implications for access to healthcare, health outcomes, and retirement savings. Figure 1 provides an overview of the choice most seniors face when they enroll in Medicare.[4] The audit study focuses on two key parts of the decision tree: Decision 1, the choice between Medicare Advantage (MA) and Traditional Medicare (plus Medigap); and Decision 2, the specific Medigap plan choice.[5]

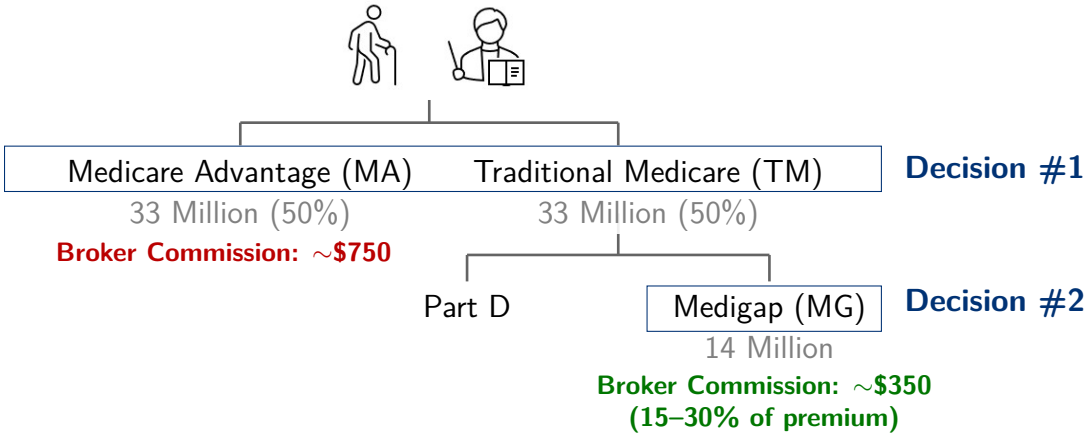
[2]Data on the use of brokers or agents is limited and mostly comes from surveys, including our own survey conducted through the Understanding America Study and described in Appendix C.4. Other surveys of broker use include Leonard et al. (2022) and Freed et al. (2023). Broker use is also expected to expand since Republican administrations have expressed support for expanding MA, which is primarily sold through brokers. See https://www.npr.org/sections/shots-health-news/2024/11/07/nx-s1-5183092/trump-election-2024-medicare-advantage.

[3]Seniors who are still working and delay Medicare coverage because they have insurance through an employer get a similar Enrollment Period when they retire and begin Part B coverage.

[4]We focus on the individual market faced by the majority of Americans. This is different from the group market, which includes a different set of non-standard (often subsidized) employer or union-sponsored retiree plans. These plans are usually managed by a former employer and not available to the general public. The choice set is also different for individuals eligible by reason of disability and under the age of 65, who have limited Medigap access, beneficiaries who are dual-eligible with Medicaid, and people with access to other supplemental care (e.g. veterans).

[5]Most beneficiaries with Traditional Medicare and Medigap also enroll in a separate Medicare Part D prescription drug plan. We do not focus on Part D in this paper since it is typically the smallest share of broker compensation: brokers are paid roughly $100 for an initial Part D enrollment.

Figure 1: Medicare Choice Landscape



Notes: Figure 1 shows the choice set a beneficiary faces upon enrolling in Medicare and the two decisions we study. Decision #1 is the choice between Medicare Advantage and Traditional Medicare; Decision #2 is the choice of Medigap plan among beneficiaries with Traditional Medicare. Dollar amounts show the approximate first-year commission a broker earns from each enrollment: a ~$750 lump sum for a Medicare Advantage enrollment and 15–30% of the annual premium (~$350) paid monthly for a Medigap enrollment. Section 2 details the full commission schedule.

Decision 1: Medicare Advantage (MA) vs Traditional Medicare + Medigap (TM+MG) The first choice all Medicare beneficiaries make is whether to enroll in a Medicare Advantage (MA) plan or Traditional Medicare (TM, also known as Original Medicare or Fee-For-Service Medicare). About half of Medicare beneficiaries have MA and half have Traditional Medicare. Beneficiaries enrolled in Traditional Medicare may also choose to enroll in a Medicare Supplement or Medigap (MG) plan, described further below, to cover out-of-pocket costs not covered by Traditional Medicare such as copays, coinsurance, or deductibles. Each option has its own costs and benefits summarized in Figure 2.

Figure 2: Medicare Plan Tradeoffs by Category

Category	Medicare Advantage (MA)	Traditional Medicare+Medigap
Premium	**Low, sometimes $0/month**	Higher (~$140/month)
Drug coverage	**Sometimes bundled in**	Separate Part D plan ($0-$50/month)
Extra benefits	**Often dental, vision, hearing, gym**	Not included
Restrictions on care	Network restrictions, referrals, prior authorization	**Any doctor that takes Medicare (nationwide coverage)**
Cost sharing (OOP)	Unpredictable; $14,000 cap	**Predictable; low (<$300)**
Switching	Underwriting required, hard to switch	**Switch to MA easily**

MA plans are offered by private insurance companies. Traditional Medicare and Medigap are managed by the federal government (specifically, the Centers for Medicare and Medicaid Services,

or CMS). Figure 2 summarizes how the two options differ. Since MA plans are not standardized, coverage, network size, and claims denials vary by plan, geography, and insurer (Biniek et al., 2024; Gaffney et al., 2024; Vabson et al., 2025).

MA and Medigap plans also differ in their flexibility to change coverage. Outside of the initial 7-month enrollment window, seniors with Traditional Medicare and Medigap can easily switch to MA without medical underwriting every year during special enrollment windows such as the Annual Enrollment Period (AEP), but seniors cannot switch from MA to a Medigap plan without medical underwriting. In practice, many seniors end up being stuck in lower-coverage MA plans or are forced to pay significantly higher premiums when they switch to Medigap (Boccuti et al., 2018; Curto, 2023; USC Schaeffer Center, 2024). For this reason, MA is sometimes described as a "one-way door." The asymmetric nature of switching between products in this market makes a poor initial plan choice difficult and sometimes impossible to reverse for seniors who would not pass medical underwriting.

Decision 2: Medicare Supplement (Medigap or MG) Plan Medicare Supplement (Medigap) plans help cover out-of-pocket costs such as copays, coinsurance, and deductibles that are not covered by Traditional Medicare. For example, Traditional Medicare covers 80% of the cost of most outpatient appointments and procedures. Medigap plans help cover the remaining cost sharing expenses. (Additional details on the market are included in Appendix B.)

A key feature of Medigap plans is that the plans are standardized, homogeneous products within a plan letter due to regulations in 1990. Plans are sold by private insurance companies, but coverage terms come from a menu set by the government, as shown in Appendix B. As explicitly stated on Medicare.gov, the only difference between Medigap plans with the same plan letter is price. CMS is the primary payer and all Medigap claims are approved by CMS. Beneficiaries essentially never interact with the private insurance company, leaving little room for differences in claims denials or customer service.[6]

About 14 million people today have a Medigap plan. Given their less restrictive coverage and higher quality of care, these plans are a popular choice for experts in the health insurance field (Ochieng et al., 2023). In interviews, several brokers stated "anyone who can afford to should have Traditional Medicare plus Medigap" and surveys of Medicare brokers found that brokers were more likely to choose Traditional Medicare plus Medigap for themselves (Leonard et al., 2023).

The Medicare market is so complex that even beneficiaries who are already enrolled in a plan struggle to correctly report their plan information. In a survey of 2,015 nationally-representative

[6]To test whether brand preferences might contribute to price dispersion among the identical products, we survey a representative group of senior Medicare beneficiaries through the Understanding America Study, described in Appendix C.4. 88% of seniors say there is no insurance company they are trying to avoid, and 61% have no positive or negative brand associations.

Medicare beneficiaries fielded through the Understanding America Study (UAS) (Appendix C.4), at least 40% of respondents describe their own current coverage in a way that does not match the Medicare plans offered. Specifically, these respondents report 1) a Medigap premium far below the minimum premium in the market; 2) a Medicare Advantage (MA) premium significantly higher than average premiums ($\geq$\$100/month); 3) a Medigap plan letter that is not part of the federal menu; or 4) not knowing which Medicare plan type they have. This is surprising for a longitudinal panel of study participants who may be positively selected. The survey results also highlight the difficulties evaluating broker quality using survey data: a beneficiary who cannot correctly identify their Medicare plan is unlikely to correctly evaluate plan attributes and plan quality.

Regulating broker incentives Medicare Advantage (MA) is regulated at the federal level by CMS; Medigap is primarily regulated by state Department of Insurance offices. The regulations that govern broker sales practices and broker commissions follow the same split. CMS sets a cap for the lump-sum commission which adjusts annually and varies by state. In California, the cap was roughly \$750 per enrollment during our audit study. Medigap commissions are 15–30% of the monthly premium; we use 20% in the structural model. Each product offers renewal commissions at half the year-1 rate through year 6. Commissions after year 6 are often much lower and have high variability.[7]

Table 1: Broker commission schedule by Medicare product

Product	Year 1 commission	Renewal (years 2-6)	Payment
Medicare Advantage	$\sim$ \$750	50% of year 1	Lump sum
Medigap	$\sim$20% of premium ($\approx$ \$350)	50% of year 1	12 monthly
Part D	\$109	\$55	12 monthly

Notes: Medicare Advantage (MA) and Part D commission maxima are set by CMS at fair-market value (FMV) under 42 C.F.R. §§422.2274(d), 423.2274(d), with renewals capped at 50% of the initial-year amount (CMS, 2026). The MA year 1 commission is the CMS fair-market-value (FMV) cap, which varies by state and year: the national rate (applicable in Texas) was \$611 in 2024, \$626 in 2025, and \$694 in 2026, while the California cap was \$762, \$780, and \$864 in those years. Most major carriers pay the cap, so there are no differential incentives across MA plans. Medigap commissions are not set by CMS but by the insurer; industry schedules pay roughly 15–30% of the annual premium in the first year and half that in renewal years, or about \$350 at the mean premium (the analysis uses 20% of the call-specific premium, with renewals at 10%) (Ali et al., 2021). There is no guarantee of renewal commissions after year 6.

Insurance Company Appointments Medicare insurance brokers are paid by the private insurance companies they are licensed, or "appointed," to work with. Studies suggest payments to Medicare insurance brokers exceeded \$10 billion in 2022 (Meyers et al., 2026). Broker commissions are set by insurance companies, but are subject to the caps described above. Brokers may or may not work with every insurance company or sell every Medicare product. Unlike in other

[7]Medigap commissions are guaranteed through year 6, but are allowed to continue after year 6. MA commissions were originally limited to years 1-6, but the 6-year guidance was dropped in a rule codified in 2021 (CMS, 2021). In practice, the amount the broker takes home from any of these enrollments may be less than the stated amounts since broker compensation structures depend on the hierarchy of their Field Marketing Organization (FMO).

intermediated markets, the appointment constraints are rarely binding since the majority of states allow brokers to obtain an appointment right before a sale is made through Just-In-Time (JIT) appointments.[8] Brokers may be appointed to sell for one or multiple insurance companies and are directly compensated by the insurance company for every beneficiary they enroll in a plan.

Government oversight of the broker market The Medicare broker market is highly regulated and closely monitored in part due to the government's financial stake in the program: the federal government spent over $1.2 trillion (3.9% of GDP) on Medicare in 2025 alone and subsidies to private insurance companies accounted for over half of this amount (Boards of Trustees, 2026). Brokers are required to pass certification exams and hold a state license to sell health insurance.[9] Limits on broker pay are set by federal and state agencies. While brokers are not subject to "fiduciary rules," state laws require insurance agents to "make reasonable efforts" to determine whether a recommendation is appropriate (NAIC, 2008, Section 21). CMS can impose sanctions on insurers who contract with brokers who violate CMS rules, including suspending federal payments or imposing monetary fines (42 C.F.R. §§ 422.750, 422.752(a)).

Direct monitoring of Medicare broker activity is, at least on paper, more extreme than other intermediated markets. CMS requires Medicare brokers to record all sales calls and store the recordings for up to 10 years.[10] CMS also requires brokers to disclose that they may not work with all available insurance companies and refer the senior to Medicare.gov or free State Health Insurance Assistance Program (SHIP) resources within the first minute of a call (Appendix B.) All broker and insurance company marketing material must be approved by regulators prior to distribution. Brokers that violate these rules can face regulatory action or lose their license.

Disciplining market forces In addition to regulatory concerns, traditional market forces such as reputation concerns, referrals, and competition might be thought to limit agency problems. With thousands of licensed brokers in each state, most of whom are licensed in multiple states, the advice market is highly competitive. Brokers are not paid by seniors per consultation and are free for seniors to consult, lowering search costs. Additionally, information on Medicare plans is freely available and must be disclosed to prospective clients on every call: a senior can check a broker's recommendation against the Medicare.gov Plan Finder tool or consult a SHIP counselor at no cost. Whether government oversight and market forces have been successful at limiting agency problems in this market remains an open empirical question.

[8]Both states in our audit study, California and Texas, allow just-in-time appointments.

[9]Brokers who sell MA or Part D plans must pass Medicare training and testing every year with a score of at least 85 percent (42 C.F.R. § 422.2274(b)).

[10]This requirement was changed to 6 years in contract year 2027 (CMS, 2026).

3 Conceptual Framework for Broker Advice

This section presents the sketch for a simple conceptual framework of the interaction between seniors and brokers. A senior enrolling in Medicare does not understand the different insurance plans and calls a broker for a plan recommendation. The broker is paid by the private insurance company if they enroll the senior in a plan. The broker earns a different commission for each plan and is paid nothing if the senior does not enroll. The senior either accepts the recommendation and enrolls in the plan (the broker closes the sale) or chooses to incur a search cost and call another broker. Appendix A proves every result. Section 6 adds structure and estimates the model.

3.1 Setup: Plans, Prices, and Broker Pay

A senior enrolling in Medicare chooses between two insurance plans, $j \in \{L, H\}$, with annual premiums $\rho_L < \rho_H$. The senior seeks a recommendation from broker b. The broker recommends one plan $j \in \{L, H\}$, not the entire menu of plan options. We also assume the broker focuses on persuading the senior that their recommendation is correct and does not explain the alternative plan. The senior then compares the recommended plan against the outside option value v_s, or the continuation value of search. The broker maximizes expected utility, or the probability of making the sale times pay from selling plan j. Selling plan j pays the broker a commission $W_j > 0$. The broker is paid W_j only if the senior enrolls in plan j.

3.2 The Consumer's Problem

The senior chooses the plan that maximizes their perceived utility. We define all utility in monetary terms. Focusing on the difference between the two plans, the net harm from plan L is defined as

$$N \equiv U_c(H) - U_c(L) = \Delta - (\rho_H - \rho_L), \tag{1}$$

where ρ_j is the plan premium and Δ refers to the fit gap, or the gap between the plan's attributes and the senior's preferences (e.g. lost utility from under- or over-insurance). We assume that at the point of sale the senior perceives plan premium ρ_j exactly and a fraction $\phi \in [0, 1]$ of the plan fit gap Δ. We define ϕ as the friction parameter, or a reduced-form object that accounts for the weight the senior puts on premiums versus plan attributes. At $\phi = 1$ the senior perceives the fit gap in full (the full-information benchmark). At $\phi = 0$ the senior perceives none of the fit gap (a pure credence good). ϕ can be thought of as information frictions, a complexity parameter that prevents seniors from fully understanding insurance attributes, or consumer biases such as overweighting plan premiums relative to plan coverage.[11] For a consumer with friction ϕ, the

[11] This framework is related to models where consumers overweight premiums relative to coverage (Abaluck and Gruber, 2011), attend to salient attributes (Bordalo et al., 2013), or do not see shrouded attributes (Gabaix and Laibson, 2006).

perceived net harm of plan L is

$$\hat{N} \;\equiv\; \hat{U}_c(H) - \hat{U}_c(L) \;=\; \phi\Delta - (\rho_H - \rho_L). \tag{2}$$

For a senior whose preferences best match plan H, we assume that

$$\Delta \;>\; \rho_H - \rho_L, \tag{3}$$

that is, the harm in monetary terms from under-insurance Δ is greater than the savings from the lower plan premium, so the net harm N in (1) is strictly positive. We assume the senior is non-strategic and does not infer anything about the broker's pay or financial incentives from the broker's recommendation.

After the senior receives a recommendation from the broker, the senior compares the recommended plan against the outside option value, or the net continuation value of additional search. The outside option value is worth v_s, a constant dollar amount that varies with senior-specific search costs, but does not vary with the broker recommendation. $s \in \{0, 1\}$ indexes the perceived competition treatment arm. We model the perceived competition treatment as a signal about lower search costs: $s = 1$ on treated calls and the treatment is the signal that $v_1 > v_0$. In the comparative statics below we treat the continuation value as a continuous variable v.

We assume a logistic taste shock on the probability that the senior accepts the recommendation or continues search. The probability that a broker recommendation in treatment arm s ends in a completed sale is then

$$P_{r,s} \;=\; \Lambda\Big(\hat{U}_c(r) - v_s\Big), \qquad \Lambda(t) \;=\; \frac{1}{1 + e^{-t}}, \tag{4}$$

where Λ is the standard logistic cumulative distribution function and $P_{r,s}$ is the sale close probability (or close probability for short). Since Λ is strictly increasing, the plan that maximizes the senior's perceived utility $\hat{U}_c(r)$ is the easier sale to close. We invert (4) to convert the probabilities back to the latent dollar index. Taking the difference between the two plans gives us

$$\log \frac{P_H}{1 - P_H} \;-\; \log \frac{P_L}{1 - P_L} \;=\; \hat{N}. \tag{5}$$

Thus $\hat{N}$, the perceived net harm from plan L, determines the gap between the two close probabilities. This implies $P_H > P_L \iff \hat{N} > 0 \iff \hat{U}_c(H) > \hat{U}_c(L)$: intuitively, the product the senior perceives to have the greater utility is the easier sale for the broker to close.

3.3 The Broker's Problem

The broker observes which plan best matches the senior's preferences and is paid W_j if the senior accepts the recommendation and enrolls in the plan and zero otherwise. Brokers choose to

recommend plan j that maximizes expected utility

$$U_b(j) \;=\; P_j \;\times\; \Big(W_j \;-\; \kappa_{b,j} N\Big) \tag{6}$$

We slightly abuse notation and use the same symbol P_j to define both the consumer's true sale close probability and the broker's perceived probability that the senior accepts the recommendation versus continuing to search. W_j is broker pay for selling plan j, and net harm N is scaled by the parameter $\kappa_{b,j}$. We refer to κ as the suitability cost. The cost is incurred only when the broker sells the plan that they do not believe matches the senior's preferences, or the "misfit plan". For a senior whose preferences match plan H the suitability cost is zero when the broker recommends plan H ($\kappa_{b,L} = \kappa_b$ and $\kappa_{b,H} = 0$). κ can be thought of as paternalism or guilt from recommending a plan that does not match the senior's preferences, lost continuation value when a poorly matched enrollee disenrolls from the plan (in which case the broker is no longer paid), regulatory liability, or the reputation costs of a poor recommendation. Figure A.6 shows the contributors to κ self-reported by brokers. The broker trades off their own pay versus senior plan suitability.[12]

3.4 Predictions under misaligned incentives

The audit observes broker recommendations under two decisions where incentives are misaligned, that is, where the plan that matches the senior's preferences pays the broker less. Table 2 describes the two decisions, both of which follow the framework and differ only in the fit gap Δ and in which plan maximizes senior utility. Predictions 1–3 generate predictions for Decision 1, where plan H matches the senior's preferences and plan L pays the broker more. Prediction 4 generates a prediction for Decision 2. Appendix A.6 proves Predictions 2–4.

Table 2: One framework, two decisions

	Decision 1: Medicare product	Decision 2: Medigap plan
Plans L, H	MA; Traditional Medicare + Medigap	Two Medigap plans with identical coverage
Plan that maximizes $U_c(j)$	plan H	plan L
Plan fit gap Δ	$\Delta > 0$	$\Delta = 0$ (coverage is identical)
Suitability cost	charged on a sale of plan L	neither plan is the misfit; not charged
What the senior loses	net harm $N = \Delta - (\rho_H - \rho_L) > 0$	markup $m = \rho_H - \rho_L > 0$
Broker commission (W_j)	$W_L > W_H$	$W_H > W_L$ (%-based commission $W_j = z\rho_j$)
Higher broker commission	plan L	plan H
Prediction tested	1, 2 and 3	4

Notes: W_j is the present value of plan j's multi-year commission stream and z is the Medigap commission percentage (Section 2). In each decision the plan that pays the broker more is the one that is less suitable for the senior. Condition (3) and the net harm N hold in the Decision-1 parameterization only; in Decision 2 both plans fit, so no sale carries the suitability cost and the cutoff (7) does not apply.

[12]Note that the broker does not directly observe ϕ, but acts on a belief about it. Nothing in the framework requires that belief to be correct, so the predictions below are signed by the broker's belief about which plan is the easier sale rather than by the senior's actual perception.

Prediction 1 (Decision 1: broker suitability cost cutoff)**.** *Brokers with a high enough suitability cost recommend the plan that best matches the senior's preferences, even though it pays them less. If no broker bore a suitability cost, every broker would recommend L.*

For Decision 1, the senior's preferences match plan H, so the suitability cost is zero when the broker recommends plan H, $\kappa_{b,H} = 0$. The expected payoffs are $U_b(H) = P_H W_H$ and $U_b(L) = P_L\Big(W_L - \kappa_b N\Big)$. The broker recommends L if and only if $U_b(L) \geq U_b(H)$. Both sides of that comparison are expected payoffs, so dividing by $P_L > 0$ leaves the inequality unchanged and transforms the sale probabilities into the ratio $R \equiv \frac{P_H}{P_L}$. We refer to R as the sale-probability ratio, the probability of a successful sale of plan H relative to plan L. R_0 is the sale-probability ratio on control calls (when $s = 0$) and R_1 the sale-probability ratio when $s = 1$.

The broker recommendation rule can be written as a function of the suitability cost

$$\text{Broker recommends } L \quad \Longleftrightarrow \quad \kappa_b \;\leq\; \kappa^* \;\equiv\; \frac{W_L - R\,W_H}{N}. \tag{7}$$

Appendix A.5 shows the full derivation. Note that the sale-probability ratio R is the only channel through which the senior's friction ϕ enters the broker problem. Equation (5) shows that $R < 1$ precisely when $\hat{N} < 0$.

Every broker whose suitability cost exceeds the cutoff, $\kappa_b > \kappa^*$, recommends plan H, the plan that matches the senior's preferences. The suitability cost varies from call to call, so κ_b is drawn on each call from a continuous cumulative distribution function F with density F'; Section 6 lets the location of F differ between call center and non-call-center brokers. Section 6 takes F logistic, which places some mass at $\kappa_b < 0$: a broker who would rather close the misfit sale. By (7), σ, the share of brokers who recommend L to a senior who is a fit for H, is the mass below the cutoff,

$$\sigma \;=\; F\Big(\kappa^*\Big), \qquad \text{and by competition treatment arm,} \qquad \sigma_s \;\equiv\; F\Big(\kappa_s^*\Big). \tag{8}$$

Note that σ is also the Decision 1 outcome, or the share of calls on which the broker recommends plan L to a senior whose preferences best match plan H.

Prediction 2 (Decision 1: recommendations respond to pay)**.** *Lowering broker pay W_L lowers σ, the share of brokers that recommend L, for every value of ϕ.*

Prediction 2 follows from differentiating (7), which gives $\partial\sigma/\partial W_L = F'(\kappa^*)/N > 0$; Appendix A.6 contains the full proof. The intuition is that raising W_L raises the suitability cost cutoff for recommending plan L by $1/N$ and the brokers who switch to recommending L are those whose suitability cost lies between the old and new cutoff, a mass of $F'(\kappa^*)$ per unit of shift. Both terms are strictly positive: F' is a density and $N > 0$ by condition (3). Experiment A observes broker recommendations when $W_L = 0$.

Prediction 3 (Decision 1: impact of perceived competition)**.** *In Decision 1, where $\Delta > 0$, the direction of the perceived competition treatment's effect on σ, the share of brokers recommending L, depends on the broker's perception of the consumer friction ϕ. In particular, the treatment raises σ when $\phi < \phi^\dagger$, lowers σ when $\phi > \phi^\dagger$, and leaves σ unchanged when $\phi = \phi^\dagger$.*

From Equation (2) the senior perceives a gain from the low-price plan, $\hat{N} < 0$, exactly when

$$\phi \;<\; \phi^\dagger \;\equiv\; \frac{\rho_H - \rho_L}{\Delta}. \tag{9}$$

We call $\phi^\dagger$ the complexity threshold: the share of the fit gap above which the plan that maximizes the senior's perceived utility $\hat{U}_c$ is also the plan that maximizes true utility U_c. Condition (3) places $\phi^\dagger$ strictly between zero and one.

The continuation value v A better outside option lowers the close probability of every recommendation, but not in the same proportion. Differentiating Equation (4) gives $-\partial \log P_r / \partial v = 1 - P_r$: each recommendation loses its sales at a proportional rate equal to its own walk-away probability. Thus the harder sale loses the larger share. Differencing the two rates gives the response of the sale-probability ratio,

$$\frac{\partial \log R}{\partial v} \;=\; (1 - P_L) - (1 - P_H) \;=\; P_H - P_L, \tag{10}$$

so relative sale probability tilts toward whichever recommendation was already easier to close.[13] Note that a rise in the continuation value leaves $\hat{N}$ unchanged: it changes how willing the senior is to accept a recommendation, but not which plan the senior prefers.

The continuation value of search enters the cutoff (7) through the sale-probability ratio R. Since W_H, R, N and $F'(\kappa^*)$ are each strictly positive, the sign is that of $P_L - P_H$, which by (5) and (9) is the sign of $\phi^\dagger - \phi$. When $\phi < \phi^\dagger$, a rise in the continuation value raises the share of brokers recommending the plan L. When $\phi > \phi^\dagger$, it lowers that share. When $\phi = \phi^\dagger$, the share does not change. Experiment B raises the continuation value from v_0 to v_1. The broker response depends on the broker's beliefs about the consumer's frictions, or where ϕ is relative to $\phi^\dagger$.

$$\begin{aligned} \frac{\partial \sigma}{\partial v} \;&=\; \frac{W_H\, R}{N}\, F'(\kappa^*)\,\Big(P_L - P_H\Big), \\ \operatorname{sign} \frac{\partial \sigma}{\partial v} \;&=\; \operatorname{sign}\Big(P_L - P_H\Big) \;=\; \operatorname{sign}\Big(\phi^\dagger - \phi\Big). \end{aligned} \tag{11}$$

Prediction 4 (Perceived competition when plans only vary in price)**.** *In Decision 2, when $\Delta = 0$ and $W_j = z\rho_j$, the broker responds to the perceived competition treatment by recommending the*

[13] Lemma 2 in Appendix A.3 derives the three close-probability derivatives in full, and Remark 1 shows that the logistic form is a normalization rather than a restriction.

lower-price plan.

A rise in the continuation value lowers the share of calls on which the higher-priced plan is recommended, and with it the average markup of the plans brokers recommend, for every $\phi \in [0, 1]$. Prediction 4 follows the same logic as Prediction 3, but with $\Delta = 0$. The proof is shown in Appendix A.6. Because the two decisions order commissions oppositely, the same movement toward the easier sale raises the broker's expected pay in Decision 1 and lowers it in Decision 2; Remark 4 works through which accounts of broker motivation that joint pattern rules out.

4 Experimental Design

Most of our evidence comes from a large-scale audit study in which senior secret shoppers request Medicare plan recommendations from randomly assigned insurance brokers. We supplement the audit data with two surveys. The first is a nationally representative panel of 2,015 current Medicare beneficiaries fielded through the Understanding America Study (Appendix C.4). The second is a survey of 157 currently licensed Medicare insurance brokers (Appendix C.5).

4.1 Audit Logistics

We randomly assign 109 secret shoppers, most of them age 65 or older, to request a Medicare plan recommendation from 554 randomly assigned insurance brokers on 637 phone calls. Data collection began in September 2024 and concluded in March 2026. This section describes the audit study logistics and protocols, including who the shoppers are, which brokers they reach, and where the consumer lives. Additional details are included in Appendix C.1.

Secret Shopper Sampling We hire and train seniors from two "secret shopping" firms.[14] Shoppers are never told the purpose of the study. Before they are assigned insurance brokers to contact, seniors must first pass an assessment demonstrating familiarity with the call script and a basic understanding of Medicare. We also collect the senior's real age, gender, race, and education level and include these characteristics in our baseline controls. Due to the individual-level random assignment of both brokers and seniors, any differences in recommendation quality across shopper characteristics can be interpreted causally.

Broker Sampling We sample from the two channels seniors actually use to find a Medicare broker. The first consists of two major Medicare call centers, which together enrolled over one million seniors in 2024 and each employ more than 1,000 brokers. Both advertise heavily on

[14] A secret shopping firm hires people to pose as regular customers in order to evaluate a business's service quality or its compliance with company policies and regulations. We use two firms due to the different strengths each firm had in reaching a different portion of our broker sample. This is further described in Appendix C.1.

television, by mail, and online, and both are named in a 2025 Department of Justice complaint under the False Claims Act (*United States ex rel. Shea v. eHealth, et al.,* No. 21-cv-11777), making the population particularly policy-relevant. The second is the nation's largest online directory of Medicare insurance brokers, which listed over 18,000 brokers as of June 2025. We match the audited brokers to a licensure database constructed from 47 state insurance department registries. This gives us the broker's tenure and the set of carriers the broker is currently appointed to sell. Table A2 in Appendix C.1 reports summary statistics for the seniors and the brokers, and Table A3 compares broker characteristics across the two firm types (call-center versus non-call-center brokers).

Geographic Context Medicare plan availability and price vary by zip code. We select zip codes in California and Texas, two states with reasonably large senior populations. We also target rural areas with a normal amount of Medigap price dispersion and with median household income and education levels that are not too far from the national average. We choose rural areas since these areas tend to have fewer provider networks to choose from and fewer MA plans in general, creating a higher bar for MA recommendations.[15] This makes MA a worse fit for our audit study senior and narrows the MA choice set a broker can steer the senior into.

4.1.1 Measuring Broker Quality

To best measure broker quality, we focus our audit study on a scenario where broker incentives and consumer preferences are misaligned. On each of the two decisions a senior makes, we assign the consumer profile so that the plan that pays the broker more is the plan that serves the senior worse. This scenario creates a conflict of interest between brokers and consumers, and is the exact scenario policymakers at CMS and the DOJ have raised concerns about.[16]

The Medigap-fit senior Our main scenario assigns the shopper preferences that make them a textbook case for Traditional Medicare plus Medigap, built from the reasons listed on Medicare.gov for not choosing Medicare Advantage (MA). The assigned preferences include wanting the flexibility to seek care in multiple states and having providers in multiple states (MA has network restrictions), not wanting to seek prior authorization for medical procedures (MA requires prior authorization), wanting to see specialists without a referral (MA requires referrals), and not expressing a preference for dental, vision, and hearing coverage (which MA has and Medigap does not).[17] We call a shopper with these preferences a "Medigap-fit" senior. Enrolling this senior in

[15]While MA plans guarantee a minimum level of coverage, beneficiaries in rural areas may have to travel further for in-network care, making network restrictions potentially more binding (Park et al., 2021).

[16]https://www.justice.gov/opa/pr/united-states-files-false-claims-act-complaint-against-three-national-health-insurance

[17]We verify these characteristics in interviews with multiple insurance brokers, against the Medicare.gov plan search tool, and against publicly available plan choice tools including the Fidelity Medicare tool and the AI-backed

MA nonetheless pays the broker about $400 more than a Medigap enrollment.

How common is a Medigap-fit consumer? The exact share of Medigap-fit consumers is difficult to estimate given the detailed data on insurance preference, current and future health risks, and financial data needed to determine which Medicare plan is most suitable for a beneficiary.[18] Our best proxy is to use broker beliefs when they are not influenced by enrollment incentives. In our Medicare broker survey, described further in Appendix C.5, we directly ask currently licensed Medicare insurance brokers what share of seniors are an unambiguous fit for each plan. Brokers report that 51% are a clear fit for MA (versus a population average of 55%), 34% are a clear fit for Traditional Medicare plus Medigap (versus 24%), and 15% are not a clear fit for either plan.

Given the institutional setting, we believe these broker survey and population estimates are reasonable proxies for the share of Medicare beneficiaries who are a strong fit for Medigap for the following reasons. First, unlike individuals enrolled in MA, individuals enrolled in Medigap are free to switch to an MA plan annually without medical underwriting during a 2-month Annual Enrollment Period. This means individuals who believe they are a poor fit for Medigap can switch to MA, suggesting beneficiaries who remain in Medigap are satisfied with the service. Second, brokers are not financially incentivized to sell Medigap over MA, so it is unlikely that individuals are incorrectly steered into Medigap. This is in line with concerns raised by the DOJ and CMS; all concerns focus on steering into MA, none reflect steering into Medigap. Finally, Medigap plans are more expensive than MA plans. Since existing studies have documented a bias towards lower-premium plans with higher out of pocket costs, we expect that the bias would run against enrolling in Medigap, once again making the true enrollment a lower bound on Medigap-fit consumers in the population (Abaluck and Gruber, 2011; Handel and Kolstad, 2015).

Outcome 1: Recommend MA Our first outcome is an indicator for whether the broker recommends Medicare Advantage (MA) over Traditional Medicare plus Medigap. To estimate the consumer surplus loss from an MA recommendation for a Medigap-fit consumer, in a separate survey we ask brokers to report the maximum additional monthly premium a senior with our audit study characteristics would pay for various MA and Medigap plan attributes. Assuming the attributes are additively separable, we sum the components to calculate the compensating variation.[19] Appendix Figure A.8 reports the decomposition.

Using brokers' own beliefs, we estimate the net loss in consumer surplus for a Medigap-fit senior in our audit study to be about $5,574 a year. Two additional checks support this interpretation.

tool Healthpilot. Appendix Figure A.3 lists the assigned preferences beside the plan attributes they map to.

[18]This would include consumer preference and behavior data related to current and future travel plans, current and future health risks, and network restrictions, specialists, and prior authorization.

[19]Appendix D.1 states the full accounting methodology, elicitation details, and the discounting and mortality assumptions for lifetime calculations.

First, we compare our broker survey estimates to network restriction costs estimated by Higuera et al. (2018), which value the plan attribute at \$84–\$390 a year. At the upper end of this range this one MA restriction alone would imply that an MA enrollment results in a net loss in consumer surplus for the average consumer. Second, in the same broker survey, we also estimate the net loss for an *average* senior who has MA to be \$2,711 (Appendix Figure A.12). The broker valuations match stated broker beliefs: in interviews, several brokers told us that "anyone who can afford to should have Traditional Medicare plus Medigap." Focus groups conducted by the Commonwealth Fund also show brokers are more likely to choose Traditional Medicare plus Medigap over MA for themselves (Leonard et al., 2023).

Outcome 2: Medigap Markup Our second outcome is the annual premium difference between the Medigap plan recommended by the broker and the low-cost plan with identical coverage available to the senior. This difference is the monetary cost of a dominated plan recommendation and requires no assumption about consumer preferences. The conflict of interest is direct: most brokers earn a percentage of the annual Medigap premium the senior pays, but each additional dollar of premium is a dollar of lost consumer surplus.

Comparing Decision 1 and Decision 2 Comparing how brokers respond to competitive pressure for both decisions sheds light on the attributes brokers perceive to be important to consumers. The two decisions differ in the structure of the choice both the consumer and broker face. In Decision 1, brokers compare two products that differ in attributes or quality (e.g. coverage, network breadth, prior-authorization requirements, out-of-pocket exposure, and price). In Decision 2, the homogeneous products differ only in price and allow us to measure the loss in consumer surplus from a recommendation in dollar terms without any assumptions about consumer preferences. How brokers respond to perceived competition helps us understand whether brokers predict consumers will respond to plan price, plan quality, or both.

Misaligned Recommendation Rate We combine the two outcomes into a single measure, which equals one if the broker recommends the plan that pays the broker more even though it fits the consumer worse: that is, if the broker recommends MA, or if every Medigap plan the broker recommends carries a markup of at least \$360 per year.[20] Table 3 collects the primary and secondary quality measures. We define the secondary measures in Appendix C.1.

[20]We pre-registered a threshold of \$240 per year (\$20 per month). After reviewing the data we judged \$20 per month too harsh and use \$30 per month, or \$360 per year, which remains an economically meaningful threshold and sits closer to the mean Medigap markup of \$377 per year.

Table 3: Summary of Measures of Broker Quality

Variable	Description	Definition
Panel A: Primary Outcomes		
Misaligned Recommendation Rate	Indicator = 1 if broker recommends a plan that pays them more, even if it is a poor fit for the consumer If broker recommends MA *or* only recommends Medigap plans with a markup $\geq$ \$360	$\mathbb{1}\{\text{Recommend MA} \vee \text{Medigap Markup} \geq \$360\}$
Recommend MA	Indicator = 1 if broker recommends MA over Traditional Medicare + Medigap (TM + MG)	$\mathbb{1}\{\text{Recommend MA}\}$
Medigap Markup	Premium difference between broker recommended plan and the low-cost identical plan	$P_{\text{rec}} - P_{\text{min}}$
Panel B: Additional Outcomes		
Medigap (%) Markup	Percentage overcharge relative to the low-cost identical plan available to the consumer	$\frac{P_{\text{rec}} - P_{\text{min}}}{P_{\text{min}}}$
Broker Markup	Premium difference between recommended plan and cheapest plan from insurance company the broker is *licensed* to sell	$P_{\text{rec}} - P_{\text{min licensed}}$

Notes: This table defines the primary and secondary outcomes of interest. "Cheapest available" refers to the lowest-cost coverage-identical plan in the consumer's zip code, whereas "cheapest licensed" restricts the choice set to plans the broker is actively appointed to sell according to the CA and TX state licensure databases.

4.2 Experimental Variation

Experiment A: Removing the Commission To evaluate whether broker recommendations are driven by incentives, we take advantage of a natural experiment in which all national Medicare Advantage insurance companies stop paying commissions for MA enrollments in our audit region. This has become a common strategy for insurance companies that mis-price their MA plans: because regulation allows MA plans to be pulled from the market only at certain points of the year, insurance companies slow enrollments by stopping MA commissions to brokers.[21] Consumers can still enroll in the MA plans, for example through Medicare.gov, but brokers are no longer paid the ∼\$750 commission for the enrollment.

Figure 3 shows the full sequence of commission changes. The key date is September 1, 2025. Before September 1, national carriers paid MA commissions in our California zip code. After September 1, the only commissionable MA plans there came from Imperial Health, a California-only carrier that is not appointed with either national call center, so a call-center broker earned nothing from an MA enrollment.[22] Non-call-center brokers could still get appointed with Imperial

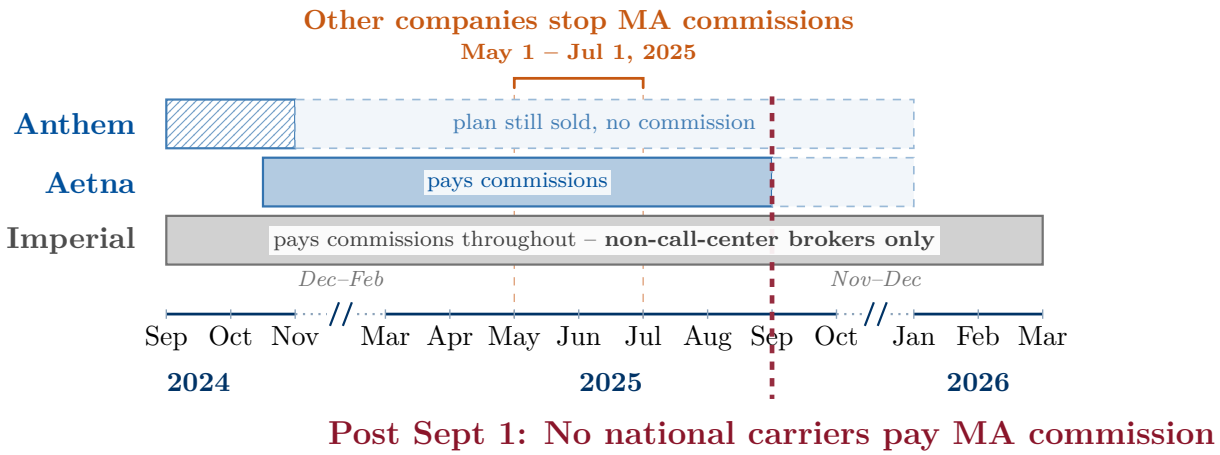
[21]The event is discussed on several Medicare broker blogs. See https://www.medicaremarketinsights.com/p/non-commissionable-plans-united-healthcare; https://www.insurance-forums.com/community/threads/california-not-paying-mapd-commission-is-this-a-national-trend.115640/.

[22]We confirm the appointments in the California Health Insurance License database. Both call centers also removed the non-commissionable plans from their internal search tools, though the plans were still available on

Health, so the commission removal binds only for call-center brokers, and this is the comparison we use to test whether removing the incentive to recommend MA reduces MA recommendations.

We collect audit data before and after September 1, holding consumer preferences fixed.[23] One unintended consequence of the natural experiment is that the change prevented us from finishing data collection for the treatment arm for California call centers. As a result, our treatment and control groups for Experiment B, described below, are not balanced on call center status.

Figure 3: MA Commission Changes Timeline



Experiment B: Signaling Perceived Competition Experiment A asks whether misaligned recommendations respond to the broker's pay. Experiment B asks whether a treatment designed to increase perceived competition can discipline advice.

The perceived competition treatment, Experiment B, instructs shoppers to inform the broker at the beginning of the call that they plan to get a few quotes and talk to multiple brokers before enrolling in a plan. Shoppers in the control group are instructed to say they want a recommendation from the broker to better understand their options, or to say nothing at all; Appendix C.3 gives the exact wording of both variations. We refer to this treatment as the perceived competition treatment, or the competition treatment for short, in the rest of the paper.

The treatment takes advantage of competitive pressure already in the broker market and makes it more salient. Collecting data from 47 state departments of insurance, we estimate that there were over 1 million currently licensed health insurance brokers in the US at the beginning of 2025.[24]

Medicare.gov. Additional details are in Appendix C.2.

[23]We allocate the post-September call center calls to California, where the commission removal binds, so that the pre-post comparison is adequately powered. Appendix C.2 explains why a paired Texas call center control series was infeasible.

[24]Collecting data from the California and Texas state department of insurance agencies, we estimate that there were over 200,000 and over 400,000 actively licensed health insurance brokers in California and Texas respectively at the beginning of the audit study.

In Texas alone, approximately 14,000 resident health insurance brokers held a Medicare product certification at the beginning of the study. This is likely a lower-bound estimate due to limited enforcement and only applies to Texas residents. Almost all brokers are (or can become) licensed in multiple states and can be easily contacted over the phone or online. Overall, the numbers suggest a large supply of brokers. However, in our Understanding America Study (UAS) survey, 66% of current Medicare beneficiaries spoke to only one broker before enrolling in Medicare.

The treatment does not change market structure or manipulate broker supply. Instead, it manipulates broker beliefs about the probability that this call ends in a sale by making consumer retention risks salient in an already competitive market. Competition acts on a broker through the risk that the senior ends the call without enrolling and buys from someone else: a senior can reach another licensed broker through any of hundreds of search results or toll-free numbers, and in interviews brokers frequently complained about competitors stealing their clients. The treatment makes that possibility salient on the call. In doing so, the treatment also tests the effectiveness of advice consumers are routinely given in intermediated markets: to seek a second opinion.[25]

We stratify randomization by state and firm type (call-center versus non-call-center brokers) to ensure balance, and assign brokers to treatment or control status before any calls are made.[26] The calls are then randomly assigned to shoppers at the individual level. We implement the treatment and control conditions in waves (at different points in time) to minimize shopper mistakes.[27] Due to the unexpected timing of the natural experiment, we are unbalanced on call center status and state. Table A4 in Appendix C.1 reports the balance table and Table A5 compares the broker and caller characteristics of compliers versus non-compliers.

Baseline controls Every reduced-form regression on the audit data shares one set of controls, which we call the baseline controls. The broker controls are the broker's gender, an indicator for high tenure status (licensed as a health insurance broker for over 4 years), and an indicator for having many insurance carrier appointments interacted with non-call-center status.[28] Caller controls include caller gender, real age and age squared, race, college education status, and caller experience (the number of audit calls the caller has previously completed).[29] We also include a call center fixed effect, a fixed effect for the caller's assigned state of residence, and an indicator

[25]Consumers shopping for a mortgage, for example, are frequently advised to bring competing quotes to a broker.

[26]The individual-level random assignment applies to non-call-center brokers. For call-center brokers we use the call center's random matching of seniors to brokers.

[27]We originally assigned both treatment and control condition calls simultaneously, but noticed problems with treatment compliance and switched to implementing the treatment in waves.

[28]We use 4 years as the cutoff for high tenure since health insurance licenses are granted in 2-3 year cycles in most states, so observing a tenure of over 4 years indicates the broker has renewed their license at least once. We only allow appointments to vary for non-call-center brokers since broker appointments are set by the call center and therefore do not vary significantly within the firm or within call center status.

[29]Race is an indicator for whether or not the caller is white. College education status is an indicator for whether the caller holds an associate's degree or higher. Both use the shopper's self-reported data.

for the Medicare Annual Enrollment Period (AEP) interacted with call center status.[30]

Regressions on the MA outcome also include an indicator for calls placed after the September 1, 2025 commission removal interacted with call center status. This is when the commission change was made salient to call-center brokers. For non-call-center brokers the incentive and plan menu change became salient on October 1, 2025, the carrier exit announcement date. We control for calls placed after October 1 interacted with non-call-center status. Regressions on a Medigap outcome also include a plan letter (N or G) fixed effect. We also cluster standard errors at the caller level. Table footnotes state any departure from the baseline set of controls.

5 Results

5.1 Baseline Quality of Broker Advice

In total, across all treatment arms, we generate over 1,692 recommendations from 637 phone calls to Medicare insurance brokers. Since some seniors in our UAS survey do call multiple brokers, we pool the results in our control group (no perceived competition) and treatment group (perceived competition treatment), to reflect the average recommendation quality seniors might receive. To measure broker recommendations when incentives are misaligned, we limit the level of baseline Misaligned Recommendation and MA Recommendation results to calls where brokers are paid the full commission for MA enrollments.[31] Table 4 summarizes the baseline broker recommendations for our main outcomes of interest.

Brokers depart from the welfare-maximizing benchmark on both decisions. For Outcome 1, whether the broker recommends Medicare Advantage (MA) or Traditional Medicare plus Medigap, we see that brokers steered the consumer toward an MA plan in 32% (SE 2) of calls. While MA plans often have lower premiums, they impose network restrictions, require prior authorization, and are not recommended for seniors who travel often, violating the consumer's stated preferences assigned in the script. We do not find evidence that the Medigap markups in Decision 2 are intentionally inflated to make Medigap look worse and to rationalize MA recommendations. We include this result in Appendix D.4.

For Outcome 2, the Medigap Markup, we find that brokers recommend a plan that is on average $377 per year (SE 9) more expensive than the optimal available option, representing a 29% markup over the low-cost identical plan available to the senior. This excess premium represents a pure

[30]Call centers recruit new brokers and hire brokers on a temporary basis during AEP, the highest-volume enrollment period of the year, so the composition of brokers answering call center calls tends to change during those months. The interaction absorbs that change.

[31]We also limit Medigap markup observations to calls where the resident state of the caller is one where the full MA commission is paid. Changing the sample to include calls where MA commissions are not paid does not materially change the Medigap markup.

Table 4: Baseline Broker Recommendations

Outcome Variable	**Mean**	**SD**	**Obs**
Panel A: Primary Outcomes			
Misaligned Recommendation (%)	52%	50	458
Recommend MA (%)	32%	47	458
Medigap Markup ($ Yearly)	$ 377.10	280.65	1,055
Panel B: Secondary Outcomes			
Medigap Pct Markup (% Yearly)	29%	23	1,055
Medigap Broker Markup ($ Yearly)	$ 168.25	281.65	1,002
Avg Medigap Premium ($ Yearly)	$ 1,764	383.04	1,055

Notes: Table 4 shows the baseline broker recommendations. Recall that for the senior preferences we assigned, the welfare-maximizing choice is Traditional Medicare plus Medigap for Outcome 1 (Recommend MA %) and the lowest-cost Medigap Plan available to the senior for Outcome 2 (the Medigap Markup). Panel A shows our primary outcomes of interest. Recommend MA (%) and the Misaligned Recommendation rate are computed on the 458 calls on which the broker earns the full MA commission, excluding the no-commission call-center calls and the post-carrier-exit calls of Experiment A. Panel B shows the secondary outcomes.

wealth transfer from the consumer to the broker (and insurance company) for a plan with identical coverage. Brokers recommend a zero markup Medigap plan (the lowest-cost plan with identical coverage available to the senior) in 6% of 1,055 baseline Medigap recommendations. Brokers recommend the low-cost Medigap plan within the choice set of insurance companies they are *currently* licensed to sell on 19% of Medigap recommendations with an average broker markup of $168 a year. Finally, brokers recommend a high-markup Medigap plan (or a Medigap plan with a markup of at least $360 a year) on 41% of calls.

Combining Outcome 1 and Outcome 2, we say that brokers provide a misaligned recommendation if they either recommend MA or recommend a high-markup Medigap plan. 52% (SE 2) of brokers provide misaligned recommendations and steer seniors toward higher-commission plans, even when the plan is less suitable for the senior.

Several precautions in our design suggest these estimates may be lower bounds on the level of broker steering. First, given the basic training we provide on the Medicare market to ensure our shoppers can correctly identify recommendations, our senior shoppers may appear more sophisticated than the average senior. Second, to ensure the Medigap recommendation, which comes second, is not contaminated by the initial Traditional Medicare versus MA recommendation, we instruct the shoppers to open the call with an overview of the two topics of conversation (MA plans and Medigap plans). This overview may signal that the shopper has a basic understanding of Medicare and increase perceived consumer sophistication. Finally, we try to limit the number of calls during AEP, when brokers are especially busy and may be more likely to make mistakes.[32]

[32]AEP is also when individuals usually change MA plans, meaning MA might be more likely to be on the broker's mind, so incorrect MA recommendations may be more common during this time.

Outcome 1: Medicare Advantage In Section 4.1.1, we show that brokers estimate that 1 year of having an MA plan would cost a Medigap-fit senior approximately $5,574 using the plan attribute valuations brokers report in our survey (Figure A.8). Since seniors cannot switch out of MA without medical underwriting, we assume that the annual loss recurs. We hold the senior in the MA plan and accumulate the loss from age 65 through the life table's terminal age of 100, weighting each year by survival and discounting at a real rate of $r = 3\%$; Appendix D.1 states the accounting identity, the elicitation, and the discounting and mortality assumptions.

We elicit the valuations from brokers rather than recover them from observed plan choices, which existing research has shown to be distorted by consumer choice frictions. Two potential concerns arise with a broker-elicited measure, and both work against our results. The first concerns broker motives: a broker inclined to rationalize their own MA steering would understate the Medigap-distinguishing protections or overstate MA's ancillary benefits, shrinking our measured consumer surplus loss toward zero. The second concerns broker competence. If brokers do not understand the different Medicare products, a concern we show in Section 5.5 to be valid, the elicited measures would be noisy. Restricting the sample to the brokers who score highest on the knowledge test would increase our MA consumer surplus loss estimates (Appendix D.1); we use the full-sample estimate to be conservative.

Over a 36-year retirement, using the broker beliefs, we estimate that an accepted broker MA recommendation would cost a Medigap-fit senior roughly $5,574 a year or $79,756 in mortality-adjusted present value dollars. For the average senior who may not be as strong a fit for Medigap as our audit study senior, the loss is $2,711 a year or $38,790 in mortality-adjusted present value dollars. Figure A.13(a) plots the lifetime loss accumulation from an MA enrollment for a Medigap-fit senior. Given the assumptions needed to estimate the welfare loss for MA, we place greater weight on the consumer surplus estimates from the dominated Medigap recommendations below.

Outcome 2: Medigap Markup To value the Medigap markup we project the year-1 markup forward over the enrollment horizon of our hypothetical 65-year-old senior with an average (10%) premium growth rate. Appendix D.2 states the full methodology.

We estimate that a dominated Medigap recommendation in our audit sample costs the senior approximately $12,000–$24,000 of forgone wealth over their lifetime under typical nominal premium growth rates of 10% and 15% a year. Table A10 reports the full set of estimates and Figure A.13(b) shows the monetary loss from the dominated plan choice over time. Figure A.14 plots the same accumulation under every premium growth rate we consider.

What Predicts Advice Quality – Seniors vs Brokers We then analyze whether senior or broker characteristics are more predictive of recommendation quality. Doing so helps us pinpoint the nature of the market failure. While we do not explicitly vary consumer characteristics to test

for discrimination, seniors are randomly assigned to call the insurance brokers, so we can interpret any correlation between recommendation quality and senior characteristics causally. Isolating these drivers is also important for designing effective policy interventions.

Table A8 in Appendix D shows the coefficients from regressions of our two primary measures of broker quality on senior and broker characteristics. Note that the caller characteristics are rarely, if ever, statistically significant. Broker characteristics, on the other hand, tend to be correlated with advice quality. Call-center brokers are more likely to recommend MA on Outcome 1, suggesting firm-level differences such as in training or incentives may be important. Brokers with higher tenure also tend to provide higher-quality advice on both outcomes. This suggests broker experience or selection may be important. We discuss this heterogeneity further in Section 5.5.

5.2 What seniors value and learn on calls

That advice quality varies with broker characteristics and not with consumer characteristics emphasizes the importance of understanding the supply-side of the market. In this section, we evaluate whether the demand-side feedback might help discipline the market.

To better understand which call attributes seniors value, we ask our secret shoppers to provide an insurance broker rating after every call within the audit study and test whether the ratings are correlated with our two quality measures.[33] Shoppers are asked the following question:

> *"On a scale from **1 (definitely would not recommend)** to **5 (definitely would recommend)**, how likely are you to recommend this insurance broker to a friend or family member selecting insurance in this state?"*

Shoppers give brokers a rating of 4 (likely to recommend) or 5 (definitely would recommend) on 79% of calls. Recall that the seniors in our audit study conduct multiple broker calls (specifically, an average of 6 calls per shopper) and must pass a multiple-choice quiz demonstrating a basic understanding of the Medicare market before making any calls, meaning our auditors may be more sophisticated and possess more Medicare knowledge than the average senior. This suggests that, even among positively selected seniors who are making multiple phone calls, the vast majority of brokers seem competent and high-quality.

Table 5 regresses the probability of a highly rated broker call on our measures of broker recommendation quality and call length. Each column reports a separate regression of an indicator for a high rating (a rating of 4 or 5 out of 5), measured in percentage points, on one of our objective measures of broker recommendation quality. We standardize each measure so each coefficient

[33] We focus on the overall recommendation rating. However, seniors were also asked to provide ratings for broker knowledge and broker friendliness. The three ratings are highly correlated. The overall recommendation rating was added to the shopper response form at a later date. For the 20 calls where the response to the overall recommendation rating is missing, we take the average of the broker knowledge and broker friendliness ratings.

reports the change in the probability of a high rating, in percentage points, from a one standard deviation increase in that measure.

The results highlight the disconnect between our objective measures of broker recommendation quality and the subjective consumer ratings. None of the four measures of broker quality spanning Decision 1 and Decision 2 shifts the probability of a high rating by an amount statistically distinguishable from zero. Call length does, however, shift the broker rating. The results in Column (5) show that a call that is one standard deviation, or 14 minutes, longer raises the probability the broker receives a high rating by 6.16 percentage points. The results suggest that seniors' ratings do not vary with objective quality metrics such as the plan recommendation, and instead reward noisy proxies for quality such as time spent on the call.[34] Appendix D.3 shows how the correlation between senior ratings and recommendation quality evolves as callers gain experience with the market. The evidence is consistent with limited caller learning.

Table 5: Broker Rating by Recommendation Quality and Call Characteristics

	(1) Pr(High Rating) (4 or 5)	(2) Pr(High Rating) (4 or 5)	(3) Pr(High Rating) (4 or 5)	(4) Pr(High Rating) (4 or 5)	(5) Pr(High Rating) (4 or 5)
Recommend MA (per 1 SD)	-2.37				
	(2.07)				
Medigap Markup (per 1 SD)		-0.76			
		(1.60)			
Medigap Broker Markup (per 1 SD)			-0.57		
			(1.57)		
Medigap Markup % (per 1 SD)				-0.70	
				(1.61)	
Call Length (per 1 SD)					6.16***
					(1.91)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055	637
R^2	0.05	0.08	0.08	0.08	0.06

Notes: Table 5 shows the relationship between our objective measures of broker recommendation quality, call length, and the senior's rating of the broker. A high rating is a 4 (likely to recommend) or 5 (definitely would recommend) on the five-point recommendation question reproduced in Section 5.2. Each column reports a separate OLS regression of the high-rating indicator, scaled to 0–100, on the row variable and controls. Table A9 reports the same regressions in raw units. Columns (1) and (5) are estimated at the call level; columns (2)–(4) are estimated on the sample of Medigap price quotes (up to one quote per plan type and state per call), so the number of observations differs across columns. We include baseline controls plus a control for assignment to the Perceived Competition treatment. Columns (2)–(4) include the Medigap plan-letter fixed effect. Standard errors are clustered at the caller level. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Consumer feedback through ratings, referrals, and regulatory action (e.g. reporting broker steering) is often the primary mechanism through which consumers might discipline low-quality intermediary advice. Seniors who rely on brokers because they have an imperfect understanding of the Medicare market do not have a benchmark against which to judge recommendation quality. This inability to recognize and punish low-quality advice, even after repeat-exposure to the market as in Appendix Figure A.15, may allow agency problems to persist. Our ratings are only a proxy

[34]Table A9 reports the same regressions in the percentage point or dollar units of each measure. The result provides empirical support for the model assumption that consumers cannot evaluate plan fit.

and do not reflect senior satisfaction with the recommendation *after* enrollment. However, the initial broker ratings are nonetheless informative given limited feedback and opportunities for consumer learning in this market.[35]

Another reason consumers may be unable to discern Medicare plan fit and identify low-quality advice is that brokers may not be educating beneficiaries or not providing them with accurate information about Medicare plans during these phone calls. To test this within the audit study, as part of the post-call survey we instruct our secret shoppers to ask and report whether the broker informs the shopper that Medigap plans are standardized or that all Medigap plans within a letter are the same. Since all shoppers are instructed to ask about Medigap plans regardless of the plan recommended in Decision 1, one might expect experts looking to educate consumers on the different Medicare plans to bring up this important fact; however, shoppers report being informed that Medigap plans are standardized on only 56% of calls.

To further understand why our secret shoppers may have difficulty discerning Medicare plan fit, we use text analysis to analyze the call transcripts from the Medicare plan consultations. We use a large language model to analyze 497 transcripts (78% of the audit calls) to better understand the content of the phone calls.[36]

Seniors may not learn with greater exposure to the broker market if brokers do not educate the seniors or if brokers make inaccurate or contradictory remarks that are difficult to verify. Figure A.18 shows the justifications brokers give for recommending Medicare Advantage (MA) and Traditional Medicare (TM) + Medigap (MG) in Decision 1. Recommendation reasons do vary along predictable product attributes: brokers cite additional benefits and lower monthly premiums as reasons for recommending MA on 66% and 75% of calls respectively, versus 9% and 34% of calls where Medigap is recommended. Each of the seven reasons, however, is cited for *both* recommendations, highlighting that the same reasons can be used to justify opposite recommendations when consumers do not understand the products. Thus, a senior who cannot verify the plan attributes learns little from the call.[37]

[35]A Medicare Advantage enrollee can change plans at most once a year and most do not change unless circumstances require it, and Medigap enrollees change plans even less often. This limits occasions where beneficiaries are forced to update their assessment of both the quality of their Medicare plan and of the broker who advised them.

[36]We use OpenAI's `gpt-4.1-mini`, queried through the Batch API with the temperature set to zero and with a fixed seed. The LLM reads the full transcript and returns a single JSON record per call. A fixed prompt instructs the model to code the call transcript along several dimensions, including identifying several statements made on the call, documenting information provision and coverage of key Medicare recommendation topics, and subjective ratings from two separate personas: 1) a first-time Medicare beneficiary and 2) a Medicare compliance auditor.

[37]For example, "predictable healthcare costs" is cited on 67% of calls recommending TM + MG and still on 23% of calls recommending MA. A broker recommending TM + MG might emphasize that an MG plan is better for someone who wants fewer unexpected bills and predictable costs since MG covers a large portion (and sometimes all) of OOP costs. Conversely, a broker recommending MA might emphasize that someone who wants fewer unexpected bills and predictable costs should have an MA plan because MA plans have a maximum OOP limit while some MG plans do not have an annual OOP limit. Both statements are factually correct, but the second statement is misleading since the beneficiaries with MA plans tend to pay more in OOP costs.

The LLM analysis further highlights that calls focus disproportionately on current costs and health status and few calls discuss future risks and uncertainty (Figure A.17 in Appendix D.7). Brokers highlight a $0 monthly premium on 57% of calls and discuss the trade-off between low premiums and low out-of-pocket costs on 59% of calls. Brokers also ask about current health status on 67% of calls. In contrast, few calls discuss future health risks and uncertainty: brokers ask about expected future health risks on 6% of calls, personal health history on 4%, and family health history on 1% of calls. The transcripts suggest that brokers are not educating consumers on the long-term health and financial implications of choosing between Medicare Advantage and Medigap. The initial enrollment period is a unique window when beneficiaries can sign up for any Medicare plan without medical underwriting. Consumers who are not fully informed about the long-term trade-offs inherent in choosing a multi-year insurance contract risk being locked into an unsuitable plan, especially if their health deteriorates.

5.3 Experiment A: Changing Incentives

Because seniors cannot distinguish suitable from unsuitable advice, little in the market punishes a broker who follows their own financial incentive. We now ask how much of the advice those incentives account for. To do so, we exploit a natural experiment where both national Medicare Advantage carriers stop paying broker commissions to reduce enrollment in mispriced plans.

To trace how MA recommendations evolve in calendar time around the commission removal, we estimate an interrupted time-series specification. Let t_i denote the calendar date of call i, and let $t^\star$ = September 1, 2025 be the event date, when all call center carriers stopped paying MA commissions.[38] We aggregate calendar dates into four-week bins ($d = 28$ days), placed so that bin 0 begins at $t^\star$: call i falls in bin $k_i = \lfloor (t_i - t^\star)/d \rfloor$, so pre-event calls have $k_i < 0$ and post-event calls have $k_i \geq 0$. We estimate

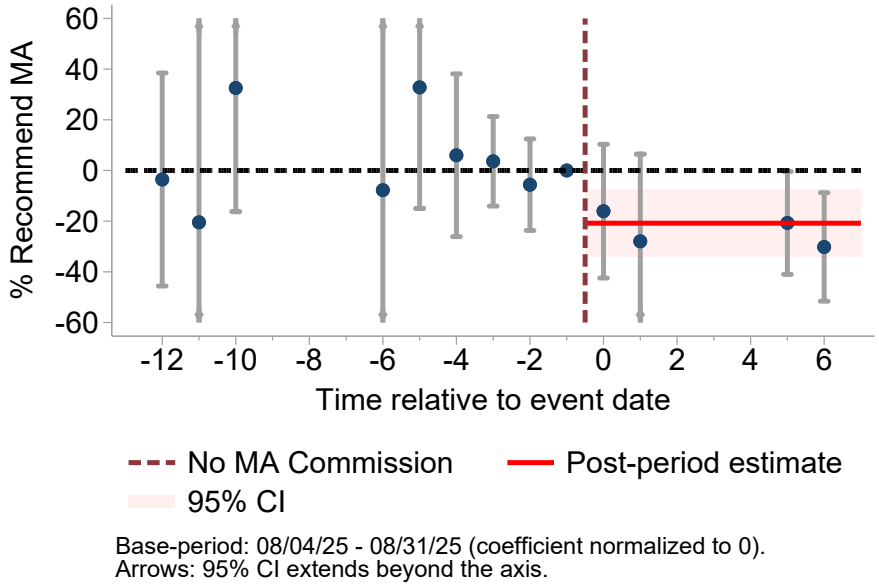
$$\text{Rec MA}_i \;=\; \alpha \;+\; \sum_{k \neq -1} \beta_k \, \mathbb{1}\{k_i = k\} \;+\; \gamma' \mathbf{X}_i \;+\; \varepsilon_i, \tag{12}$$

where the sum runs over the bins observed in the sample and bin -1 (the four weeks ending August 31, 2025) is the omitted reference period, so each β_k measures the difference in the MA recommendation rate between bin k and the four weeks immediately before the commission removal. The sample is restricted to the no perceived competition arm (that is, to calls where the shopper was not assigned to inform the broker that they were talking to competitors); this ensures the variation

[38] Recall that there are differences in the timing of when the incentive removal became salient to call-center versus non-call-center brokers. For call-center brokers, the plans were immediately removed from the internal plan search systems as soon as the company stopped paying commissions. For non-call-center brokers, who do not always have separate plan search platforms and often use Medicare.gov to check which plans are available, the plan commission removal was not salient until October 1, when Anthem and Aetna also announced plans to leave the MA market in the county of interest beginning January 1, 2026. Since non-call-center brokers could still contract with a carrier that continued to pay MA commissions, Imperial Health, the commission removal applies only to call center brokers.

in the outcome is not contaminated by the experimentally introduced perceived competition. $\mathbf{X}_i$ is the vector of baseline controls, with the September 1, 2025 indicator omitted because the event it marks is what the bins trace. Figure 4 plots the estimated coefficients $\widehat{\beta}_k$.

Figure 4: MA Recommendation Rates Over Time



Notes: Figure 4 plots the estimated coefficients $\widehat{\beta}_k$ from equation (12) with 95% confidence intervals: the difference in the MA recommendation rate (in percentage points) between each four-week calendar bin and the omitted reference bin, the four weeks ending August 31, 2025. The sample is the no perceived competition arm of the audit. The vertical line marks September 1, 2025, when all call center carriers stopped paying MA commissions; the horizontal line shows the pooled estimate from equation (13). Baseline controls. Standard errors are clustered at the caller level.

To measure the pooled impact of the MA commission change, we estimate:

$$\text{Rec MA}_i \;=\; \alpha \;+\; \beta_1\,\text{Call Center}_i \;+\; \beta_2\,\text{No MA Commission}_i \;+\; \gamma'\mathbf{X}_i \;+\; \varepsilon_i, \tag{13}$$

where $\text{No MA Commission}_i \equiv \mathbb{1}\{\text{Call Center}\}_i \times \mathbb{1}\{\text{CA}\}_i \times \mathbb{1}\{t_i \geq \text{September 1, 2025}\}$ and CA_i indicates that the shopper's assigned primary state is California. The indicator equals one exactly when the broker answering call i earns no commission from an MA enrollment: call center carrier appointments are set at the company level, and on September 1, 2025 every carrier available to call center brokers stopped paying MA commissions in our California county, while non-call-center brokers could still get appointed with additional companies that continued to pay. MA plans themselves remained available for consumers to enroll in throughout (Section 4). The coefficient of interest, β_2, measures the change in the MA recommendation rate for call center brokers once the MA commission is removed; because every post-September-1 call in the estimation sample is a call center California call, β_2 is identified from the before–after comparison for call center calls, conditional on covariates. $\mathbf{X}_i$ contains the controls from equation (12) other than the call center

indicator, which now enters separately.[39]

Table 6: Natural Experiment: MA Incentives Removed

	(1) Recommend MA (%)
No MA Commission	-20.80***
	(6.78)
Counterfactual Mean (No Commission Change)	44.39
Controls	Yes
Observations	384
R^2	0.14

Notes: Table 6 reports the OLS estimates of equation (13) on the no perceived competition arm of the audit sample. The dependent variable is the MA recommendation rate or share of calls on which the broker recommends MA over TM + MG. The coefficient on "No MA Commission" is the change in that share for California call center brokers after carriers stopped paying MA commissions on September 1, 2025. The counterfactual mean is the share of calls on which these brokers would have recommended MA over the same months had the commission remained, and is the base against which the text expresses the estimate as a percentage. Baseline controls; the September 1, 2025 commission removal enters as the coefficient of interest rather than as a control. Standard errors are clustered at the caller level. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Table 6 shows that removing MA commissions lowers the share of calls on which a California call center broker recommends MA by 21 percentage points (SE 7). Absent the change, these brokers would have recommended MA at a counterfactual rate of 44% of calls over the same months. The commission removal therefore lowers the MA recommendation rate by 47% relative to the counterfactual mean. Crucially, the non-commissionable plans and every other MA plan remained in the market despite the commission change. Consumers could still enroll in the Anthem and Aetna plans through the end of 2025 and in the Imperial Health MA plans throughout 2025 and 2026, and a broker who checked Medicare.gov would still see all three MA plans available. Administrative enrollment data point the same way: in the audit county, enrollment shifted away from the Medicare Advantage plans that stopped paying commissions and toward the plans that continued to pay them (Appendix E).

The second important takeaway from the natural experiment is that removing commissions does not *completely* eliminate MA recommendations even when recommending MA 1) is unsuitable for the senior, and 2) does not earn the broker a commission. We find that call center brokers still recommend MA on 24% of calls, even after commissions are removed. This suggests MA recommendations are partially driven by broker information frictions such as limited product knowledge, limited training, or a genuine belief that MA is the best-fit plan for seniors in this scenario. We discuss this further in Section 5.5.

[39]The AEP × Call Center indicator is identified off the 2024 and 2025 enrollment windows; the specification cannot include a smooth calendar-time trend or a stand-alone post-September-1 indicator, either of which would be collinear with the treatment in this sample.

5.4 Experiment B: Changing Perceived Competition

Experiment A shows that misaligned recommendations respond to the broker's pay and decomposes the causes of broker steering; Experiment B tests whether competitive pressure can discipline brokers and improve advice quality. Our main regression specification is:

$$Y_{is} = \alpha + \beta_1 \mathbb{1}(\text{Perceived Comp}_{is}) + \beta_2 \text{Call Center}_{is} + \gamma' \mathbf{X}_{is} + \epsilon_{is} \tag{14}$$

where Y_{is} is the recommendation outcome provided by broker i to senior s. Perceived Comp_{is} is an indicator variable for whether the senior s was assigned the competition line for the call with broker i. Call Center_{is} is an indicator variable for broker firm (that is, whether broker i is reached through a call center phone number). Finally, $\mathbf{X}_{is}$ is the vector of baseline controls. Additional robustness checks, including specifications that control for caller experience and calendar time and the instrumental-variable estimates, are in Appendix D.

Our main coefficient of interest is β_1, which measures the change in broker recommendation quality in response to the perception of increased competition. Our main specification reports the Intent to Treat (ITT) in Table 7, where we regress the main outcomes of interest on treatment assignment. The IV results are similar in magnitude and are shown in Appendix D.6.

Table 7: Impact of Perceived Competition Assignment on Advice Quality

	(1) Recommend MA (%)	(2) Medigap Markup ($)	(3) Medigap Broker Markup ($)	(4) Medigap % Markup
Perceived Competition Assignment	11.33**	-84.44***	-84.11***	-7.10***
	(4.92)	(19.95)	(17.24)	(1.65)
Generalized Lee Bounds		[-168, -25]	[-153, -14]	[-17, -2]
Control Mean	28.13	407.25	217.69	31.47
Controls	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055
R^2	0.16	0.16	0.16	0.23

Notes: Table 7 reports the treatment effect of perceived competition assignment on our main measures of broker recommendation quality. Baseline controls. Column (1) adds the indicator for the September 1, 2025 commission removal interacted with call center status and the indicator for the October 1, 2025 announcement that major Medicare Advantage carriers would leave the market, interacted with non-call-center status; Medicare Advantage plans remained available for enrollment throughout. Columns (2)–(4) add the Medigap plan-letter fixed effect. To account for selection on the Medigap outcome we report the Semenova generalized Lee bounds 95% Confidence Region (Semenova, 2025). Standard errors are clustered at the caller level. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Focusing on Outcome 1, column (1) shows that the competition treatment *increases* the rate of Medicare Advantage (MA) recommendations by a statistically significant 11 percentage points (SE 5). Recall that we assign consumer preferences so the senior is a strong fit for Traditional Medicare plus Medigap according to Medicare.gov, even though brokers are paid more for enrolling seniors in Medicare Advantage.

This result may seem counterintuitive; we often expect competition to increase efficiency and

improve consumer outcomes and one might expect the threat of external verification to lead brokers to improve their recommendation quality. On Decision 1 the treatment raises the share of brokers recommending the higher-commission product.

The Medigap outcome, however, implies the opposite result. Columns (2) to (4) of Table 7 show that across all the ways to measure the Medigap markup, the competition treatment causes brokers to recommend lower-cost Medigap plans. Since Medigap products are standardized and homogeneous within plan letter, these recommendations save the consumer money and *improve* consumer welfare. Since brokers are paid a percent of the Medigap premium, this implies brokers respond in a way that *lowers* their pay. On this decision the treatment lowers the mean recommended markup by $84.44 a year, which is a dollar-for-dollar increase in consumer surplus for the senior since the plans have identical coverage.

Together, the results show that increasing perceived competition causes brokers to compete on price by recommending the *lower-price plan*, even when it reduces consumer welfare. In Decision 1 (the MA outcome), when the decision is between heterogeneous products, brokers respond to perceived competition by recommending MA, which is lower in price than TM + Medigap, even though it does not match the consumer's preferences. Conversely, in Decision 2, when the broker is choosing between homogeneous products, the competition treatment causes brokers to recommend *lower-price plans with identical coverage*, increasing consumer surplus. Furthermore, the Medigap results suggest that brokers do possess the ability to provide high-quality recommendations, but supply them strategically, withholding the cheaper coverage-identical plan from consumers who give no sign that they will compare quotes.

Taken together, the two decisions also rule out the most natural alternative reading. The signal does not push brokers toward higher commissions: it raises MA recommendations, which pay more, but also lowers the Medigap markup, which pays less. A broker answering competitive pressure by maximizing expected commissions would have moved the same way on both margins; these move in opposite directions. When a sale becomes less certain, brokers recommend lower-premium plans to close it, whether or not the plan fits the consumer and whether or not it maximizes their own pay.

The results also highlight that competitive pressure and consumer search may have unintended consequences in complex markets with asymmetric information where quality is difficult to verify. In complex markets where consumers do not fully perceive product quality and do not understand intermediary incentives, competition may actually *reduce* consumer welfare, as in Outcome 1. The mechanism is that a broker who expects a sale to be less certain moves toward whichever recommendation the senior is more likely to accept, and which recommendation that is depends on the utility the consumer perceives. Medicare Advantage is the easier sale whenever the senior perceives less than 0.25 of the harm from an unsuitable plan. This threshold is the premium

Medicare Advantage saves the consumer divided by the harm from an unsuitable plan that it costs the consumer. In this case, the share of brokers recommending the unsuitable plan rises when the consumer signals that they will talk to multiple brokers and compare quotes. In Decision 2 nothing is hidden: the plans are coverage-identical, the cheaper plan is always the easier sale, and the same force lowers the markup of the recommended plan. Appendix A states the model and proves both results.

Finally, we test whether the result is an artifact of caller learning or the timing of the treatment waves by controlling for caller experience and calendar time trends. Appendix Table A11 shows that the results are robust to these specifications.

Discussion of Alternative Explanations The perceived competition treatment can signal several attributes of the caller at once: that the consumer is sophisticated, that they are price sensitive, that their search costs are low, or simply that the sale is now less likely to close. One limitation of the treatment is that we are unable to precisely identify the mechanisms; that is, we cannot fully separate a change in what the broker infers about the consumer (sophistication, education, income, etc.) from a change in competitive pressure that impacts the broker's incentive to make the sale. Below, we discuss the leading alternative explanations in turn.

The first potential confound is that planning to talk to multiple brokers and get multiple quotes signals that the consumer is informed or more educated. However, one would expect brokers to give *better* advice to shoppers they believe to be sophisticated. This should improve recommendation quality on every margin, including the product margin of Decision 1, and we would see *fewer* MA recommendations for our Medigap-fit senior. We find the opposite (Table 7): sophistication predicts the wrong sign and cannot account for the MA result, though it may reinforce the Medigap price response.

Another credible alternative explanation is that the perceived competition treatment signals price sensitivity. That is, a consumer who searches and solicits multiple quotes may be revealing that they are income constrained or care mainly about price.[40] This explanation rationalizes the results of both Outcome 1 (MA) and Outcome 2 (Medigap); a broker accommodating a price-sensitivity preference would steer consumers toward the cheapest premium in both markets and would think they are improving consumer welfare, even when there are no other changes to income and shoppers are explicitly instructed to say they can afford any plan.

Recall that this would require brokers to rationalize a level of price sensitivity where an average premium savings of \$120/month (\$1,440/year) is enough to overcome an estimated consumer surplus loss of \$5,574/year for a Medigap-fit consumer or \$2,711/year for the average senior.

[40] A similar interpretation is that brokers know consumers have a bias towards lower premiums and that brokers are catering to rather than correcting these biases as in Anagol et al. (2017) and Mullainathan et al. (2012).

These numbers are based on the stated beliefs of brokers themselves.[41] Because brokers who answer more of the survey's knowledge-check questions correctly report larger consumer surplus losses (Appendix D.1), we view these valuations as lower bounds.

We also test for price sensitivity within the audit study calls using a large language model (LLM) analysis. For each call where we have a transcript, we ask the LLM to rate the shopper's perceived price sensitivity on a one-to-five scale.[42] We might expect the LLM to rate the callers who deliver the perceived competition line as more price sensitive. Columns (1)–(3) of Table 8 show that the rating does not move: assignment to treatment changes the price sensitivity rating by -0.04 points (SE 0.05) against a control mean of 2.87 and a control standard deviation of 0.57. Column (4) shows that the impact of the perceived competition treatment on the MA outcome remains, even when limited to the subset of calls with call transcripts.

Table 8: Perceived Competition and Caller Price Sensitivity

	(1) Caller Price Sensitivity (1-5)	(2) Caller Price Sensitivity (1-5)	(3) Caller Price Sensitivity (1-5)	(4) Recommend MA (%)
Perceived Competition Assignment	-0.04	0.08	0.00	12.34**
	(0.05)	(0.06)	(0.04)	(5.46)
Control Mean	2.87	2.87	2.87	29.43
Controls	No	Yes	Yes	Yes
Caller Experience	None	Linear	Fixed Effects	Linear
Observations	497	497	497	497
R^2	0.00	0.45	0.74	0.17

Notes: Table 8 reports the effect of perceived competition on how price sensitive the caller appeared, as coded by a large language model from the call transcript (Appendix D.7). The outcome in columns (1) through (3) is a one-to-five rating of the caller's overall price sensitivity; in column (4) it is an indicator for whether the broker recommended Medicare Advantage, expressed in percentage points. All four columns report the intent to treat, regressing the outcome on assignment to the perceived competition treatment. Column (1) includes no controls, so it reports the difference in means that random assignment identifies. Column (2) adds the baseline controls. Column (3) replaces the linear control for caller experience with caller experience fixed effects, as in Appendix D.5. Column (4) uses the baseline controls of column (2). Instrumenting delivery of the perceived competition line with assignment leaves the same pattern for the rating: the treatment on the treated estimates are -0.05 (SE 0.06) without controls and 0.12 (SE 0.08) with them. The estimation sample is the 497 calls with a transcript rating, from 98 callers. Standard errors are clustered at the caller level. $^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$.

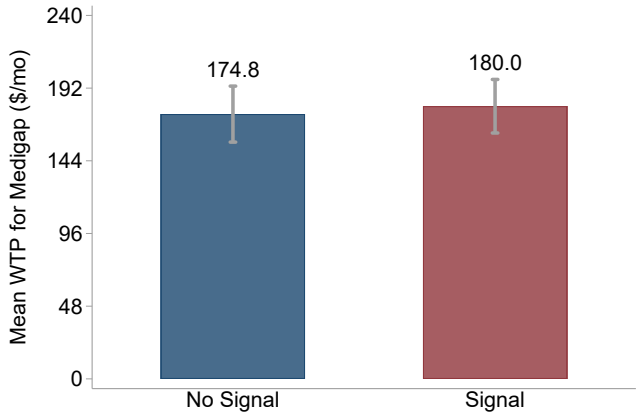
Finally, we directly test whether the treatment signals greater price sensitivity in our broker survey. We include an experiment where we hold consumer type fixed and show each broker two identical Medigap-fit prospective client vignettes that differ only in whether they deliver the perceived competition line. We also randomize the perceived competition line across the two vignettes. Figure 5 shows the maximum willingness to pay for a Medigap plan reported by the broker split by perceived competition signal. Recall that if the MA recommendation is driven

[41]We expect the survey population to be representative. If anything, the survey population may have a lower opportunity cost of time and be negatively selected.

[42]Ratings are available for 497 of the 637 calls. Coverage is 82% among calls assigned to control and 72% among calls assigned to treatment; the 10.8 percentage point gap (SE 5.3) falls to 4.1 percentage points (SE 5.2) once we control for call center status, since call center calls are 21.9 percentage points more likely to be transcribed.

by genuine concerns over the price of the Medigap premium, the difference in WTP would have to be larger than the average Medigap premium, or at least $120/month. We do not find large differences in WTP in response to the signal.

Figure 5: WTP for Medigap by Perceived Competition Treatment



Notes: Figure 5 shows the average WTP for a Medigap plan ($/mo) by competition signal, or by perceived competition treatment arm, from our broker survey. Grey bars are ±1 SE. Figure 5 shows the results from the within-broker randomization. Figure A.5 in Appendix C.5 shows the results from the across-broker randomization. The results are consistent across both specifications.

5.5 Evidence of Broker Information Frictions

This section presents evidence that brokers may not have perfect information on both of the two main Medicare plan types: Medicare Advantage (MA) and Medigap. These broker information frictions motivate our two broker type model described in Section 6.[43]

First, we show evidence from the audit call transcripts and broker survey. Brokers say they only sell either MA or Medigap on 131 of 497 transcribed calls. The MA-only brokers are concentrated at call-centers: 32 of the 33 calls where brokers say they only sell MA come from a call-center broker in a sample where call-centers make up 48% of calls.[44] This share is likely a lower bound since a broker who is only knowledgeable about one product is unlikely to volunteer this information. We see the same pattern in our survey of currently licensed Medicare insurance brokers: 18% of brokers

[43] Anagol et al. (2017) find similar evidence of intermediary information frictions among life insurance agents in India and Linnainmaa et al. (2021) find that financial advisors also show biases in their own investment accounts by holding high-fee funds, chasing returns, and being under-diversified.

[44] Whether a broker sells one product or both is the endogenous choice of the firm and the broker. Both audit states license MA and Medigap under a single health line of authority (LOA). No publicly recorded certification differentiates brokers who sell MA or Medigap. Insurance company appointments are attached to the LOA rather than a specific product (Tex. Ins. Code §§4001.201–4001.206). This means a broker appointed to sell health insurance for an insurance company can sell any of that carrier's health products, including both MA and Medigap plans. Federal regulation requires agents selling MA to be trained and tested annually. However, these product-specific requirements are not publicly recorded and it is unclear whether they are enforced. Medigap is regulated at the state level and does not have additional training requirements in most states.

only sell one of the two Medicare products (15% only sell MA and 3% only sell Medigap).[45]

The second piece of evidence is the result from the natural experiment that a large share of brokers continue to recommend MA when it is not a good fit for the senior and when it does not pay a commission (Section 5.3). We create a proxy for broker knowledge from the coded indicator for whether or not the broker stated on the call that Medigap plans are standardized. While this measure is endogenous, callers are explicitly told to ask about Medigap regardless of the initial recommendation, giving all brokers a chance to state this important fact. Including baseline controls, brokers without the knowledge proxy flag recommend MA on 37% of calls (versus 22% for brokers with the knowledge proxy). The results are shown in Appendix Figure A.7.

A third piece of evidence, also highlighted in Section 5.1, is that high-tenure brokers tend to provide better advice.[46] Table A13 in Appendix D.7 further highlights the broker knowledge disparities by showing that high-tenure brokers are more likely to ask key diagnostic questions needed to provide a high-quality recommendation than low-tenure brokers. The content of the transcripts is endogenous, meaning we cannot distinguish between a broker who provides a bad recommendation because they do not *know* which diagnostic questions to ask and a broker who *chooses* not to ask the questions so they can provide a bad recommendation. Nonetheless, the differences by broker tenure are consistent with broker learning or positive selection over time.

Finally, the results from our broker survey show that many Medicare brokers demonstrate important gaps in Medicare knowledge. In our survey we ask brokers to answer three multiple-choice knowledge-check questions phrased as hypothetical questions beneficiaries might ask them. These three questions are built from facts every licensed Medicare broker should know and that are clearly stated on Medicare.gov.[47] Performance is strikingly low for a licensed professional population: only 61% correctly identify that switching from MA to Medigap requires medical underwriting, 48% know that Traditional Medicare has no out-of-pocket cap, and only 39% that same-letter Medigap plans are standardized.

6 A Structural Model of Broker Advice

The two experiments establish that broker recommendations respond to financial incentives and the probability of losing a sale (perceived competition) and provide evidence consistent with broker information frictions. In this section we use a structural model to quantify the relative importance

[45]Brokers believe single-product specialization is far more common than their own licensure implies: on average they believe 28% of brokers sell only MA and 22% sell only Medigap, together half the market and roughly three times the 18% implied by their own licensure. The two sources also disagree about which of the two specializations is more common, and we do not adjudicate between them. The argument here turns on the share of seniors who reach a single-product broker, not on which product that broker carries.

[46]High-tenure brokers are brokers actively licensed for more than four years, as specified in the baseline controls.

[47]Figure A.10 shows the results. The full questions and answers are shown in Appendix Figure A.9.

of these forces and to estimate consumer surplus under different counterfactual policies. The estimated model is based on the conceptual framework in Section 3, but adds a misinformed broker type to represent brokers who, whether due to asymmetric training or information frictions, do not know that the Medigap plan matches the audited senior's preferences.

The model is deliberately parsimonious: its purpose is to isolate the supply-side factors governing the broker's recommendation decision and to recover the parameters needed to simulate counterfactual policies. We model the broker recommendation decision for our two main audit study outcomes: Decision 1 (Medicare Advantage versus Medigap) and Decision 2 (which Medigap plan to recommend). Broker pay (in PV terms to reflect the multi-year commission streams) is calibrated from the regulations in Table 1. Baseline sale probabilities and the net harm of an unsuitable recommendation are calibrated from the broker survey. The remaining parameters are estimated from the audit study; these include commission sensitivity β, the Decision 1 suitability cost location by firm type (non-call-center and call-center) $\bar{K}_{1,ncc}$ and $\bar{K}_{1,cc}$, the misinformed broker share λ, the sale-probability ratio on treated calls R_1, the perceived competition treatment deflator η, the Decision 2 suitability cost κ_2, and the appointment cost a_{appt}.

6.1 Setup

By design, every audited senior is a "Medigap-fit" senior, or a senior whose assigned preferences match the attributes of Medigap plans (Section 4.1.1). On call i, broker b makes the two recommendations observed in the audit study. As in Section 3.3, the broker trades off pay against a *perceived* suitability cost of recommending the "misfit plan," or the plan the broker believes does not match the senior's preferences (Decision 1), or against the suitability cost of recommending a plan that overcharges the senior for identical coverage (Decision 2).

In Section 3, we use plan j to index both the Decision 1 and Decision 2 recommendations to highlight the symmetry in the perceived competition result. In this section they are estimated using different data and are kept separate: $r \in \{\text{MA}, \text{MG}\}$ indexes the Decision 1 plan recommendation and j indexes the Decision 2 Medigap recommendation. Table 9 lists the model parameters and the source of identifying variation.

Decision 1 Broker b recommends the plan r that maximizes expected utility, which is the probability of closing the sale times net pay, or pay net of the cost $\kappa_1 N_r$:

$$\max_{r \in \{\text{MA},\text{MG}\}} \; P_r \cdot \Big(W_r - K_1\Big), \tag{15}$$

P_r is the broker's perceived probability that the sale closes (or that the senior accepts recommendation r). $K_1 \equiv \kappa_1 N_r$ is the suitability cost the broker incurs when selling the misfit plan. κ_1 is the weight the broker places on the senior's harm and N_r is the annual net harm in Equation (1),

calibrated from the broker survey.[48] W_r is the broker commission for recommendation r.

Broker types We assume that broker b carries a latent type $\tau_b \in \{I, M\}$. The type governs which plan recommendation incurs the suitability cost: the informed type (I) knows Medigap plans match the preferences of the audited senior and believes MA plans to be the unsuitable plan. The misinformed type (M) believes MA to be the suitable plan and Medigap to be the unsuitable plan. The broker type is determined by information frictions: a broker of the misinformed type may not be fully informed or fully trained to sell both products, or may face inertia or other frictions that make it difficult to sell the other Medicare product.[49] Section 5.5 presents reduced-form evidence of the two broker types. The type probability is a logit in call-center status and log licensure tenure, both fixed before the call, with intercept γ_{ncc} (the index for a non-call-center broker of average tenure), call-center shift γ_{cc}, and tenure coefficient γ_{ten}. λ is the share of the misinformed brokers.

The informed broker knows that Medigap is the suitable product and incurs the suitability cost on an MA sale. Comparing the two payoffs in (15) gives each broker type a cutoff rule based on $K_{1,i} = \kappa_1 N_r$, the suitability cost (in dollars) of selling the plan the broker's type believes to be the unsuitable plan. Thus, the informed broker recommends MA if and only if $P_{\text{MA}}(W_{\text{MA},i} - K_{1,i}) \geq P_{\text{MG}} W_{\text{MG},i}$. Dividing both sides by P_{MA}, we have

$$\begin{aligned} I \text{ (Informed) Recommends MA iff:} \quad & K_{1,i} \;\leq\; W_{\text{MA},i} - R_{s_i}\, W_{\text{MG},i}. \\ M \text{ (Misinformed) Recommends MA iff:} \quad & K_{1,i} \geq \left(W_{\text{MG},i} - W_{\text{MA},i}/R_{s_i}\right). \end{aligned} \tag{16}$$

where $R_{s_i} \equiv P_{\text{MG}}/P_{\text{MA}}$ is the sale-probability ratio on call i, which the perceived competition treatment $s_i \in \{0, 1\}$ moves from R_0 to R_1. Since the broker compares two plan recommendations r, only the relative sale close probabilities, R_{s_i}, determine which product pays more in expectation; the level of each sale close probability alone is less important.[50]

For the informed broker type I, the right-hand side of Equation (16) is the "expected net pay gap from recommending MA," or the MA commission net of the Medigap commission, discounted by the relative sale-close probability. The informed type I only recommends MA if the suitability cost (incurred on an MA recommendation) is less than or equal to the expected net pay gap. Conversely, the misinformed type M incurs the suitability cost on a Medigap sale and recommends MA iff the suitability cost is larger than the expected net pay gap from recommending Medigap.

[48]The harm enters the broker's problem only through the product $\kappa_1 N_r$, so the audit identifies the dollar cost K_1 and not κ_1 and N_r separately.

[49]The misinformed type brokers can also be thought of as "MA-default brokers".

[50]The audit records recommendations, not enrollments, so the data cannot identify the true sale probability levels. This transformation highlights why the model does not need the levels: R_0 on control calls is calibrated from the broker survey and R_1 on treated calls is estimated from the perceived competition treatment (Section 6.2).

The suitability cost varies across calls around its location, $K_{1,i} = \bar{K}_{1,i} + \varepsilon_i$, where ε_i is drawn independently on each call from a logistic distribution with mean zero and scale $1/\beta$. The informed broker recommends MA when $\varepsilon_i \leq (W_{\text{MA},i} - R_{s_i} W_{\text{MG},i}) - \bar{K}_{1,i}$. The misinformed broker recommends MA when $\varepsilon_i \geq (W_{\text{MG},i} - W_{\text{MA},i}/R_{s_i}) - \bar{K}_{1,i}$. By symmetry of the logistic distribution, we have

$$\begin{aligned} \Pr(r_i{=}\text{MA} \mid I) &= \Lambda\Big(\beta\Big[\Big(W_{\text{MA},i} - R_{s_i} W_{\text{MG},i}\Big) - \bar{K}_{1,i}\Big]\Big), \\ \Pr(r_i{=}\text{MA} \mid M) &= \Lambda\Big(\beta\Big[\bar{K}_{1,i} - \Big(W_{\text{MG},i} - W_{\text{MA},i}/R_{s_i}\Big)\Big]\Big), \end{aligned} \tag{17}$$

where $\Lambda(t) = 1/(1+e^{-t})$ and s_i indicates assignment to the perceived competition treatment. All commissions are measured in present-value thousands of dollars. The term ε_i is variation in the suitability cost across calls, not a shock to the recommendation: it enters only the sale of the plan the broker's type believes to be the unsuitable plan. β is the commission sensitivity, or log-odds of an MA recommendation per present-value \$1,000 of commission.[51] We assume that commission sensitivity β is common across brokers (including informed, misinformed, call-center, and non-call-center brokers). The location $\bar{K}_{1,i}$ is the commission gap at which an informed broker type is as likely to recommend MA as Medigap. Equation (17) is a logit with slope β on the commission gap and intercept $-\beta\bar{K}_{1,i}$, so the dollar location is the intercept divided by the slope. We allow the suitability cost location to vary by call-center status: $\bar{K}_{1,i}$ equals $\bar{K}_{1,ncc}$ for non-call-center (ncc) brokers and $\bar{K}_{1,cc}$ for call-center (cc) brokers.

Decision 2 Since broker type is a belief about which product (MA versus Medigap) matches the senior's preferences, we assume that broker type does not enter broker utility for Decision 2. Broker b recommends a Medigap plan j from the senior's choice set according to:

$$u_{ij} = \beta\Big[e^{-\eta\, s_i m_j}\, z\mathcal{M}_{\text{MG}}\big(\rho_{\min,i} + m_j\big) \;-\; \kappa_2\, m_j \;-\; a_{\text{appt}}\mathbb{1}\{j \notin \mathcal{P}_i\}\Big] \;+\; \varepsilon_{ij}, \tag{18}$$

where ε_{ij} is i.i.d. extreme value and the bracketed term multiplied by β is the present-value commission dollars of plan j net of the Decision 2 suitability cost κ_2 and appointment cost a_{appt}.[52] Medigap commissions are a percentage z of the plan premium, so a higher-markup Medigap plan pays the broker more. The Decision 2 suitability cost κ_2 penalizes the Medigap markup: the broker incurs the cost per dollar of premium the senior pays above the low-cost coverage-identical plan. We assume a separate suitability cost for Decision 1 and Decision 2 (κ_1, κ_2).

The appointment cost a_{appt} reflects the cost of getting licensed to sell plans outside of $\mathcal{P}_i$, the portfolio of insurance carriers the broker on call i is already appointed to sell. The cost may include

[51] We report the scale of the logistic distribution of K_1 in dollars as $\mu_K \equiv \$1{,}000/\beta$ (its standard deviation is $\pi\mu_K/\sqrt{3}$): a \$1,000 change in either commission gap shifts the log-odds of an MA recommendation by $\$1{,}000/\mu_K$.

[52] Since all beneficiaries who enroll in Medigap must also enroll in Part D or pay a penalty, the calibrated Medigap commission also includes the present value commission stream from a Part D enrollment. The commission is suppressed in Equation (18) since the value is the same for every Medigap plan recommendation (Appendix F).

licensing fees, which can be waived by insurance companies, time costs, or general effort costs.[53] Note that in this setting appointments are not a binding constraint on the broker's choice set due to "just-in-time" appointments: brokers can still recommend plans outside their choice set and do so within our audit study on 29% of Medigap recommendations.[54]

To account for the broker response to the perceived competition treatment $s_i = 1$, the competition treatment deflator, or "the deflator", discounts the expected Medigap commission on treated calls $e^{-\eta s_i m_j}$. The deflator accounts for broker beliefs that the sale probability on a treated call could decrease (positive η) or increase (negative η) per dollar of markup m_j.

We assume the same commission sensitivity term β multiplies both Decision 1 and Decision 2. The assumption implies a present-value dollar of commission converts to the same utility in both decisions. The Medigap recommendation in our audit study data identifies only the net markup slope $\alpha \equiv \beta(z\mathcal{M}_{\text{MG}} - \kappa_2)$, the coefficient on the markup m_j in Equation (18), and the scaled appointment cost from the broker's recommendation given the consumer's choice set. Separating the costs relies on the assumption that the commission sensitivity β is the same in Decision 1 and Decision 2.[55] The Decision 2 model fit does not depend on this assumption. Appendix F discusses which parameters and counterfactuals are sensitive to the assumption.

6.2 Identification and Estimation

Table 9 summarizes each parameter of the estimated model and its source of identifying variation in the audit study.

We estimate the model by maximum likelihood. By assumption, broker type is a fixed attribute across calls. Let $\mathcal{C}_b$ be the calls broker b answered and $\mathcal{D}_2(i)$ the broker's Medigap plan recommendations on call i. Broker b's contribution to the likelihood is therefore:

$$\mathcal{L}_b \;=\; \Big[\lambda(x_b)\prod_{i\in\mathcal{C}_b}\Pr(r_i \mid M) \;+\; \big(1-\lambda(x_b)\big)\prod_{i\in\mathcal{C}_b}\Pr(r_i \mid I)\Big] \;\times\; \prod_{i\in\mathcal{C}_b}\prod_{\ell\in\mathcal{D}_2(i)}\Pr\big(j_{i\ell}\big)^{w_{i\ell}}, \tag{19}$$

where $\Pr(r_i \mid \tau)$ is the Decision 1 probability in (17) evaluated at the observed recommendation, $\lambda(x_b) = \Lambda\big(-(\gamma_{ncc} + \gamma_{cc}\, cc_b + \gamma_{\text{ten}}\,\text{ten}_b)\big)$ is the probability that broker b is the misinformed broker type, $\Pr(j_{i\ell})$ is the Decision 2 conditional-logit probability from (18) given the Medigap recom-

[53] Appendix F explains why κ_2 and κ_1 are distinct objects, and why the appointment cost is a cost the broker can avoid rather than a binding constraint.

[54] Just-in-time (JIT) appointments allow brokers to obtain appointments right as they are about to enroll a beneficiary in an insurance company's plan.

[55] The plan-choice data cannot test this assumption. In our model we assume that the Medigap commission is a fixed percentage of the premium. This means that within a senior's choice set every dollar of markup raises the commission by the same amount and the data identify α, but not its two components. The only payment that varies separately from the premium is a carrier bonus, which we observe only for plans the broker is already appointed to sell and which is paid once rather than as a commission stream.

Table 9: Parameters of estimated model and identifying variation

Parameter	Economic interpretation	Identifying variation
Estimated		
β	Commission sensitivity	Variation in broker pay (including cross-time variation and Experiment A). Experiment A: $W_{\text{MA}} \to 0$ (call-centers), $W_{\text{MA}} \to \frac{1}{3}W_{\text{MA}}$ (non-call-centers)
$\bar{\lambda}$	Misinformed broker share	Share that recommend MA at $W_{\text{MA}} = 0$ (call-centers); non-call-centers extrapolated from $\frac{1}{3}W_{\text{MA}}$ using β and logistic shape; repeat calls within broker
$\bar{K}_1$	Decision 1 suitability cost location	Level of MA recommendations unexplained by commission gap given β and $\bar{\lambda}$
R_1	Sale-probability ratio when $s_i = 1$	Experiment B, Decision 1 given calibrated R_0
η	Treatment discount (when $s_i = 1$) on sale probability per \$ Medigap markup	Experiment B, Decision 2
κ_2	Suitability cost of higher markup plan	Within-menu Medigap plan choices
a_{appt}	Cost of obtaining appointment	Implied cost from plans chosen outside broker's portfolio
Calibrated		
W_{MA}, W_{MG}	Present value commission stream	CMS commission schedule
N^{MA}, N^{MG}	Net harm from misfit sale	Broker survey; $N^{\text{MA}} = N^{\text{MG}}$ in main model
R_0	Sale-probability ratio when $s_i = 0$	Broker survey

mendation plan $j_{i\ell}$ and the senior's choice set. $w_{i\ell} = 1/|\mathcal{D}_2(i)|$ re-weights the recommendations so each call counts once in Decision 2 even if the call has multiple Medigap recommendations.[56] We estimate ten parameters $\Theta = (\beta, \bar{K}_{1,ncc}, \bar{K}_{1,cc}, \gamma_{ncc}, \gamma_{cc}, \gamma_{\text{ten}}, R_1, \eta, \kappa_2, a_{\text{appt}})$ by maximum likelihood. Standard errors and intervals are estimated from a caller-cluster block bootstrap over 1000 resamples. See Appendix F for details.

Misinformed broker share Approximately 18% of brokers in our audit study are estimated to be misinformed. Consistent with the reduced-form results, the share is heavily concentrated among call center brokers: 37% of call-center brokers are estimated to be misinformed versus 9% of non-call-center brokers.[57] Note that the non-call-center share rests on an extrapolation from the one-third-pay calls; Appendix F re-estimates the model with a calibrated share of misinformed non-call-center brokers and finds similar results.[58]

[56]Decision 2 sits outside the bracket because (18) does not vary with broker type. A call where the broker recommends MA in Decision 1 but also recommends a Medigap plan in Decision 2 still contributes a Decision 2 factor (Appendix F).

[57]We assume a two-type model given the reduced-form evidence of broker information frictions in Section 5.5. We also test for a one-broker-type model in which the suitability cost is drawn independently on each call. The one-type model is rejected with likelihood ratio 29.7 (Appendix F).

[58]This exercise uses the share of non-call-center brokers who report that they are only licensed to sell MA in the broker survey (13.3%). The data do not reject the calibrated restriction (likelihood ratio 0.56, $p = 0.454$). This version of the calibrated model changes the gains in consumer surplus from pay parity from \$510 to \$466 per senior per year and moves no other policy in Figure 6 by more than \$11. (See Tables A14 and A15).

Table 10: Structural Model Parameter Estimates

Parameter	Estimate	(SE)	95% interval
Commission sensitivity β (per PV-$1,000)	1.01	(0.35)	[0.59, 1.98]
Suitability-cost location, non-call-center $\bar{K}_{1,ncc}$ (PV $)	3,558	(674)	[2,684, 5,107]
Suitability-cost location, call-center $\bar{K}_{1,cc}$ (PV $)	2,803	(487)	[2,147, 4,026]
Misinformed share $\bar{\lambda}$	0.18	(0.04)	[0.12, 0.27]
Perceived competition deflator η	0.60	(0.48)	[0.17, 1.96]
Decision 2 suitability cost κ_2 (PV $ per $1 of annual markup)	2.35	(0.58)	[1.53, 3.72]
Appointment cost a_{appt} (PV $)	1,758	(607)	[888, 3,064]
Log-likelihood	-1,579.0		
Observations (calls + plan rows)	1,692		

Notes: Parameters are estimated using maximum likelihood on 637 audited calls. Bootstrap standard errors and 95% percentile intervals are estimated from 1000 caller-cluster block-bootstrap refits (Appendix F). Dollar parameters are in present-value (PV) commission dollars. The Decision 2 suitability cost is per dollar of annual markup. κ_2 and a_{appt} are converted at the Decision 1 commission sensitivity β. Calibrated commission streams are valued over 6 years (to reflect renewal commissions) at a discount factor of 0.94 per year. Since all renewal commissions are 50% of year-1 commissions the multiplier rescales all present value commissions by the same factor. The perceived competition treatment sale-probability ratio R_1 lies at the lower limit of its parameter space, zero, where a point estimate is not interpretable and a Wald standard error is zero by construction. We report its profile-likelihood bound $R_1 \leq 0.50$ instead. Appendix F discusses inference at the boundary, and Appendix F.1 shows that the model is able to reproduce the perceived competition treatment's effect on Decision 1 at this value.

The Decision 1 suitability cost The estimated location of the dollar suitability cost for non-call-center brokers is $\bar{K}_1 = \$3,558$ (SE $674) and for call-center brokers is $2,803 (SE $487) (Table 10).[59] The estimates imply there are costs that discipline brokers against recommending the higher-commission MA plans; on average a broker is indifferent between recommending Medigap and MA when MA pays $3,283 more in PV-commission dollars, or 2.1 times the $1,578 sale-probability-ratio-discounted commission gap the informed broker type faces on those calls.

The Decision 2 suitability cost The estimated Decision 2 suitability cost is κ_2=$2.35 (SE $0.58) of PV Medigap commission per $1 annual Medigap markup. To interpret this cost a Medigap plan that is $1,000 a year more expensive than a coverage-identical plan pays the broker $617 more in present-value commission, but the broker behaves as if this $1,000 markup costs $2,351, or 3.8 times the commission the broker earns.

Appointment Cost The estimated appointment cost, or the cost of becoming licensed to sell plans for a specific insurance company, is $a_{\text{appt}} = \$1,758$ (SE $607) of present-value commission.

Decomposing Broker Advice Table A17 in Appendix F.2 reports how the share of brokers recommending MA to a Medigap-fit senior changes when each force in the model is switched off, holding the others at their estimated values.

[59] We also estimate the model with one suitability-cost location for both call-center and non-call-center brokers (that is, $\bar{K}_{1,cc} = \bar{K}_{1,ncc}$; see Appendix F).

Together, we estimate that 38%–45% of MA recommendations are driven by commissions and that information frictions account for 43% of recommendations. In the estimated model, brokers recommend MA on 34% of commission-paying calls. With no pay for either product, that share falls by 13.0 percentage points. Removing only the MA commission, the change carriers made for call-center brokers in Experiment A, lowers the share by 15.3 percentage points. Under zero MA commissions, changing every misinformed broker to the informed type would reduce the share of MA recommendations by 14.7 percentage points. Note that this does not account for the role incentives may have played in shaping which products brokers are trained to sell. Table A18 in Appendix F.2 reports the same exercise for the mean recommended Medigap markup.

The suitability cost receives a large weight in both decisions. Removing the suitability cost from both decisions and holding commissions, broker types, and choice sets constant raises the share of brokers recommending Medicare Advantage to a Medigap-fit senior from 34% to 88% of calls, a change of +53.5 percentage points. Similarly, the mean recommended Medigap markup would increase from $386 to $691 a year.

7 Counterfactuals and Welfare

The following counterfactuals assume the broker recommendations are accepted by the senior. A Medigap-fit senior who follows the recommendation of a broker in our audit study loses up to $2,154 a year or $30,811 in present-value wealth over their lifetime from the broker recommendation. Two recommendations generate this loss: the senior loses $1,899 in expectation per year when the broker recommends Medicare Advantage to the senior (Decision 1), and $255 in expectation per year when the broker recommends a Medigap plan with a markup (Decision 2).

Each counterfactual policy changes one feature of the supply-side environment or the senior's search behavior and recalculates the broker recommendation probabilities implied by the model under this change. We value each averted Medicare Advantage (MA) recommendation at the net annual harm elicited from the broker survey less the expected Medigap markup the senior pays when recommended Medigap instead. We report every result per Medigap-fit senior per year. Once again, this assumes the recommendation is accepted. Additional details and a discussion of the partial-equilibrium assumption are included in Appendix F.2.

7.1 Policy Simulations

We examine two sets of policies: 1) market changes that governments or regulators can impose; and 2) changes to consumer search that consumers can implement alone, such as announcing plans to seek recommendations from multiple brokers or consulting only highly-rated brokers. We focus on two changes to broker commissions: equalizing the MA and Medigap commissions, which

we call pay parity, and capping the Medigap markup at 25% above the low-cost identical plan. Appendix F.2 shows the full set of counterfactual policies such as changes to the composition of brokers in the market.

Supply-side policies improve broker advice Figure 6 shows the annual consumer surplus gain for a Medigap-fit senior under four policy counterfactuals. Changing the market rules recovers the largest share of consumer surplus. We estimate that pay parity raises consumer surplus by $510 per senior per year, recovering 24% of lost consumer surplus, and a markup cap set at 25% of the cheapest coverage-identical premium is worth $127 (6% of lost consumer surplus).

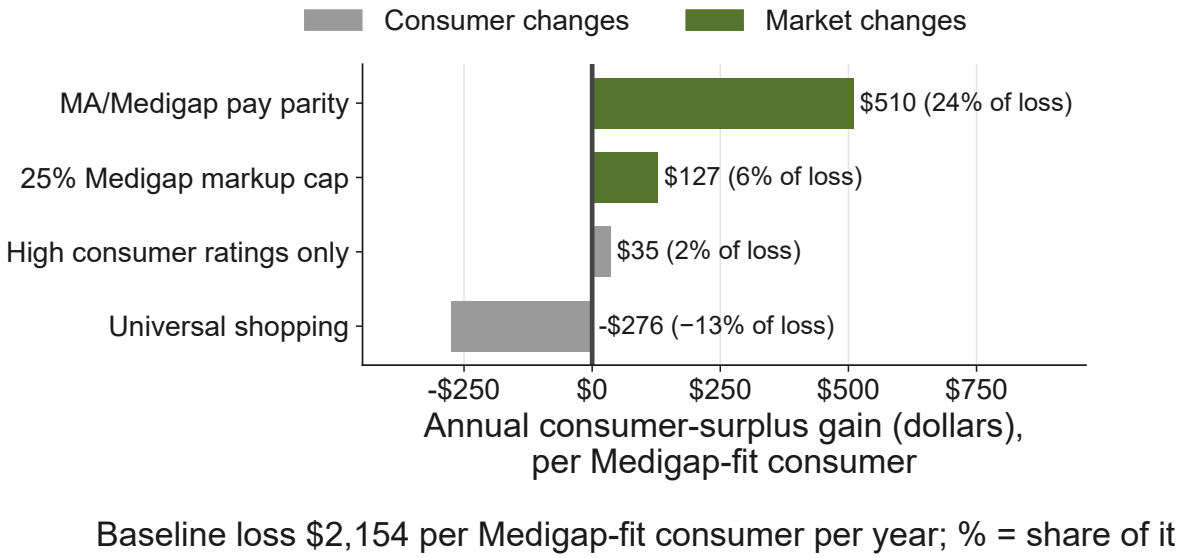


Figure 6: Annual Consumer Surplus Gain by Policy

Notes: Change in annual consumer surplus per Medigap-fit senior under each counterfactual, colored by who can implement the policy: changes a consumer can make in grey, changes to market rules in green, changes to broker composition in navy. The baseline loss is $2,154 a year. The rating screen limits the broker composition to brokers who receive a rating of a 4 or 5. The universal-shopping bar is measured against the audit's observed mix of treated and control calls, the baseline every bar in the figure shares; Section 7.1 values the same policy as the all-control to all-treated contrast. Appendix F.2 reports the full set of counterfactuals, including changes to broker composition.

Limitations of consumers as a disciplining mechanism Two policies rely on the consumer rather than on the market's rules or its brokers: soliciting advice only from brokers with high consumer ratings, and telling every broker that the senior intends to solicit multiple recommendations. Both recover less lost consumer surplus than any supply-side policy in Section 7.1.

The "high consumer ratings" only counterfactual restricts the pool of brokers to brokers who receive a high rating, or a rating of a 4 or 5 (Section 5.2). The counterfactual recovers $35 per senior per year, or 2% of the loss. The screen lowers the share of brokers recommending MA to a Medigap-fit senior from 34% to 33% of calls, a reduction of 0.7 percentage points. The screen only lowers the share of misinformed-type brokers from 17% to 16%. The screen recovers little because

shopper ratings carry almost no information about broker and recommendation quality.

The "universal shopping" counterfactual, where all seniors tell brokers they plan to solicit multiple recommendations, has a *net negative* impact on consumer surplus. Moving every call from the control arm to the treated arm raises the share of brokers recommending MA from 31% to 39% of calls and lowers the average Medigap markup by $56 a year. Valuing each additional Medicare Advantage recommendation at the survey-elicited net annual harm less the expected Medigap markup the senior would otherwise pay, the net impact is a change in consumer surplus of $-386 per Medigap-fit senior per year.[60] The treatment moves the recommendation toward the attribute the senior can directly observe and verify, the premium, and away from the coverage attributes the senior cannot; the net impact is a reduction in consumer surplus.

7.2 Scaling to the national population

A back-of-the-envelope calculation allows us to scale the consumer surplus loss from enrolling in a misfit plan to the population level. We assume that Medicare beneficiaries fall into one of three plan-fit types: 34% unambiguous Medigap-fit, 51% unambiguous MA-fit, and 15% ambiguous-fit (not a clear fit for either plan). The shares reflect the broker beliefs elicited in our survey. We assume there is a consumer surplus loss from enrolling in a plan that does not match the senior's true type. A Medigap-fit senior loses the net annual harm of $5,574 if a broker enrolls them in Medicare Advantage and loses the Medigap markup otherwise. An MA-fit senior loses the over-insurance harm that brokers report in the survey, or $1,243 a year, plus the Medigap markup if the broker enrolls them in Medigap. In the broker survey, brokers recommend a Medigap plan for a senior profile designed to be an "ambiguous fit" for both products 58% of the time.[61] We assume that the ambiguous-fit senior would only lose the Medigap markup on the 58% of calls where brokers enroll this type of senior in Medigap and assume no loss from under- or over-insurance.

We multiply the loss per advised senior by the estimated 4.1 million, the number of people who turn 65 each year (Fichtner, 2024). After excluding beneficiaries with full Medicaid benefits, whose cost sharing Medicaid pays, and beneficiaries with employer or union plans, we estimate that 2.81 million seniors each year are new Medicare enrollees facing the choices in our audit study (Appendix F.3). For this simple exercise, we assume that seniors enroll in Medicare when they turn 65 and 50% use a broker to enroll in Medicare. This gives us an estimate of 1.41 million broker-advised Medicare enrollees per year. The 50% assumption is conservative; in our UAS survey of current Medicare beneficiaries 56% of non-Medicaid and non-employer beneficiaries enrolled in Medicare through a broker (Appendix Table A7).

[60] Appendix F.2 reports the same policy against the audit's observed mix of treated and control calls, the basis of Figure 6; the valuation that uses the reduced-form treatment effects in place of the model's fitted effects; and the band that the boundary on the treated sale-probability ratio places around the point estimate.

[61] This rate comes from a broker survey question where we ask brokers to provide a plan recommendation for a senior vignette designed to be a fit for either MA or Medigap.

Since the audit study only measures recommendations to a Medigap-fit senior, for an MA-fit senior we assume that an informed broker applies the estimated Decision 1 rule with the roles of the two products reversed, and that a misinformed broker believes that a Medigap plan matches the senior's preferences. Appendix F.3 reports the baseline loss under alternative assumptions.

Figure 7 reports lost consumer surplus from broker steering for the new Medicare enrollee population at baseline and under various counterfactual policies. We estimate the total consumer surplus loss for new 65-year-old enrollees using a broker to be about $1,257 million of consumer surplus a year. About 82% of lost consumer surplus falls on Medigap-fit seniors; based on our assumptions, the remaining 18% is from MA-fit and ambiguous-fit seniors enrolled in Medigap.

Our estimates suggest pay parity increases total consumer surplus for these seniors by $204 million a year. Adding the 25% Medigap markup cap policy increases consumer surplus by another $141 million a year. Of this amount $35 million comes from seniors who are steered into Medigap, where the markup cap now applies. Together, the two policies recover $345 million a year, or 27% of baseline consumer surplus loss. The gains in consumer surplus accrue to Medigap-fit seniors and ambiguous-fit seniors: MA-fit seniors lose $10 million a year because pay parity increases the share of brokers who recommend Medigap to an MA-fit senior.

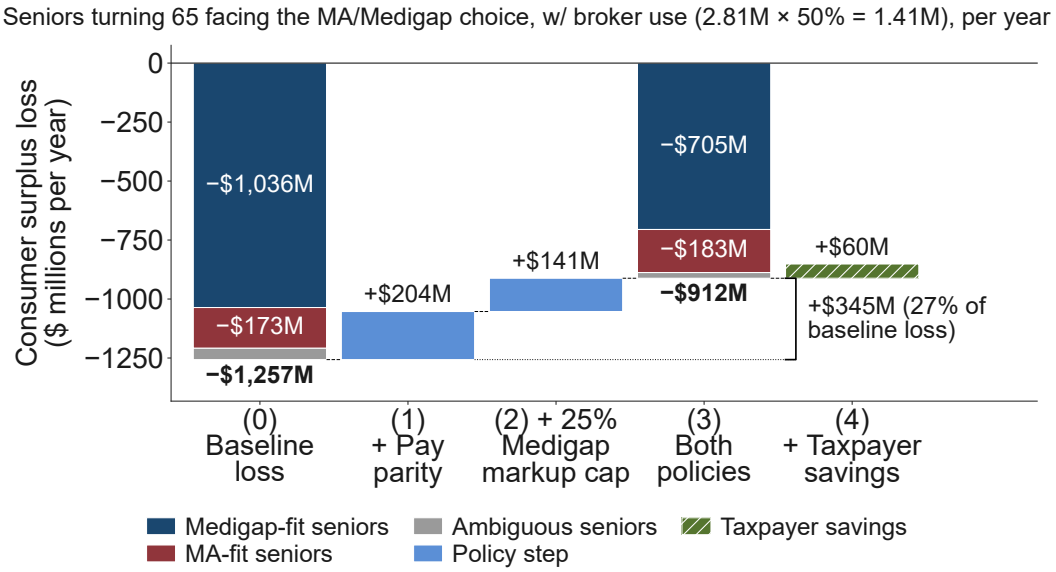


Figure 7: Consumer Surplus Loss Among Seniors Turning 65 Who Face the Medicare Advantage–Medigap Choice and Use a Broker, and the Share Two Market Policies Recover

Notes: Figure 7 estimates the annual consumer surplus loss for the 1.41 million seniors who turn 65 in one year who choose between Medicare Advantage and Medigap in the individual market and use a broker (Appendix F.3). Lost consumer surplus is measured against each senior holding the plan that matches their true "type" and against a $0 Medigap markup. A negative value is a loss. Bars (0) and (3) split the loss by the senior's plan-fit type using the broker survey estimates as the "true" fit type. Bar (1) shows the surplus change from equalizing Medicare Advantage and Medigap commissions. Bar (2) shows the additional change from capping the Medigap markup at 25% of the lowest-priced coverage-identical plan once pay parity is in place. The cap alone is worth $106 million. The bracket shows the gain from bar (0) to bar (3). Bar (4) is the reduction in federal Medicare spending under both policies, valued at $846 per Medicare Advantage enrollee per year, not consumer surplus. Bar (0) contains no taxpayer cost. The full set of assumptions is stated in Appendix F.3. Premiums, commission schedules, choice sets, and broker compositions are held fixed.

The final bar in Figure 7 incorporates the federal savings (or taxpayer savings) generated by the two policies. A 2025 Medicare Payment Advisory Commission report estimated that Medicare spends 20% or $84 billion more on beneficiaries enrolled in Medicare Advantage than it would spend on these same beneficiaries if they were enrolled in Traditional Medicare (Medicare Payment Advisory Commission, 2025). After subtracting the costs of additional Medicare spending (a response to the more comprehensive Medigap coverage, estimated in Cabral and Mahoney (2019)), the net taxpayer savings is approximately $846 per enrollee per year. At this estimate, pay parity and the Medigap markup cap would lower federal spending by $60 million a year for the 1.41 million broker-using seniors who turn 65 each year.

8 Conclusion

More than half of the nearly 70 million people on Medicare enroll through an insurance broker, with the broker directly enrolling beneficiaries in the plan they recommend. The initial enrollment decision is difficult to reverse, so the recommendation a senior receives on one phone call can shape their insurance coverage and premiums for years. These stakes make the quality of expert advice consequential, yet the same complexity that creates demand for brokers can also make their advice difficult for consumers to evaluate.

In the first large-scale audit study of Medicare insurance brokers we show that brokers steer seniors toward the higher-commission, worse-fit product on 52% of calls when incentives are misaligned. In the Medigap market, these agency problems generate a directly measurable cost: the average recommendation costs a senior more than $377 per year, or a 29% markup over an otherwise identical plan. Compounded over time, a senior who follows this advice at age 65 loses more than $12,000 in lifetime wealth. These are likely conservative estimates given our decision to base our seniors in rural areas and to avoid enrollment periods when brokers are most busy.

Two experiments and a structural model identify two sources of low-quality advice: misaligned incentives and broker information frictions. A natural experiment in which carriers stopped paying Medicare Advantage (MA) commissions reduced MA recommendations by 21 p.p. Yet brokers continued to recommend MA even when MA recommendations did not generate commissions and did not match senior preferences. Additional evidence from recommendation differences by broker tenure, call transcripts, and a broker survey is consistent with broker information frictions rather than beliefs that MA matches senior preferences. Using the structural model to quantify the components of advice, we estimate that the MA commissions alone account for 45%, and broker information frictions for 43%.

Our perceived competition treatment shows why competition alone may not discipline these distortions. The experimental results show that increasing competitive pressure by signaling that a

consumer plans to solicit multiple recommendations reduces Medigap markups, but also increases steering toward Medicare Advantage. Competitive pressure therefore disciplines the attribute consumers can verify, the premium, and not the attributes they cannot, such as plan coverage and network restrictions. Consistent with this interpretation, senior ratings of brokers are uncorrelated with our objective measures of recommendation quality, and this relationship does not strengthen with caller experience. Thus, the consumer-lever counterfactuals show little benefit from consumer-directed mechanisms: screening brokers using high ratings recovers only a small fraction of the consumer surplus loss, while encouraging all consumers to solicit multiple recommendations lowers consumer surplus.

The findings illustrate a broader challenge in designing markets with complex choices: when consumers cannot evaluate the products themselves or expert advice, policies that rely on consumer choice and competition may fail to address conflicts of interest created by misaligned incentives. Instead, policies should target the root causes of low-quality advice directly, by aligning compensation and ensuring that intermediaries have sufficient information and training to evaluate *all* the products they are licensed to sell.

One implication concerns the growing use of AI-based decision support in advice markets. Our results suggest why such technologies may be especially valuable in settings like Medicare. If programmed correctly, AI technology can help align incentives and reduce the information frictions that generate poor advice even when financial incentives are aligned. At the same time, AI does not by itself resolve an agency problem when the intermediary's incentives remain misaligned. If technology makes brokers better informed, it may instead increase the effectiveness of incentive reform by giving brokers the information needed to act on aligned incentives. Thus, the gains from AI in expert advice markets may depend not only on the quality and incentives of the technology itself, but also on the incentives faced by the experts using this technology.

Overall, our results highlight the importance of well-designed markets and incentives, especially in complex markets with information frictions and limited feedback. More broadly, consumer information frictions can reinforce supply-side frictions: when consumers have difficulty evaluating product quality or expert advice, competitive pressure may provide little discipline and firms may have weaker incentives to invest in training and expertise when markets do not reward high-quality advice. The same information frictions that create demand for expert intermediaries can therefore weaken the market forces that would discipline agency problems, allowing distortions to persist even in competitive markets.

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Appendix

A A Model of Broker Advice

This appendix proves the predictions of Section 3, adds the second broker type that Section 6 estimates, and derives the consumer surplus accounting that Section 7 uses. We use the same notation as Section 3 and cite the relevant equations instead of restating them.

A.1 The Environment

The game is the one in Section 3.1. One senior and one broker meet once. The senior enrolls in one of two plans $j \in \{L, H\}$ with annual premiums $\rho_L < \rho_H$. The broker recommends one plan, and the senior either enrolls in it or walks away and continues searching. Neither party moves again. Section A.7 maps the game onto the two decisions in the audit by setting parameters, and every result below holds for both.

Utility is in dollars per year. The senior pays the premium of the chosen plan and bears the fit gap $\Delta \geq 0$ only on the low-price plan, so $U_c(L) = -\rho_L - \Delta$ and $U_c(H) = -\rho_H$. The high-price plan carries no fit gap by construction, so Δ measures the harm of the low-price plan relative to the high-price plan. The difference $U_c(H) - U_c(L)$ is the net harm N of equation (1). Condition (3) describes the senior we construct in the audit: the coverage the low-price plan lacks is worth more than the premium it saves, so the high-price plan is the plan that fits, or the plan that matches the senior's preferences, and the low-price plan is the misfit plan. We impose the condition only where stated. Decision 2 sets $\Delta = 0$ (Section A.7). There the net harm is negative and the low-price plan is the plan that fits, which is why we carry Δ as a free parameter.

Selling plan j pays the broker $W_j > 0$ and nothing if the senior does not enroll (Section 3.1). W_j is the present value (PV) of the full commission stream from the sale, the first-year payment plus the renewal payments the broker earns while the enrollee stays in the plan. The present-value convention matters for the magnitudes in Section 6.2 and not for any sign below, which requires only $W_j > 0$. Appendix F gives the term structure and the horizon we use to construct the streams.

A.2 Perceived Fit and the Complexity Threshold

The senior perceives the premium exactly and a share $\phi \in [0, 1]$ of the fit gap (Section 3.2), so the senior acts on

$$\hat{U}_c(L) \;=\; -\rho_L - \phi\,\Delta, \qquad \hat{U}_c(H) \;=\; -\rho_H \;=\; U_c(H). \tag{20}$$

The perception wedge falls on the low-price plan only. The senior overvalues the low-price plan by $(1-\phi)\Delta$, the part of the fit gap the senior does not perceive, and values the high-price plan correctly. The difference $\hat{U}_c(H) - \hat{U}_c(L)$ is the perceived net harm $\hat{N}$ of equation (2).

Lemma 1 (The complexity threshold)**.** *Let $\phi^\dagger = (\rho_H - \rho_L)/\Delta$ be the complexity threshold of equation* (9)*. Then $\hat{U}_c(L) > \hat{U}_c(H)$ if and only if $\phi < \phi^\dagger$. Under condition* (3)*, $\phi^\dagger \in (0,1)$.*

Proof. By (20), $\hat{U}_c(L) > \hat{U}_c(H)$ if and only if $\phi\Delta < \rho_H - \rho_L$. Condition (3) gives $\Delta > \rho_H - \rho_L > 0$, so dividing by Δ gives $\phi < \phi^\dagger$, and the same two inequalities place $\phi^\dagger$ strictly between zero and one. □

When $\phi < \phi^\dagger$ the senior perceives the misfit plan to have higher utility than the plan that fits because the perceived harm $\phi\Delta$ is smaller than the premium the misfit plan saves. When $\phi \geq \phi^\dagger$ the senior correctly perceives the plan that fits to have higher utility. Lemma 1 shows that the first case is never empty under condition (3), because the threshold lies strictly below one. The threshold compares comprehension with the stakes. Two seniors who perceive the same share of the fit gap can fall on different sides of it, and what places a senior below it is a large premium gap relative to the fit gap. The model does not assume which case holds. Section 6.2 estimates the treated sale-probability ratio R_1 against the calibrated R_0, and the comparison tells us the case (Section A.10).

At the calibrated premium gap of $1,855 per year and the gross coverage harm of $7,429 per year, which includes the value of the option to switch back to Medigap later under medical underwriting, the threshold is $\phi^\dagger = 0.25$ (Section 5.4). Using the survey's point-of-sale attribute valuations instead, $6,087 per year, which exclude that option, raises the threshold to 0.30. We use the harm with the switch option because its net value, after the premium the senior saves, is the harm that enters the broker's suitability cost in Section A.5 and the consumer surplus accounting in Section A.9. Under either harm the threshold is well below one.

We assume the senior does not infer anything about the broker's pay from the recommendation (Section 3.2). This is the most restrictive assumption in the model. In the credence-goods model of Dulleck and Kerschbamer (2006), the consumer's suspicion that a recommendation of the cheap plan serves the expert is what deters the expert from making it, and our assumption switches that inference off. We impose it because the audited senior is a first-time enrollee with no prior relationship with the broker, and because Section 5.2 shows that seniors do not rate advice by its quality even after many calls. Allowing the inference would attenuate every effect derived below, and a strong enough inference would restore the canonical prediction. Note that the inference has the same sign as $\phi \geq \phi^\dagger$, so the sign of the Decision 1 response to Experiment B cannot distinguish the two. The results below hold under this assumption and do not test it.

A.3 The Senior's Acceptance Decision

Assumption A.1 (The continuation value)**.** *The senior who hears a recommendation of plan r either enrolls and obtains $\hat{U}_c(r)$ or walks away and obtains the continuation value v_s, where $s \in \{0,1\}$ indexes the perceived competition treatment (Section 3.2). The perceived competition treatment raises the continuation value, $v_1 > v_0$, and changes nothing else.*

Assumption A.1 is the model's entire representation of Experiment B. A senior who tells the broker about plans to solicit multiple recommendations tells the broker that the senior's cost of continuing to search is low, which is what a higher continuation value means. Nothing on the broker's side of the model changes with s, and every result below uses only the inequality $v_1 > v_0$, not the size of the gap or the search process behind it.[62]

Under Assumption A.1 the close probability of a recommendation of plan r is $P_r(s) = \Lambda(\hat{U}_c(r) - v_s)$, equation (4). Since Λ is strictly increasing, the plan the senior perceives as better is the easier sale. Lemma 1 says which plan that is, and Lemma 3 below shows that this ranking is the only feature of the senior's problem that enters the broker's decision.

Lemma 2 (Responses of the close probabilities to the continuation value)**.** *For each plan r,*

$$\frac{\partial P_r}{\partial v} \;=\; -P_r\Big(1 - P_r\Big) \;<\; 0, \tag{21}$$

$$-\frac{\partial \log P_r}{\partial v} \;=\; 1 - P_r, \tag{22}$$

and for any two plans r and r', $\partial \log(P_r/P_{r'})/\partial v = P_r - P_{r'}$, which for $r = H$ and $r' = L$ is equation (10)*.*

Proof. Write $t_r = \hat{U}_c(r) - v_s$, so $P_r = \Lambda(t_r)$ and $\partial t_r/\partial v = -1$. Since $\Lambda' = \Lambda(1-\Lambda)$, the chain rule gives (21). Dividing by P_r gives (22), and subtracting (22) for r' from (22) for r gives the third expression. □

A higher continuation value lowers the close probability of every recommendation, but the level decline $P_r(1-P_r)$ is largest at $P_r = \frac{1}{2}$ and is not ordered by which recommendation is the harder sale. The proportional decline is. Equation (22) says that recommendation r loses its sales at the proportional rate $1-P_r$, the probability that the senior walks away from it, so the recommendation with the lower close probability loses sales at the faster rate. Equation (10) then says that the ratio of the two close probabilities tilts toward whichever recommendation was already easier to close,

[62] The inequality follows from a stationary sequential search problem in which the senior pays a cost to contact each broker, draws a recommendation, and continues until the senior accepts one. The continuation value is then the reservation value of continued search, which falls with the contact cost, so a signal that the contact cost is low raises it.

by the gap between the two close probabilities. Every comparative static below works through this derivative.

Remark 1 (The logistic form is a normalization for the signs but not for the range of R). Under a general distribution for the taste shock, the term $1 - P_r$ in (22) becomes the reverse hazard rate of the shock distribution. The reverse hazard rate is monotone in the close probability whenever the shock density is log-concave (Bagnoli and Bergstrom, 2005), which holds for the normal, extreme-value, uniform, and Laplace distributions among others, so every sign derived below survives any of them. The scale normalization is also free, because only ratios of sale close probabilities enter the broker's decision (Lemma 3). The normalization does not extend to the range of the sale-probability ratio. The limit of R as the continuation value rises depends on the tail of the shock distribution, and Remark 2 states what the logistic tail implies.

Remark 2 (The logistic tail places a floor on the sale-probability ratio). Write $a = \hat{U}_c(H)$ and $b = \hat{U}_c(L)$. Under (4), $\Lambda(a-v)/\Lambda(b-v)$ converges to $e^{a-b} = e^{\hat{N}}$ as the continuation value v rises without bound. By (5), this limit is the odds ratio of the two close probabilities, $[P_H/(1-P_H)]/[P_L/(1-P_L)]$, which does not depend on v. Since R is decreasing in v when $P_H < P_L$ (Lemma 3), the logistic form bounds the sale-probability ratio below by that odds ratio at every continuation value. At the survey close probabilities that calibrate R_0, $P_{\text{MA}} = 0.69$ and $P_{\text{MG}} = 0.42$ (Section 6.1), the floor is 0.33, against $R_0 = 0.61$. We do not impose this floor, for two reasons. First, it is a property of the tail of the shock distribution, which the audit's recommendation data do not identify. Under a normal shock the same limit is zero, and under a shock with power tails it is one. The sign results of this appendix need only log-concavity (Remark 1), and the floor is the one result that depends on the tail. Second, R_1 is the broker's belief about how a senior who announces plans to solicit multiple recommendations closes, relative to a senior who says nothing (Section 3.3), and no audited senior accepts or rejects a recommendation. The floor constrains the senior's acceptance probabilities. It constrains the broker's belief only if that belief obeys the tail of the acceptance shock. Section 6.2 therefore reports R_1 against the bound $R_1 \geq 0$, which holds for any shock distribution because R is a ratio of probabilities. The 95% profile-likelihood region for R_1 runs from that bound to 0.50 and contains the floor, so the audit rejects neither the floor nor its negation.

A.4 The Sale-Probability Ratio

The senior's problem enters the broker's problem through one number, the sale-probability ratio $R(s) \equiv P_H(s)/P_L(s)$ of Section 3.4. Substituting (20) into (4),

$$R(s) \;=\; \frac{\Lambda\Big(-\rho_H - v_s\Big)}{\Lambda\Big(-\rho_L - \phi\Delta - v_s\Big)}.$$

For every hundred seniors who would enroll after a recommendation of the low-price plan, $100R$ would enroll after a recommendation of the high-price plan. The perception parameter ϕ appears in the denominator only, because the fit gap attaches to the low-price plan alone.

Lemma 3 (Position and derivative of the sale-probability ratio)**.** *(i)* $R(s) < 1$ *if and only if* $\phi < \phi^{\dagger}$*, and* $\partial \log R/\partial \phi = \Delta(1 - P_L) > 0$*.*

(ii) $\partial \log R/\partial v = P_H - P_L$*, equivalently* $\partial R/\partial v = R\,(P_H - P_L)$*.*

Proof. Since Λ is strictly increasing, $P_H < P_L$ if and only if $\hat{U}_c(H) < \hat{U}_c(L)$, which by Lemma 1 holds if and only if $\phi < \phi^{\dagger}$, and dividing by $P_L > 0$ gives $R < 1$. For the derivative in (i), ϕ enters only the denominator, through the argument $t_L = -\rho_L - \phi\Delta - v_s$ with $\partial t_L/\partial \phi = -\Delta$, so by the same step as (22), $\partial \log R/\partial \phi = -\partial \log P_L/\partial \phi = \Delta(1 - P_L)$. Part (ii) is equation (10), and the level form follows from $\partial R/\partial v = R\,\partial \log R/\partial v$. □

Part (i) states what the senior's incomplete perception of the fit gap does to the broker's problem. The share of the harm the senior does not perceive raises the close probability of the misfit plan and leaves the close probability of the plan that fits unchanged, so to the broker the friction appears as the fact that the wrong plan is the easier sale. Part (ii) states what competitive pressure does. Each close probability falls at its own walk-away rate $1 - P_r$, so the sale-probability ratio declines at the rate $(1 - P_H) - (1 - P_L)$. When $\phi < \phi^{\dagger}$ the high-price plan is the harder sale, its walk-away rate is the larger one, and the sale-probability ratio falls as the continuation value rises.

Remark 3 (Only the ratio matters). The broker's decision, derived next, compares two expected payoffs, each a close probability multiplied by a net payment. Dividing both sides of that comparison by P_L leaves the close probabilities entering only through R. This matters for estimation. The audit records recommendations and never enrollments, so no close-probability level can be recovered from the audit data, and none is needed for the predictions, for estimation, for the counterfactuals, or for consumer surplus on the audited calls.

A.5 The Broker's Problem

The broker's problem is the one in Section 3.3. The broker observes which plan fits the senior, makes one recommendation, is paid W_j if the senior enrolls in plan j and nothing otherwise, and bears the suitability cost κN on a completed sale of the plan the broker perceives as the misfit (equation (6)). The cost parameter κ is drawn on each call from a continuous distribution F on the real line with density F'.

The suitability cost scales with the net harm the broker believes the sale does, which is why N multiplies κ. The broker bears the cost only on a completed sale. A senior who does not enroll

does not cancel the plan, file a complaint, or leave the broker feeling guilty about an unsuitable recommendation. This timing is what removes the close-probability levels from the broker's rule (Proposition 1). Because κ is unobserved, the probability of a low-price recommendation on any one call is $F(\kappa^*)$, and taking F logistic turns that probability into the logit that Section 6 estimates. The logistic function in that logit describes the variation in the suitability cost across calls, not error in recording the recommendation. It also absorbs any influence on the Decision 1 recommendation that the commission streams do not capture, such as sales contests or ease of enrollment, which is why the distribution places mass on negative values.

For a senior whom the high-price plan fits, the low-price plan is the misfit and the broker's expected payoffs are $U_b(L) = P_L(s)(W_L - \kappa N)$ and $U_b(H) = P_H(s)W_H$. Each element of the senior's problem enters the broker's in one place. The share of the harm the senior does not perceive raises P_L and nothing else, so the broker profits directly from what the senior cannot see. The continuation value enters both payoffs, so competitive pressure moves the broker's choice only through the two plans' relative chances of closing. The senior's realized welfare enters the broker's payoff only through the suitability cost, which is charged only on a sale of a plan that does not fit. With $\Delta = 0$ the two plans have identical coverage, neither is the misfit, and the suitability cost does not arise.

Proposition 1 (The recommendation rule and the share of misfit recommendations)**.** *The broker recommends the low-price plan L if and only if $\kappa \leq \kappa^*(s) \equiv (W_L - R(s)W_H)/N$, the cutoff of equation* (7)*, and the share of brokers who recommend the low-price plan to this senior is $\sigma(s) = F(\kappa^*(s))$, equation* (8)*.*

Proof. The broker recommends L when $P_L(W_L - \kappa N) \geq P_H W_H$. Dividing by $P_L > 0$ gives $W_L - \kappa N \geq R(s)W_H$, and dividing by $N > 0$ gives the cutoff. Since κ has a continuous distribution F, the probability that a randomly drawn broker recommends L is $F(\kappa^*)$. □

Proposition 1 is Prediction 1: brokers with a suitability cost above the cutoff recommend the plan that fits, and with no suitability cost every broker would recommend L whenever $W_L > RW_H$. Note that the timing of the cost matters here. If the broker bore the suitability cost at the moment of the recommendation instead of at the sale, the comparison would be $P_L W_L - \kappa N \geq P_H W_H$, the cutoff would be $(P_L W_L - P_H W_H)/N$, and it could not be evaluated without both close-probability levels. Because the rule is a cutoff on κ, dividing the comparison by P_L, by P_H, or by any other positive number leaves the cutoff and the share $F(\kappa^*)$ unchanged, so no normalization is needed.

Throughout the rest of this appendix, $\sigma(s)$ is the probability that a randomly drawn broker recommends the low-price plan L, which under condition (3) is the misfit plan, and every comparative static is a statement about that probability. Lemma 4 writes σ_2 for the share of brokers

recommending the higher-priced plan in Decision 2, where $\Delta = 0$ and the low-price plan is the plan that fits.

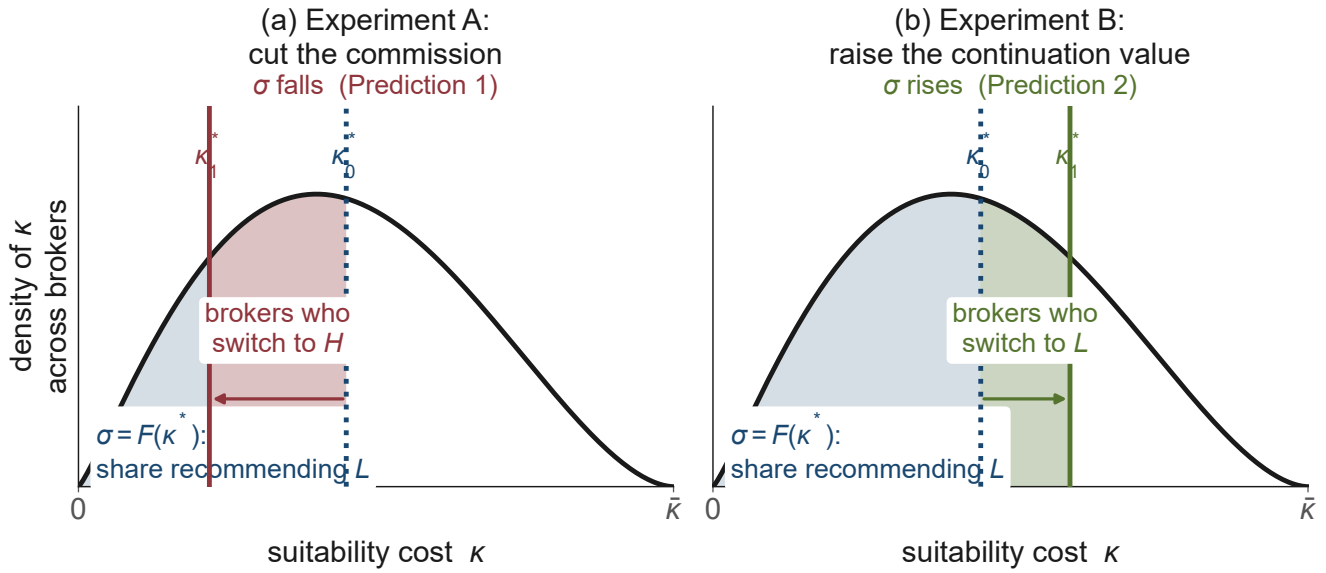


Figure A.1: The cutoff rule and the two experiments

Notes: Figure A.1 draws the density of the suitability cost κ across brokers. The shaded region is the mass below the cutoff $\kappa^* = (W_L - RW_H)/N$ of (7), which by (8) is the share σ of brokers who recommend the misfit low-price plan. The two experiments move the cutoff in opposite directions. Removing the misfit plan's commission (Experiment A, $W_L \to 0$) lowers the cutoff and shrinks the shaded mass. Raising the continuation value when $\phi < \phi^\dagger$ (Experiment B) lowers the sale-probability ratio R, raises the cutoff, and expands the shaded mass. The band between κ_0^* and κ_1^* in each panel is the set of brokers whose recommendation each experiment changes. The parameters are chosen to make the two cutoff shifts visible and are not the estimates of Section 6.2. The figure shows the directions of Predictions 2 and 3, not their magnitudes.

A.6 Predictions

Section 3.4 states three further predictions. Each differentiates the cutoff (7) with respect to one quantity the audit moves. We prove each prediction and extend Prediction 4. We then show how each prediction maps to the corresponding result in Section 5.

Proof of Prediction 2. Differentiating (7) with respect to W_L, holding R, W_H and N fixed, gives $\partial\kappa^*/\partial W_L = 1/N$, so by (8), $\partial\sigma/\partial W_L = F'(\kappa^*)/N > 0$. Differentiating with respect to the commission on the plan that fits gives the companion result $\partial\sigma/\partial W_H = -RF'(\kappa^*)/N < 0$. Neither expression contains ϕ or compares the close probabilities, so neither sign depends on which side of the complexity threshold the senior is on. □

Prediction 2 corresponds to Experiment A. The commission removal set the Medicare Advantage commission, which is W_L in Decision 1 (Section A.7), to zero while holding the senior, the plans, and the brokers fixed, and Section 5.3 reports that the share of brokers recommending Medicare Advantage fell. Since the prediction holds on both sides of the threshold, Experiment A shows that recommendations respond to commissions. It also rules out the explanation in which every

misaligned recommendation reflects limited broker knowledge, since a broker who recommends MA by mistake has no reason to change the recommendation when the pay changes. It cannot separate the explanations of Experiment B that Prediction 3 distinguishes.

Proof of Prediction 3. In (7) the two commissions and the net harm are fixed by the market, and Assumption A.1 moves only the continuation value, so $R(s)$ is the only term in the cutoff that depends on v and $\partial\kappa^*/\partial v = -(W_H/N)\,\partial R/\partial v$. By (8), $\partial\sigma/\partial v = F'(\kappa^*)\,\partial\kappa^*/\partial v$, and substituting $\partial R/\partial v = R(P_H - P_L)$ from Lemma 3(ii) gives equation (11). In (11), W_H, R, N, and $F'(\kappa^*)$ are all strictly positive, so the sign of $\partial\sigma/\partial v$ is the sign of $P_L - P_H$, which by Lemma 3(i) is the sign of $\phi^\dagger - \phi$. □

Table A1: The response to competition by the senior's perceived harm, for a senior whom the high-price plan fits

	Perception below the complexity threshold $P_L > P_H$, i.e. $R < 1$, i.e. $\phi < \phi^\dagger$	Perception at or above the threshold $P_L \leq P_H$, i.e. $R \geq 1$, i.e. $\phi \geq \phi^\dagger$
Commission on the misfit plan rises	share recommending it rises	share recommending it rises
Continuation value rises	share recommending it rises	share recommending it falls

Notes: Table A1 reports the sign of the change in the share of brokers who recommend the misfit low-price plan, for the senior whom the high-price plan fits. The first row is Prediction 2 and holds in both columns. The second row is Prediction 3 and changes sign across them. The classifying condition $P_L \gtrless P_H$ and the threshold $\phi^\dagger = (\rho_H - \rho_L)/\Delta$ are properties of one senior's problem, not of the market as a whole. Corollary 1 treats the senior whom the low-price plan fits.

When $\phi < \phi^\dagger$ the misfit plan is the easier sale. Raising the continuation value lowers the log close probability of the harder sale by more than that of the easier sale, so the sale-probability ratio falls, the cutoff rises, and the share of brokers recommending the misfit plan rises. A senior who announces an intention to compare quotes therefore raises the probability of being recommended the misfit plan. This reverses the second-opinion result of Pitchik and Schotter (1987) and Wolinsky (1993), in which a consumer who can seek another opinion makes the expert less likely to recommend the wrong product, and it is consistent with the life-insurance evidence in Anagol et al. (2017). It corresponds to the Outcome 1 result of Experiment B (Section 5.4). When $P_L = P_H$ exactly, the derivative is zero and the share does not respond to the continuation value, while the Decision 2 response of Prediction 4 remains.

Corollary 1 (The classification is senior-specific)**.** *Consider instead a senior whom the low-price plan fits, so that* $U_c(L) = -\rho_L$ *and* $U_c(H) = -\rho_H - \Delta'$ *with* $\Delta' > 0$*. For that senior,* $P_L > P_H$ *for every* $\phi \in [0, 1]$*.*

Proof. This senior's perceived utilities are $\hat{U}_c(L) = -\rho_L$ and $\hat{U}_c(H) = -\rho_H - \phi\Delta'$. Since $\rho_L < \rho_H$ and $\phi\Delta' \geq 0$, $\hat{U}_c(L) > \hat{U}_c(H)$ for every ϕ, and Λ is strictly increasing. □

For this second senior the inequality at the head of the first column of Table A1 holds for every ϕ and signals no friction. The cheaper plan is the better plan for that senior, so its being the easier sale is not a distortion, and because the cheaper plan also pays the broker more in Decision 1, the broker's pay and that senior's interest point the same way. A higher continuation value raises the share of brokers who recommend the low-price plan to that senior, and that plan is the plan that fits. The distinction between the two cases has content only for a senior whose fitting plan is the costlier one, which is the senior the audit constructs and the case in which the broker's pay conflicts with the senior's interest. The misinformed type in Section A.8 is a broker who has mistaken the audited senior for this second senior.

Corollary 2 (Aligned pay does not reverse the competition response)**.** *Suppose the misfit plan's pay advantage is removed, $W_L \leq RW_H$. Then $\kappa^* \leq 0$, so the share of brokers recommending the misfit plan is at most the mass of brokers with a suitability cost at or below zero, and the magnitude of $\partial\sigma/\partial v$ falls with it, because* (11) *is evaluated where the density $F'(\kappa^*)$ is small. The sign of $\partial\sigma/\partial v$ is unchanged.*

Proof. $\kappa^* = (W_L - RW_H)/N \leq 0$ follows from $W_L \leq RW_H$ and $N > 0$. The statements about σ and its derivative follow from (8) and (11), whose sign does not depend on where the density is evaluated. □

Corollary 2 separates the commission from the competition response. Cutting the misfit plan's commission lowers both the share of brokers recommending it and the sensitivity of that share to the continuation value, because it moves brokers away from the cutoff at which the continuation value acts. No commission schedule changes the sign of the competition response. Only the senior's perception crossing $\phi^\dagger$ does. The audit contains the aligned-pay case directly, since the commission removal of Experiment A sets $W_L = 0$, and Proposition 3 gives the model's prediction there.

Lemma 4 (Decision 2: the recommended markup)**.** *Let $\Delta = 0$ and $W_j = z\rho_j$ for a commission rate $z > 0$. Then, for every $\phi \in [0, 1]$:*

(i) the broker recommends the higher-priced plan if and only if $R(s) \geq \rho_L/\rho_H$;

(ii) $P_H < P_L$, so the cheaper plan is the easier sale;

(iii) a broker who recommends the cheaper plan at continuation value v recommends the cheaper plan at every $v' > v$;

(iv) the share of brokers recommending the higher-priced plan is non-increasing in the continuation value, and strictly decreasing whenever a positive density of choice sets is at the switching threshold, and the mean markup of the recommended plans falls with it.

Parts (i) and (iv) are Prediction 4.

Proof. (i) With $\Delta = 0$ neither plan is the misfit, no sale carries the suitability cost, and the payoffs are $U_b(L) = P_L z \rho_L$ and $U_b(H) = P_H z \rho_H$. The broker recommends the higher-priced plan when $P_H z \rho_H \geq P_L z \rho_L$, and dividing by $z P_L \rho_H > 0$ gives $R(s) \geq \rho_L / \rho_H$. (ii) With $\Delta = 0$, (20) gives $\hat{U}_c(L) = -\rho_L > -\rho_H = \hat{U}_c(H)$ for every ϕ, so $P_H < P_L$. (iii) Let $D(v) = \log R(v) - \log(\rho_L / \rho_H)$, so that by (i) the broker recommends the higher-priced plan if and only if $D(v) \geq 0$. By Lemma 3(ii) and part (ii), $\partial D / \partial v = P_H - P_L < 0$, so once D falls below zero it stays below zero. (iv) Across choice sets, the share of brokers recommending the higher-priced plan is $\sigma_2(v) = \Pr\{D(v) \geq 0\}$. By (iii) the set of choice sets on which $D(v) \geq 0$ shrinks as v rises, so σ_2 is non-increasing, and strictly decreasing whenever the distribution of $D(v)$ across choice sets has positive density at zero. The markup of a recommendation is $\rho_H - \rho_L$ when the higher-priced plan is recommended and zero otherwise, so the mean recommended markup falls with σ_2. □

Prediction 4 corresponds to Outcome 2 of Experiment B. Section 5.4 reports that the treatment lowered the mean annual markup of the Medigap plans brokers recommended. Two features of the result matter for how we use it. First, it holds for every value of ϕ, so it serves as the control experiment for Prediction 3: the same increase in the continuation value, applied to a decision in which nothing is hidden, lowers the recommended premium on either side of the complexity threshold. Second, the Decision 2 response exists only because Medigap commissions are proportional to the premium. With a flat commission, part (i) would read $R \geq 1$, which by part (ii) never holds, and no marked-up plan would ever be recommended. The framework keeps two plans per choice set. The estimated model in Section 6 uses the senior's full choice set, but does not change the intuition behind part (iv).

Remark 4 (The joint pattern rules out commission maximization). Predictions 3 and 4 describe one movement on both decisions: competitive pressure moves the recommendation toward the plan the broker expects the senior to accept. Because the two decisions order the commissions oppositely (Table 2), that movement raises the broker's expected pay in Decision 1 and lowers it in Decision 2. This rules out the simplest alternative explanation of the Experiment B results, in which the broker recommends whichever plan pays the largest commission while the caller is still on the phone. Such a broker would move toward Medicare Advantage and toward the higher-priced coverage-identical plan. In the audit, brokers move toward Medicare Advantage and toward the cheaper plan. Both explanations are commission-motivated. They differ in whether the broker protects the probability of the sale or its size, and neither decision alone separates them. One alternative explanation survives the pair: a broker who reads the announcement as evidence that the caller weighs price heavily would also move toward the cheaper option on both decisions. The two explanations attribute the same signs to different primitives, the senior's willingness to walk away versus the broker's belief about the caller's price sensitivity, and Section 5.4 separates them directly by testing whether the treatment changes the price sensitivity brokers perceive.

A.7 The Two Audit Study Decisions

The two decisions in the audit are the game of Sections A.1–A.5 under the two parameterizations in Table 2. In Decision 1 the broker chooses between recommending Medicare Advantage and recommending Traditional Medicare plus a Medigap plan. Setting $\Delta = \Delta^{\text{MA}} > 0$, the gross under-insurance harm of Medicare Advantage for the audited senior, and $W_L > W_H$, everything in Sections A.5 and A.6 applies without modification. In Decision 2 the broker chooses among coverage-identical Medigap plans in the senior's choice set. Setting $\Delta = 0$ and $W_j = z\rho_j$ gives Prediction 4. Two further consequences follow. First, the suitability cost does not arise, because the plans are coverage-identical and neither is the misfit, so in the framework Decision 2 carries no cost of recommending the wrong plan and in the estimated model it does not depend on the broker's type. Second, the broker's rule contains no parameters once z cancels (Lemma 4(i)). The estimated model does give Decision 2 a cost of its own, the Decision 2 suitability cost κ_2 in equation (18), which penalizes the premium the senior pays above the cheapest coverage-identical plan and not a loss of coverage. Appendix F explains why the two suitability costs are distinct parameters.

A.8 Broker Types

The conceptual framework described above assumes the broker knows which plan matches the senior's preferences. Experiment A shows that this is not always true. The estimated model in Section 6 accounts for this with a latent broker type.

Assumption A.2 (Broker types)**.** *Each broker b has a fixed type $\tau_b \in \{I, M\}$. The informed type I knows that the high-price plan fits the audited senior. The misinformed type M believes that the low-price plan fits the audited senior, so the plan that type perceives as the misfit is the high-price plan. $N^{MA} = \Delta^{MA} - (\rho_H - \rho_L)$ for the informed type, the gross under-insurance harm less the premium the senior saves on MA as in* (1)*, and $N^{MG} = \Delta' + (\rho_H - \rho_L)$ for the misinformed type, the over-insurance harm that type perceives plus the premium the senior pays above MA. The probability of being misinformed, $\lambda(x_b)$, is a logistic function of broker characteristics fixed before the call.*

The two types believe they are doing different damage, so the assumption lets each type's suitability cost scale with its own perceived harm. A broker of the informed type who recommends Medicare Advantage to the audited senior perceives the under-insurance harm $N^{\text{MA}} = \$5,574$ per year (the survey's gross coverage harm of \$7,429 per year, including the forgone option to switch back to Medigap under medical underwriting, net of the \$1,855 premium the senior saves). This is the net harm the consumer surplus accounting uses. A broker of the misinformed type recommends MA on most calls whether or not MA pays a commission (83% of no-MA-commission calls at the estimates), although that broker's recommendation still responds to both commissions

(equation (17)). The harm cost this type perceives from a Medigap sale may be better described as a perceived cost of effort or an information cost of recommending a Medigap plan. For the senior in Corollary 1, where the low-price plan fits, this net harm is $N^{\mathrm{MG}} = \Delta' + (\rho_H - \rho_L)$. The main specification sets $N^{\mathrm{MG}} = N^{\mathrm{MA}}$ (Appendix F).

Proposition 2 (The two cutoffs are mirror images)**.** *Under Assumption A.2, a broker of the informed type recommends the low-price plan if and only if* $\kappa \leq \kappa_I^* \equiv (W_L - R(s)W_H)/N^{MA}$*, the cutoff* (7) *with* $N = N^{MA}$*, and a broker of the misinformed type recommends the low-price plan if and only if*

$$\kappa \;\geq\; \kappa_M^* \;\equiv\; \frac{W_H - W_L/R(s)}{N^{MG}}. \tag{23}$$

Proof. For the informed type the perceived misfit is the low-price plan, and Proposition 1 applies with $N = N^{\mathrm{MA}}$. For the misinformed type the perceived misfit is the high-price plan, so $U_b(L) = P_L W_L$ and $U_b(H) = P_H(W_H - \kappa N^{\mathrm{MG}})$. The misinformed type recommends the low-price plan when $P_L W_L \geq P_H(W_H - \kappa N^{\mathrm{MG}})$. Dividing by $P_H > 0$, using $P_L/P_H = 1/R(s)$, and rearranging gives $\kappa N^{\mathrm{MG}} \geq W_H - W_L/R(s)$, and dividing by $N^{\mathrm{MG}} > 0$ gives (23). □

In the dollar units of Section 6.1, the informed type's rule is equation (16), and the two rules are the two lines of equation (17). The two rules mirror each other in two respects. First, the inequality reverses, because a higher suitability cost pushes the informed type away from the low-price plan and pushes the misinformed type toward it. Second, the sale-probability ratio multiplies W_H in the informed type's rule and enters as $1/R$ in the misinformed type's rule, because each type divides the payoff comparison by the close probability of the plan it perceives as the misfit.

Proposition 3 (The no-MA-commission floor)**.** *Set* $W_L = 0$*, the commission-removal case of Experiment A. Then:*

(i) the informed type's cutoff is $\kappa_I^* = -R(s)W_H/N^{MA} < 0$*, so the share of brokers of the informed type recommending the low-price plan is* $F(\kappa_I^*)$*, the mass of brokers with a suitability cost below a strictly negative cutoff, which is small when* F *places little mass below zero;*

(ii) the misinformed type's cutoff is $\kappa_M^* = W_H/N^{MG}$*, which does not contain the sale-probability ratio;*

(iii) the perceived competition treatment therefore has no effect on the share of brokers of the misinformed type recommending the low-price plan, and the observed share of brokers recommending the low-price plan at zero commission is $\lambda\Big(1 - F(\kappa_M^*)\Big) + (1-\lambda)F(\kappa_I^*)$*.*

Proof. Setting $W_L = 0$ in κ_I^* and in (23) gives (i) and (ii), since R, W_H, N^{MA} and N^{MG} are strictly positive. The sale-probability ratio enters (23) only through $W_L/R(s)$, which is zero when $W_L = 0$: the close probability of the Medigap sale multiplies both the commission from that sale and the

cost the misinformed type attaches to it, so it cancels. For (iii), the misinformed cutoff does not depend on s, and by Assumption A.1 the treatment moves nothing else in the misinformed type's problem. With probability λ the broker is of the misinformed type and recommends the low-price plan when $\kappa \geq \kappa_M^*$, which has probability $1 - F(\kappa_M^*)$, and with probability $1 - \lambda$ the broker is of the informed type and recommends it with probability $F(\kappa_I^*)$. □

Proposition 3 is what makes the misinformed share identifiable, and it states how the two types differ at zero commission. At $W_L = 0$ the informed type gives up the commission on the plan that fits and bears the suitability cost on the plan that does not, so the informed type's probability of recommending the low-price plan is close to zero. The misinformed type does not recommend the plan that pays because the misinformed broker believes this plan is unsuitable for the senior. Thus the misinformed type's index does not have to be negative. The observed rate at which brokers recommend MA when it pays nothing is therefore approximately the misinformed share of the population multiplied by the misinformed type's MA recommendation rate, which is high, but below one.

At the estimated parameters, brokers of the misinformed type recommend MA on 83% of no-MA-commission calls and brokers of the informed type on 3%. Since the informed rate of MA recommendations is small, the observed rate of 23% can be interpreted as a rough lower bound on the misinformed share. Section 6.2 recovers λ from this floor through the distribution of the suitability cost.

A.9 Consumer Surplus

The predictions sign the broker's responses. Converting them into consumer surplus for the senior requires one further assumption.

Assumption A.3 (Recommendation compliance)**.** *For the consumer surplus accounting, the senior enrolls in the plan the broker recommends.*

This is a reasonable approximation in this market because the broker enrolls the beneficiary during the call (Section 4), so the recommendation and the enrollment are the same event far more often than in advice markets where the consumer transacts elsewhere. It is also the reason the close-probability levels, which Remark 3 shows are neither identified nor needed for the predictions, are not needed here either.

Write σ for the share of brokers recommending Medicare Advantage in Decision 1, σ_2 for the share recommending the higher-priced Medigap plan in Decision 2, ρ_c for the premium of the cheapest coverage-identical Medigap plan in the senior's choice set, and m for that choice set's markup. Under Assumption A.3 the audited senior's expected utility is $\sigma\, U_c(L) + (1 - \sigma)(-\rho_c - m\sigma_2)$. With probability σ the senior is recommended Medicare Advantage and receives the utility

of the misfit plan. With probability $1-\sigma$ the senior is recommended Medigap, pays the cheapest coverage-identical premium, and pays the markup on the share σ_2 of those calls on which the broker recommends a higher-priced plan. Measured against the benchmark in which the senior receives the cheapest plan that fits, which gives $-\rho_c$, the expected loss is

$$\mathcal{L} \;=\; \sigma\, N \;+\; (1-\sigma)\, m\, \sigma_2, \tag{24}$$

where N is the net harm of (1) evaluated at $\rho_H = \rho_c$. The first term is the misallocation loss: the share of calls on which the senior is steered into the wrong product, multiplied by the dollars that error costs the senior per year. The second is the overpayment loss: the share of calls on which the senior is recommended the right product but a needlessly expensive version of it, multiplied by the annual overpayment. Section 7 evaluates (24) at the estimated parameters, per Medigap-fit senior per year.

Remark 5 (The model does not sign the consumer surplus effect of perceived competition). Differentiating (24) with respect to the continuation value,

$$\frac{\partial \mathcal{L}}{\partial v} \;=\; \frac{\partial \sigma}{\partial v}\big(N - m\,\sigma_2\big) \;+\; (1-\sigma)\, m\, \frac{\partial \sigma_2}{\partial v}. \tag{25}$$

When $\phi < \phi^{\dagger}$ the first term is positive, because Prediction 3 gives $\partial\sigma/\partial v > 0$ and the net harm exceeds the mean markup at the calibrated values. The second term is negative by Prediction 4. Which term dominates depends on the estimated parameters, which is why we report the consumer surplus effect of Experiment B as a counterfactual in Section 7 and not as a prediction.

Remark 6 (Transfers and misallocation). The two terms of (24) enter the two surplus measures differently. The markup is a transfer from the senior to the broker and the insurance company, so it enters the consumer surplus loss in full and cancels from total surplus. Commissions are transfers from insurance companies to brokers and also cancel from total surplus. The misallocation term σN enters both. A Medigap-fit senior steered into Medicare Advantage loses coverage value and the option to switch back later, net of the premium gap the senior saves, and no other party gains that amount. A policy that acts only on Decision 2 therefore raises consumer surplus and not total surplus, while a policy that acts only on Decision 1 raises both.

A.10 Mapping to the Structural Model

The objects in this appendix enter the estimated model in Section 6 as follows. The suitability cost κ and its distribution F become the Decision 1 suitability cost, a logistic distribution with location $\bar{K}_1$ in PV dollars and scale $1/\beta$, and Table 9 lists the estimated and calibrated parameters. The net harm N does not enter the likelihood. It values each averted MA recommendation in the consumer surplus accounting (Remark 6). We note two features of the mapping. First, no parameter of the

senior's problem is estimated. The perception parameter ϕ, the continuation value v_s, and the close probabilities enter the broker's decision only through the sale-probability ratio, which is what the estimation treats. Second, the direction of the competition response is not imposed. We calibrate the baseline sale-probability ratio R_0 from the broker survey and estimate the treated ratio R_1 against it, so the question of which side of the complexity threshold the senior is on, the two columns of Table A1, becomes the testable statement $\hat{R}_1 < R_0$.

A.11 Model limitations and simplifying assumptions

For tractability, we make three simplifying assumptions. First, we assume a broker makes a single recommendation to a single senior and the game ends, meaning reputation and repeat business enter only through the suitability cost κ. Second, we assume insurance companies set premiums and commissions outside the model. Thus, the counterfactuals estimated in Section 7 are partial equilibrium. Finally, we assume seniors do not learn about policy changes such as commission changes or update their beliefs in response to them. Section 5.2 suggests this is a better approximation in this market than a fully informed senior.

B Medicare Market Details

Medicare Supplement (Medigap) Plans Traditional Medicare covers 80% of the cost of most outpatient appointments and procedures. Beneficiaries are responsible for the remaining 20%. There is no limit to the 20% coinsurance expenses (that is, no maximum out-of-pocket limit, or MOOP). In exchange for a fixed monthly premium, Medigap plans help fill these "gaps" by paying for a portion of these cost-sharing expenses. Figure A.2 shows the standardized Medigap plans by plan letter.

Broker disclosure requirements Brokers must disclose that they do not represent all insurers and all Medicare plans and tell the senior that information on all available options is available from Medicare.gov, 1-800-MEDICARE, or the free State Health Insurance Assistance Program (SHIP) (42 C.F.R. § 422.2267(e)(41); CMS, 2023). Brokers are also required to review each client's doctors, drugs, and costs before enrollment (CMS, 2025b). Note that these are the rules in place during our audit, which ran from September 2024 to March 2026, some of which were amended in Contract Year 2027. For example, brokers are no longer required to mention SHIP resources and starting in Contract Year 2027 call recordings must be stored for up to 6 years instead of the original 10 years (CMS, 2026).

Figure A.2: Standard Medicare Supplement (Medigap) Plans

Medigap Benefit	Plan A	Plan B		Plan D		Plan G*	Plan K	Plan L	Plan M	Plan N
Part A coinsurance and hospital costs up to an additional 365 days after Medicare benefits are used up	✓	✓		✓		✓	✓	✓	✓	✓
Part B coinsurance or copayment	✓	✓		✓		✓	50%	75%	✓	✓***
Blood (first 3 pints)	✓	✓		✓		✓	50%	75%	✓	✓
Part A hospice care coinsurance or copayment	✓	✓		✓		✓	50%	75%	✓	✓
Skilled nursing facility care coinsurance	×	×		✓		✓	50%	75%	✓	✓
Part A deductible	×	✓		✓		✓	50%	75%	50%	✓
Part B deductible	×	×		×		×	×	×	×	×
Part B excess charge	×	×		×		✓	×	×	×	×
Foreign travel exchange (up to plan limits)	×	×		80%		80%	×	×	80%	80%
Out-of-pocket limit**	N/A	N/A		N/A		N/A	$6,940 in 2023	$3,470 in 2023	N/A	N/A

Source: Medicare.gov

C Study Details

C.1 Audit Study Logistics

Shopper Recruitment and Screening We hire and train seniors from two secret shopping firms. Secret shoppers are all senior citizens and most are over 65 years old and therefore Medicare-eligible. We use two firms because each could reach a different part of our broker sample: one firm could mask shopper phone numbers, which suited it to contacting call center brokers, while the other could pre-schedule calls, which suited it to contacting brokers drawn from the large online directory.

Shoppers are never told the purpose of the study. Before a shopper is assigned any broker to contact, the shopper must pass an assessment demonstrating familiarity with the call script and a basic understanding of Medicare. The content of that training is described under the Caller Training section below.

We also collect demographic information on each shopper's real age, gender, race, and education level, and we include these consumer characteristics in our baseline controls. Potential discrimination is not our primary treatment of interest, but we use the natural variation in shopper characteristics to test whether recommendation quality differs across seniors. Because we randomly assign both brokers and seniors at the individual call level, any correlation between recommendation quality and shopper characteristics carries a causal interpretation. We do not

Figure A.3: Medicare Plan Characteristics vs Senior Preferences

Medicare Advantage (MA) Pros	Senior Preferences
Low cost: ▪Plans usually less than $100/month ▪Some low-income beneficiaries quality for $0 premium plans	▪Main priority is having good insurance coverage ▪Willing to spend more money for good coverage ▪No SSI or food stamps *(does not quality for extra discounts)*
Drug Plan: Most plans include Medicare drug coverage	▪Not a priority - no regular prescriptions (only multivitamins)
May include dental, vision, and hearing	▪Not a priority - have pretty good vision and hearing
Medicare Advantage (MA) Cons	**Senior Preferences**
Prior authorization: May need approval from your plan before it covers certain services or supplies.	▪Do not want to deal with prior authorization
Seeing specialists: May need a referral before seeing a specialist.	▪Want to be able to see a specialist if needed
Network restriction: in many cases, you can only use doctors and hospitals in the plan's network (for non-emergency or non-urgent care).	▪Do not want to worry about network restrictions ▪Live in a rural area area *(more limited networks)*
Travel: maybe harder to get coverage outside of where you live due to network restrictions	▪Spend 4-5 months of the year in a rural area of a different state

Notes: Figure A.3 lists the preferences and characteristics assigned to the "Medigap-fit" shopper beside the plan attributes on which Medicare Advantage and Traditional Medicare plus Medigap differ. Each assigned preference corresponds to a reason Medicare.gov gives for not choosing Medicare Advantage.

enforce balance on gender and race, given the cost of recruiting a balanced sample.

Caller Training We also provide the seniors with some basic facts about the Medicare market that many Medicare-eligible seniors may already know (for example, the definition of Traditional Medicare and Medicare Advantage, what a Red White and Blue card is, who is eligible for Medicare). We also have shoppers pass a comprehension check quiz to ensure they are familiar with the script and to ensure the shoppers would be able to accurately document the broker recommendations. The goal of the pre-call assessment and script was to keep the content of the calls as consistent as possible, aside from our experimentally assigned treatments.

Broker Sampling and the Licensure Database We sample brokers from two sources commonly used by seniors. First, we sample from two major Medicare call centers. Combined, the two call centers in our study enrolled over one million seniors in Medicare plans in 2024, a figure taken from the public financial records of the companies. Each employs over 1,000 brokers and runs extensive television, mail, and internet advertising campaigns. Second, we sample from the nation's largest directory of Medicare insurance brokers, which included over 18,000 brokers as of June 2025. We stratify the sampling to ensure a mix of brokers from both sources in both the treatment and control groups. As noted in the text, unforeseen changes in broker commissions forced us to halt data collection for call center brokers prematurely, which left a smaller call center treatment group.

We sample brokers licensed in both California and Texas, the two states in which we assign our

Table A2: Broker and Senior Characteristics

	Obs.	Mean	SD	Q1	Q3
Broker Characteristics					
Call-Center Brokers	637	0.40	0.49	0.00	1.00
Broker - Female	637	0.43	0.50	0.00	1.00
# Company Appointments (Group)	624	8.42	4.29	5.00	12.00
Broker Tenure (yrs)	634	9.40	8.10	3.78	13.11
High Tenure (4+)	637	0.70	0.46	0.00	1.00
Consumer Characteristics					
Caller - Female	637	0.83	0.38	1.00	1.00
Caller - Real Age	637	71.18	6.28	67.00	74.00
Caller Race - White	637	0.81	0.39	1.00	1.00
College Degree	637	0.59	0.49	0.00	1.00

Table A3: Call Center v Non-Call-Center Brokers

Broker Characteristics	All Mean (SE)	Call-Center Brokers Mean (SE)	Non-Call-Center Brokers Mean (SE)	Difference Δ	Reg P-value (cntrl)
# States Registered In	23.4	32.5	17.4	15.1	0.00
	(0.6)	(0.7)	(0.6)		
# Insurance Comps Appointed To (Group)	8.4	12.8	5.4	7.4	0.00
	(0.2)	(0.1)	(0.1)		
Broker Tenure (yrs)	9.4	5.8	11.8	-6.0	0.00
	(0.3)	(0.4)	(0.4)		
High Tenure	70.0%	51.8%	82.4%	-30.6	0.00
	(1.8)	(3.1)	(2.0)		
% Broker Female	43.3%	42.8%	43.7%	-0.9	0.77
	(2.0)	(3.1)	(2.5)		
N (Medigap Recs)	1,055	373	682		
N (Calls)	637	257	380		

Notes: Table A3 compares broker characteristics for brokers reached through a call center and brokers reached through the online broker directory. The first three columns report means with standard errors in parentheses, and the Difference column is the call center mean less the non-call-center mean. The final column reports the p-value on the call center indicator from a controlled regression of the row characteristic.

shoppers residency. Because the majority of brokers are licensed in more than one state, we are not concerned that this procedure yields a selected sample. Between 33% and 37% of California brokers in the large broker directory are also licensed in Texas, and between 18% and 19% of Texas brokers in the directory are also licensed in California, depending on the exact date the broker data was pulled.

To understand which broker characteristics correlate with advice quality, we construct a broker licensure database from insurance licensure data covering 47 states and match our audit brokers to it. The database gives us broker tenure, that is, how long the broker has held a health insurance license; firm of employment, including call center versus non-call-center status; and the number of companies the broker is licensed to sell. In several states, including both California and Texas, an insurance broker must be appointed with a company before selling that company's

plan. Appointments are filed with the state Department of Insurance, which allows us to observe a broker's current appointments. Both California and Texas also allow just-in-time appointments, so a broker can become appointed quickly before making a sale.

Alternative Broker Sampling Strategies We also considered several alternate broker sampling strategies. One approach would be to sample from the set of all licensed insurance brokers; however, we believe there are several problems with this approach. First, turnover in this occupation is high, and the set of licensed brokers at a given time is likely not representative of the set of brokers who are actively enrolling seniors in Medicare plans. Additionally, in almost every state Medicare brokers do not need special licensure, and are licensed under the broader "Health" or "Accident and Health" categories. This means many of the brokers in the health licensure database do not enroll consumers in Medicare, so sampling from the set of all licensed health insurance brokers would have a very high failure rate. We also considered sampling from Medicare insurance brokers identified through Google search results. Many of these search results directed us to the large Medicare insurance broker database that we ultimately decided to use. In a small pilot, we found that many of the remaining brokers contacted using this method either did not enroll individuals in Medicare or had a high nonresponse rate (defined as no response after three contact attempts across different workdays).

Geographic Choice We intentionally choose two states with different regulations regarding when seniors can change their Medigap plans: California has a "Birthday Rule," an increasingly common policy that allows individuals who have a Medigap plan to change plans every year around their birthday, while Texas does not have a Birthday Rule. Birthday Rules are increasingly common; as of December 2025, 10 states have birthday rules and more are planning to adopt them. In theory, the additional flexibility of having a Birthday Rule should make it easier to ensure the senior always has the cheapest plan: if the plan increases in price, which plans can do once a year, a broker should recommend that the senior switch to a cheaper plan. In practice, however, the two states look very similar, suggesting that plan recommendations do not change with switching legislation. We pool the results and include state fixed effects as a control.

Since Medicare plan coverage varies by zip code, we intentionally select zip codes in counties that (1) are not in urban areas, (2) have a normal amount of Medigap price dispersion, and (3) have demographic characteristics that do not deviate significantly from national averages on income and education.

Focusing on less urban areas, especially for an audit consumer who has providers in the rural areas in more than one state, raises the potential consumer surplus loss from an MA plan since rural areas tend to have fewer provider networks to choose from and result in longer travel and greater difficulty accessing care. It also means the zip codes we choose offer fewer MA plans than

urban areas do, which further limits the choice set for a consumer who seeks flexible care.

Call Script and Survey Instruments Each broker is asked to make two decisions, which correspond to the two main decisions faced by seniors enrolling in Medicare. The first decision asks brokers to choose between Medicare Advantage (MA) and Traditional Medicare plus a Medicare Supplement (Medigap) plan. The second decision asks brokers to recommend a Medigap plan for the provided zip code. These two decisions correspond to our two main outcomes of interest.

The calls proceed as follows. First, the senior explains that they have just become Medicare-eligible and are interested in discussing both Medicare Advantage (MA) and Medigap plans. We intentionally ask seniors to highlight that they want to discuss both Medicare plan types at the beginning of the call so the broker is prepared to provide two recommendations. This also reduces the risk that brokers interpret the discussion of a different plan as a rejection of their plan recommendation. Seniors then provide their assigned characteristics and coverage preferences and ask the broker whether they would recommend MA or TM + Medigap. Whether or not the broker recommends MA at this point in the conversation is our first primary outcome (Outcome 1).[63]

Seniors are specifically instructed *not* to explicitly accept or reject the first plan recommendation before redirecting the conversation to ask about Medigap plans. Seniors ask the broker for a Medigap plan recommendation for the two most popular Medigap plans, Plan N and Plan G. These Medigap plan recommendations are our second main outcome of interest. We also intentionally design the scenario so the senior travels back and forth between two states, CA and TX, for months at a time. This allows seniors to ask about the Medigap plan premiums in each state, a design which gives us up to four Medigap plan recommendations per call.

The shopper collects the name of the insurance company that provides the Medigap plan and the quote of the monthly premium provided by the broker. To ensure that we use consistent and accurate pricing, we match the Medigap plan recommended by the broker to the Medigap search tool on Medicare.gov to determine the true premium available to the senior for the recommended plan without improperly applied discounts.[64] We compare this premium to the lowest-cost premium available to the senior for a plan with identical coverage to calculate the Medigap markup, or Outcome 2.[65]

[63]As specified in the pre-analysis plan, responses where seniors select "Other" and describe the recommendation as ambiguous or explain that the broker said it was the senior's decision are coded as not recommending MA to be conservative. After reviewing the data, we found this to be appropriate since all responses where the seniors select "Other" correspond to cases where the broker initially talked about MA, but ultimately recommended Traditional Medicare plus Medigap.

[64]As specified in the pre-analysis plan, we use the publicly available Medicare.gov premium to avoid comparisons where brokers apply household discounts or short-term promotional pricing that the senior is not eligible for when quoting the premium in order to make the sale. We use a combination of the insurance company name and the premium to match. Additional details about the plan matching process are provided in the pre-analysis plan.

[65]We code non-standard Medigap plans as missing. Non-standard Medigap plan recommendations are rare and only happen 1% of the time in our sample.

One potential confound is that the Medigap plan recommendation comes *after* the initial Medicare Advantage versus Traditional Medicare recommendation. To mitigate this concern, we instruct all seniors to remain intentionally vague about their decision. Specifically, seniors are instructed to not say yes or no to a recommendation. Instead, they thank the broker, say they will think about it more, and move on to the next question. We also have seniors outline the topics of discussion at the beginning of the call so redirecting the conversation to Medicare Supplement is not seen as a rejection of the broker's recommendation. This may signal senior sophistication and a strong understanding of the Medicare market, which may bias the broker recommendations toward better recommendations. For this and other reasons discussed in Section 5.1, we believe our figures may represent an *underestimate* of unsuitable recommendations.

To try to keep calls as standardized as possible, we assign consumer preferences and design a call script for the secret shoppers to follow when contacting brokers. The call scripts are based on several real conversations between Medicare insurance brokers and beneficiaries. The script was developed after interviewing multiple Medicare beneficiaries and currently and previously licensed Medicare insurance brokers.

Secondary Outcome Definitions We report two secondary measures alongside the two primary outcomes. The first is the Medigap Percent Markup, which expresses the Medigap Markup as a share of the annual premium of the cheapest coverage-identical plan available to the consumer. This measure gives us a sense of the overpayment in relative terms and allows us to compare the Medigap markup to the markup found in other markets.

The second is the Medigap Broker Markup, which asks how much of the markup survives once we hold the broker's own choice set fixed. It compares the premium of the recommended Medigap plan to the lowest premium coverage-identical plan available to the senior that the broker is currently licensed to sell.[66] The measure is coded as missing when the broker does not name a specific Medigap plan on the call, when the broker cannot be identified, and when appointment data are unavailable.[67]

Data Collection We have surveyors fill out an information collection form within three business days of completing the call, although most shoppers fill out the form within 24 hours. If shoppers do not submit the survey instrument within 8 hours they get a message reminding them to submit the form. Data collection requires surveyors to fill out a form recounting the recommendations

[66]We collect these data from the publicly available state department of insurance licensure databases. California and Texas both require the broker to be appointed with an insurance company before selling that company's plan, and appointments are reported to the state Department of Insurance. Depending on match quality, we match insurance companies at the parent or the subsidiary level.

[67]California's appointment records appear less complete than Texas's, and active Medicare brokers are often missing appointments there. Brokers can also become appointed with a company after deciding to make a sale. For both reasons we treat this as a secondary measure.

made on their calls and explanations or statements made by the broker during the call. We also collect a call recording, which is converted to a call transcript, when possible.

C.2 Natural experiment: plan availability and design constraints

When we began data collection, three insurance companies offered MA plans in our California zip code: two national carriers, Anthem and Aetna, and one California-focused carrier, Imperial Health. In October 2024, Anthem announced that it would stop paying MA commissions in several California regions, including our zip code, effective November 1, 2024. Since this was too late to remove the plans from the market for 2025, Anthem plans remained in the market through December 31, 2025.

Between May and July 2025, several large insurance companies, including Anthem Blue Shield of California and UnitedHealthcare, announced similar commission cuts for MA plans in other California counties. These changes drew significant attention from state governments and from Medicare broker blogs and forums.[68] While the changes between May and July 2025 did not directly affect the MA plans in our California audit zip code, the prevalence of commission cuts throughout California may have made the previous Anthem commission change and future cuts more salient to brokers.

Finally, Aetna announced that it would stop paying MA commissions after September 1, 2025. At this point, no national insurance carriers, and as a result no carriers that contracted with call centers, paid broker commissions within our audit study region.

Throughout the audit study window at least one MA plan was always available for enrollment. Prior to September 1, 2025, there was also at least one national carrier that paid broker commissions: Medicare.gov publishes the available plans for the upcoming plan year on October 1 and the Annual Enrollment Period opens October 15, meaning brokers could earn a commission on an Aetna enrollment from October 15, 2024 onward. Anthem's commission cut took effect on November 1, 2024, but Anthem's commissions ran through October 31.

Both the Anthem and Aetna plans remained available and continued to accept new enrollments through December 31, 2025. This means it was still possible for call center brokers to enroll consumers in those plans through the end of the year: the Anthem plan was available through December 2025 and the Aetna plans from October 1, 2024 through December 2025. MA plans were still available to consumers after January 1, since two of the four Imperial Health Medicare Advantage plans remained available for the entire duration of the audit study. A broker who checked Medicare.gov would have seen all three carriers' plans.

[68] See https://jrreport.wordandbrown.com/2025/07/01/blue-shield-of-california-cuts-medicare-advantage-broker-commissions/; https://www.beckerspayer.com/payer/unitedhealth-cuts-some-medicare-advantage-broker-commissions/;

We confirmed that Imperial Health does not work with the national call centers by checking the California Health Insurance License database, which requires companies and individuals to be appointed with any carrier whose products they sell. Neither of the two major call centers we sample from was licensed to sell Imperial Health products at the time of the commission change or in the three months after.

One further consequence of the commission removal is worth noting for interpretation. If an MA carrier leaves the market unexpectedly, its enrollees receive a new enrollment window in which they may choose any MA plan without medical underwriting or the risk of denial. This means enrolling in an MA plan that will leave the market in 1-2 months may still be the best choice for the senior; the senior can receive the benefits of this (potentially under-priced) MA plan for a few months, then enroll in whichever Medicare plan they choose without medical underwriting once the plan leaves the market.

An ideal design would have paired the post-September California call-center series with a Texas call-center control series of equal size; however, one limitation of the field setting made this infeasible. Call-center brokers required shoppers to provide a phone number during each call (and often required the number to be verified by text). Each call with the shopper's phone number was then logged in the shared firm-wide system and each additional call raised the risk that a call center would become suspicious.[69] Thus, our pool of roughly one hundred shoppers set a ceiling on the number of call-center calls we could place during the post-MA commission removal window. We allocated all of these calls to California, where the commission removal binds, so that the pre–post comparison at the center of the natural experiment would be adequately powered. Dividing these calls between the two states would have left the analysis underpowered.[70]

C.3 Experiment B: Additional Details

At the beginning of the phone call, shoppers assigned to the perceived competition treatment say one of the following:

> *"I want to get a few quotes from different brokers before making a decision"*
>
> *"I want to shop around before I choose a plan"*
>
> *"I want to get a few recommendations from other brokers before I choose a plan"*

Shoppers in the control group are instructed to say that they want a recommendation from the broker to better understand their options, or to say nothing at all. The signal is delivered at the start of the call, usually in the first sentence, to make it salient.

[69] We considered using Google Voice numbers, but the secret-shopper firm we partnered with advised against it since, in their experience, these phone numbers were often screened or blocked by similar firms.

[70] The seven Texas call center calls in September occurred before the secret shopper firm had a chance to cancel the calls that had already been assigned.

The treatment does not vary market structure or manipulate market-level broker supply. It works by lowering the broker's perceived probability of making the sale, which makes the risk of losing the client salient in a market that is already competitive. Finding a Medicare broker is inexpensive for a consumer: an online search returns hundreds of results and multiple toll-free numbers that a beneficiary can call at almost any hour to reach a licensed broker. Brokers are aware of this at baseline. In interviews, several complained about competitors taking their clients, and many Medicare beneficiaries receive cold calls from brokers seeking their business.

Randomization is stratified by state and firm type, and brokers are assigned to treatment or control before any calls are placed. Calls are then randomly assigned to shoppers at the individual level. We implement the treatment and control conditions in waves, at different points in time, to reduce shopper confusion and mistakes: when we originally assigned both conditions simultaneously, we found compliance problems, because our senior shoppers had difficulty remembering which condition applied to a given call. Because the arms were fielded at different times, Appendix D.5 reports specifications that flexibly control for caller experience and calendar time.

Table A4: Competition Assignment - Balance Table of Broker and Consumer Characteristics

	All Mean (SE)	Competition Assigned Mean (SE)	Not Assigned Mean (SE)	Difference Δ	Reg P-value (strata)
% Competition Treated	41.6%	89.7%	9.9%	79.8	.
	(2.0)	(1.9)	(1.5)		
Broker Characteristics					
% Call-Center Brokers	40.3%	22.1%	52.3%	-30.2	.
	(1.9)	(2.6)	(2.6)		
# States Registered In	23.4	20.9	25.1	-4.2	0.61
	(0.6)	(0.9)	(0.7)		
# Insurance Comps Appointed To (Group)	8.4	7.1	9.3	-2.2	0.95
	(0.2)	(0.3)	(0.2)		
Broker Tenure (yrs)	9.4	10.4	8.7	1.7	0.92
	(0.3)	(0.5)	(0.4)		
High Tenure	70.0%	75.5%	66.4%	9.1	0.78
	(1.8)	(2.7)	(2.4)		
Broker Female	43.3%	41.5%	44.5%	-3.0	0.44
	(2.0)	(3.1)	(2.5)		
Caller Characteristics					
Caller Female	83.0%	84.2%	82.3%	1.9	0.63
	(1.5)	(2.3)	(2.0)		
Caller White	81.0%	87.0%	77.1%	9.9	0.26
	(1.6)	(2.1)	(2.1)		
Caller Black	6.9%	5.9%	7.6%	-1.6	0.54
	(1.0)	(1.5)	(1.4)		
Caller Real Age (yrs)	71.2	71.5	71.0	0.5	0.95
	(0.2)	(0.4)	(0.3)		
Caller Above Med Age	50.5%	55.3%	47.4%	7.9	0.53
	(2.0)	(3.1)	(2.6)		
Caller College Degree	58.9%	57.7%	59.6%	-1.9	0.95
	(2.0)	(3.1)	(2.5)		
Caller # Calls	20.0	27.8	14.8	13.0	0.10
	(1.3)	(2.4)	(1.3)		
N (Medigap Recs)	1,055	427	628		
N (Calls)	637	253	384		

Notes: Table A4 shows the balance table of broker and consumer characteristics by competition assignment.

Table A5: Competition Compliers vs Non-Compliers - Broker and Senior Characteristics

	All Mean (SE)	Competition Correct Mean (SE)	Competition Incorrect Mean (SE)	Difference Δ	Reg P-value (strata)
% Competition Treated	41.6%	39.6%	59.4%	-19.8	.
	(2.0)	(2.0)	(6.2)		
Broker Characteristics					
% Call-Center Brokers	40.3%	39.1%	51.6%	-12.5	.
	(1.9)	(2.0)	(6.3)		
# States Registered In	23.4	23.3	24.8	-1.5	0.93
	(0.6)	(0.6)	(1.7)		
# Insurance Comps Appointed To (Group)	8.4	8.3	9.2	-0.9	0.95
	(0.2)	(0.2)	(0.6)		
Broker Tenure (yrs)	9.4	9.6	7.5	2.2	0.11
	(0.3)	(0.3)	(0.7)		
High Tenure	70.0%	70.0%	70.3%	-0.3	0.38
	(1.8)	(1.9)	(5.8)		
Broker Female	43.3%	42.6%	50.0%	-7.4	0.34
	(2.0)	(2.1)	(6.3)		
Caller Characteristics					
Caller Female	83.0%	83.2%	81.3%	2.0	0.94
	(1.5)	(1.6)	(4.9)		
Caller White	81.0%	82.7%	65.6%	17.1	0.11
	(1.6)	(1.6)	(6.0)		
Caller Black	6.9%	6.8%	7.8%	-1.0	0.99
	(1.0)	(1.1)	(3.4)		
Caller Real Age (yrs)	71.2	71.4	69.1	2.3	0.17
	(0.2)	(0.3)	(0.5)		
Caller Above Med Age	50.5%	52.2%	35.9%	16.2	0.16
	(2.0)	(2.1)	(6.0)		
Caller College Degree	58.9%	59.0%	57.8%	1.2	0.92
	(2.0)	(2.1)	(6.2)		
Caller # Calls	20.0	21.3	7.9	13.4	0.19
	(1.3)	(1.4)	(1.4)		
N (Medigap Recs)	1,055	980	75		
N (Calls)	637	573	64		

Notes: Table A5 shows the balance between the compliers and non-compliers on broker and senior characteristics.

C.4 Understanding America Study (UAS) Senior Survey and Additional Results

To evaluate the external validity of our audit study design, we survey a nationally representative panel of 2,015 Medicare beneficiaries through the Understanding America Study (UAS). The goal of the survey is to better understand broker use and Medicare enrollment and to test the external validity of our audit study design. Our survey population includes seniors who are 65 years of age or older and individuals with disabilities who have Medicare. We re-weight the survey responses to reflect the US population. This helps adjust for a slight oversampling of respondents from California, where the Center for Economic and Social Research at the University of Southern California (USC), which administers the study, is based.

Similar to previous surveys, we find that the majority of beneficiaries used a broker to enroll in Medicare. Table A7 reports broker use by Medicare coverage group: among beneficiaries who report neither Medicaid eligibility nor access to an employer plan, the population to which Section 7.2 scales the structural estimates, 56% used a broker. Figure A.4 highlights the different distribution channels that seniors use to search for and match with brokers. 51% of beneficiaries find their broker in existing networks (through word of mouth or prior relationships) and 49% of beneficiaries find their brokers through new networks (advertisements/TV/mailers, cold outreach where the broker reached out to the senior directly, online search, or "other" methods). We take this as evidence that our sampling method of finding brokers through commonly advertised distribution channels is not unusual.

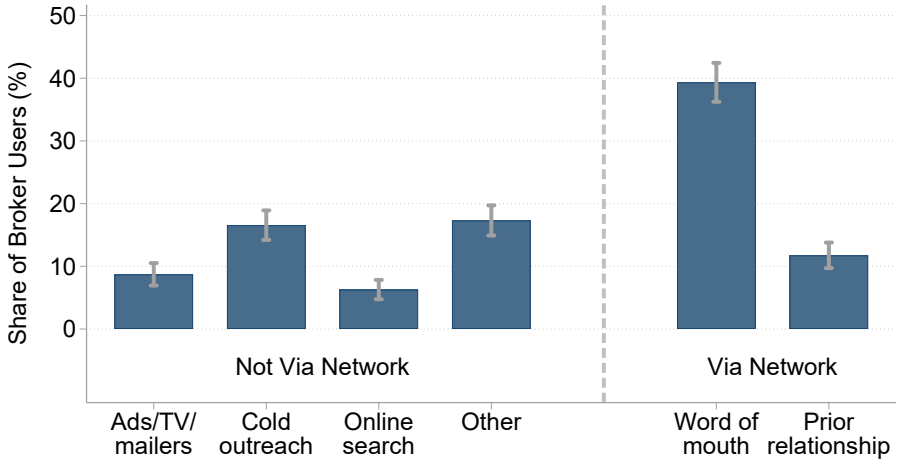
We also ask whether seniors talk to multiple brokers before enrolling in a Medicare plan. Of the seniors who enrolled in Medicare using a broker, 66% of seniors spoke to only one broker before enrolling in a plan and 7% "do not know" how many brokers they spoke with. While this does not directly test whether *telling* a broker you will speak to their competitors is unusual, the survey results at least suggest that speaking with multiple brokers before enrolling itself is not unusual, but may still be unusual enough to cause brokers to change their behavior.

The evidence of limited search is striking given how low the search costs of finding a broker are. Brokers heavily advertise to seniors via mailers, cold-calls, online advertisements, and even in-person visits.[71] A broker can easily be reached by calling a call center number at almost any hour of the day or by scheduling an automated appointment online. Additionally, most initial Medicare consultation appointments are about 30 minutes. This limited search, even though contacting multiple brokers is cheap and easy, provides evidence that our shoppers' signal that they plan to contact multiple brokers is both realistic and meaningful: it is the kind of thing a real senior might say, yet salient enough to potentially change a broker's recommendation.

[71] Multiple brokers we spoke to reported competitors showing up at their clients' doorsteps and frequenting community centers to try to find clients.

Finally, the results from our UAS survey highlight one of the benefits of using our audit study evidence over noisy survey evidence. We find that even among our sample of positively selected seniors who participate in a longitudinal research survey panel, at least 40% of our survey respondents answered questions about what Medicare plan they have or how much they pay in premiums incorrectly.[72] Our audit study methodology addresses these limitations by directly asking brokers to recommend plans in a specially designed "textbook" scenario where we assigned senior preferences, rather than relying on self-reported and potentially noisy survey data.[73]

Figure A.4: Broker Distribution Channel (Broker Users Only)



Notes: Figure A.4 shows the reweighted share of Medicare beneficiaries who found their broker through each distribution channel. Distribution channel options included advertisements/TV/mailers, cold outreach (the broker reached out to the senior directly), online search, word of mouth, prior relationship, or other. Error bars show the 95% confidence intervals. Note that this may underestimate brokers found through distribution channels such as ads/TV, since many seniors who speak to brokers through them believe they are speaking to Medicare representatives and do not realize they are speaking to brokers.

[72]The survey questions were carefully vetted by the Understanding America Study team and tested for clarity before the survey was conducted. We consider the following beneficiaries to be confused about what plan they have: beneficiaries who report having Medigap plans with premiums that are ≤$20 per month (Medigap plans tend not to have monthly premiums this low, meaning the respondent likely has Medicare Advantage), beneficiaries who report having Medicare Advantage plans that are over $100/month (MA plans are unlikely to have a premium that high), beneficiaries who report having Medigap plan letters that do not exist or who do not know what Medigap plan letter they have, and beneficiaries who answer "do not know" when asked which Medicare plan they have.

[73]We consider a respondent confused about what plan they have if they fall into any of the following categories: 1) The respondent answers "Do not know" to the question about what Medicare plan they currently have. 2) The respondent reports a monthly Medigap premium ≤ $20 per month (Medigap plans tend not to have monthly premiums this low, meaning the respondent likely has Medicare Advantage). 3) The respondent selects "13 - Other" or "14 - Do not know" in response to the question about what Medigap Plan letter they have (the list of Medigap plan letters is exhaustive). 4) Respondents report an MA monthly premium ≥ $100/month (MA plans are unlikely to have a premium that high).

Table A6: Summary of UAS Survey Results

Panel A. Respondents unable to correctly identify current Medicare coverage

Survey Response	**Share**	**Base**
Medicare but does not know the specific type	10%	All respondents (2,015)
Medigap plan letter of "Other" or "Do not know"	32%	Medigap enrollees (719)
Medigap premium $\leq$ \$20/month	14%	Medigap enrollees (705)
Medicare Advantage (MA) premium $\geq$ \$100/month	34%	MA enrollees (1,042)
Fails at least one of the four tests above	$\geq$ 40%	All respondents (2,015)

Panel B. Broker use and brand preferences

Survey Response	**Share**	**Base**
Used a broker to enroll	51%	Answered item (1,900)
Consulted only one broker before enrolling	66%	Broker users (1,011)
Does not know how many brokers were consulted	7%	Broker users (1,011)
Has neither a preferred nor an avoided insurer	61%	Answered both items
Has no insurer they prefer not to work with	88%	Answered item (1,889)
Found broker through friends, family, or a prior relationship, household income $\leq$ \$60,000	58% (SE 2.37)	Broker users
Found broker through friends, family, or a prior relationship, household income $>$ \$60,000	69% (SE 2.58)	Broker users

Notes: Table A6 summarizes the results from a survey of Medicare beneficiaries fielded through the Understanding America Study (UAS) in July 2025. MA = Medicare Advantage. All shares are re-weighted to match the US population. Each row's denominator is the set of respondents who answered the question (excluding missing observations). Panel A shows a lower bound for the share of respondents who were unable to correctly identify their current Medicare coverage. Each row shows a specific response inconsistency (e.g. selecting a Medigap plan letter of "Other" or "Do not know" from an exhaustive list of Medigap plans). Panel B's brand preference rows compare respondents who report a preferred or not preferred insurer against respondents who report caring only about what the plan covers. The referral rows report the share of broker users who found their broker through word of mouth or a prior relationship, split at a household income of \$60,000. Standard errors treat the weighted sample as i.i.d.

Table A7: Broker Use by Medicare Coverage Group (UAS)

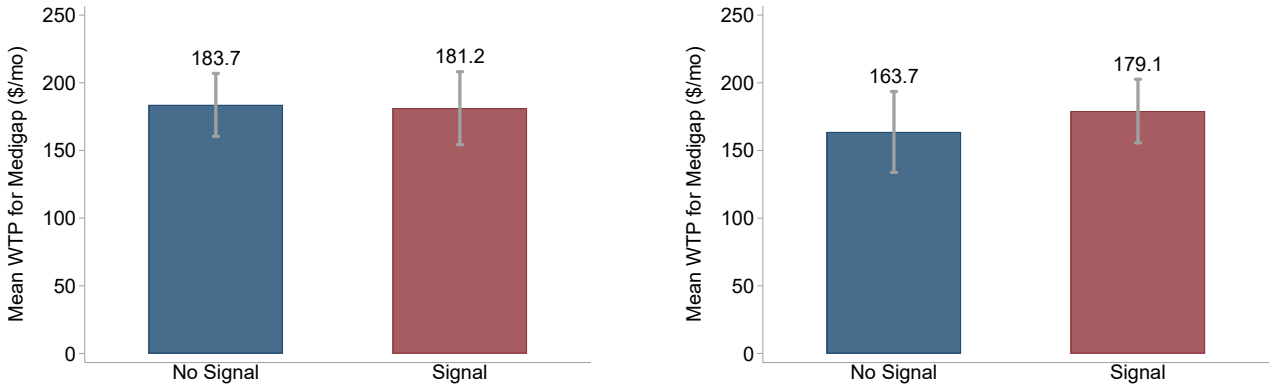
Medicare population	Broker use	n	Share of reweighted sample
All	51.3% (SE 1.5)	1,875	100.0%
Medicaid-eligible	57.9% (SE 4.7)	182	12.0%
Employer access	20.8% (SE 3.1)	335	16.0%
Remaining	56.4% (SE 1.8)	1,371	72.8%

Notes: Table A7 reports the share of respondents to our Understanding America Study (UAS) survey of July 2025 who say an agent helped them enroll in their current Medicare plan ("Not sure" counts as no), by the respondent's answers to a select-all item asking whether they are Medicaid-eligible or have access to Medicare plans through a current or prior employer. All shares use the survey's post-stratification weights: each broker-use share is a weighted mean with a linearized standard error in parentheses, and the last column is the group's share of the total weight. Medicaid eligibility is self-reported and does not separate full-benefit from partial-benefit dual eligibles; the employer row records access to an employer plan, not enrollment in one. The two groups overlap: 13 respondents report both and appear in both rows, so the three group rows sum to more than 100%. The Remaining row is every respondent who reports neither, and it includes the 76 respondents who report military or TRICARE health benefits. Among Remaining respondents aged 65 to 69, 61% used a broker (SE 3.1). The 1,875 respondents are those who answered the broker item and carry a positive weight. Section 7.2 scales the structural estimates to the population the Remaining row describes.

C.5 Broker Survey and Additional Survey Results

We supplement our audit study data with a survey of 157 currently licensed Medicare insurance brokers. To recruit brokers, we hire a Business-to-Business (B2B) market research firm to recruit a sample of licensed Medicare insurance brokers through their partner firms. To distribute the survey, the firm contacts insurance brokerage firms within its existing network of partner firms and asks the insurance brokerage firms to distribute the survey through their networks. The firm's survey portal automatically assigns each respondent a unique ID to prevent duplicate responses.[74] We use the broker survey results to calibrate structural model parameters and as an additional test of the mechanisms contributing to broker advice.

Figure A.5: WTP by Perceived Competition Signal, by Vignette



(a) Vignette 1

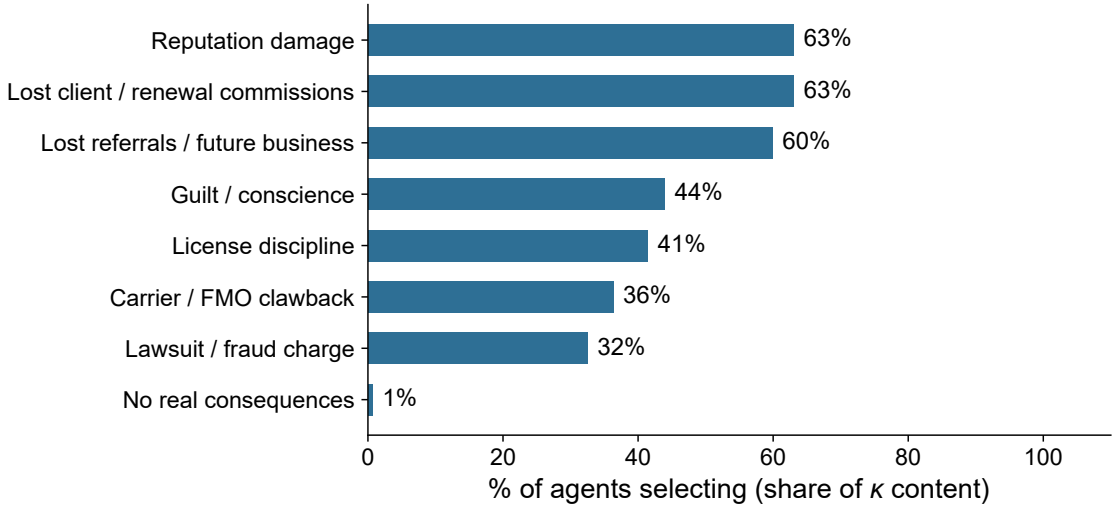
(b) Vignette 2

Notes: Figure A.5 shows the average maximum WTP for a Medigap plan ($/mo) by competition signal, using across-broker variation. Both Vignette 1 and Vignette 2 describe Medigap-fit seniors similar to the one in the audit study. These figures are the across-broker equivalent of Figure 5. Grey bars are ±1 SE.

Figure A.6 shows the self-reported consequences of a low-quality recommendation reported by currently licensed Medicare brokers in our broker survey. These can be interpreted as the contributors to the suitability cost κ in Section 3.3.

[74]We also supplement data collection using the online survey platform Prolific and by directly messaging the brokers in our audit study and asking them to participate in our survey, although the majority of the responses are from the B2B provider.

Figure A.6: What a poor-fit recommendation costs the broker



Notes: Figure A.6 shows the share of brokers who select each response to the question "Think about the typical broker in your market. If a broker gives a client a Medicare recommendation that turns out to be a poor fit (e.g., the client overpays or has coverage problems), what are the worst things that can happen to the broker?" Respondents could select any number of options, so shares do not sum to 100 percent. Bar labels show the abbreviated survey options. "Carrier / FMO clawback" stands for "Carrier/FMO action (broker is dropped, or commissions is [*sic*] clawed back)." FMO stands for field marketing organization. "License discipline" refers to disciplinary action by a regulator where the broker's license is suspended or revoked. The figure omits the write-in option "Other (please specify)."

D Additional Results and Robustness Checks

This appendix reports the full regression tables behind two exhibits in Section 5.

Table A8: Predictors of Recommendation Quality

	(1) Misaligned Rec (%)	(2) Rec MA (%)	(3) Medigap Markup	(4) Medigap Broker Markup	(5) Medigap % Markup
Caller - Female	-3.28	-4.69	16.12	26.00	3.76
	(4.74)	(4.41)	(29.83)	(29.26)	(2.49)
Caller - Real Age	4.72	1.64	33.95	21.08	2.17
	(6.46)	(5.98)	(36.35)	(39.50)	(2.67)
Caller - Real Age^2	-0.04	-0.01	-0.25	-0.17	-0.02
	(0.04)	(0.04)	(0.27)	(0.30)	(0.02)
Caller Race - White	2.78	-6.41	34.89	25.08	2.45
	(4.96)	(5.45)	(22.88)	(24.28)	(1.77)
Caller - College Degree	-0.45	9.19*	-29.81	-24.24	-3.27*
	(3.39)	(5.40)	(23.07)	(22.90)	(1.80)
Caller Experience (# Calls)	-0.11*	-0.12	0.78	0.96	0.04
	(0.07)	(0.08)	(0.54)	(0.63)	(0.04)
Call-Center Brokers	0.11	16.81***	-4.71	178.07***	-0.50
	(5.00)	(4.85)	(19.17)	(19.25)	(1.46)
Broker - Female	-1.87	3.23	-36.02**	-32.74	-3.19**
	(3.77)	(3.72)	(17.74)	(21.90)	(1.36)
High # Ins Comps	-16.17***	-9.54**	-61.52***	71.20***	-4.82**
	(3.96)	(3.91)	(20.99)	(21.89)	(1.96)
High Tenure (4+ yrs)	-11.58***	-10.28**	-77.04***	-45.96**	-6.47***
	(4.34)	(4.38)	(19.37)	(21.73)	(1.54)
State/Plan/AEP FE	Yes	Yes	Yes	Yes	Yes
Observations	637	637	1,055	1,002	1,055
R^2	0.11	0.16	0.16	0.16	0.23

Notes: Table A8 shows the full table of recommendation quality by broker and senior covariates. Each column is one regression on the baseline controls, and the reported coefficients are that regression's broker and caller controls. Standard errors, in parentheses, are clustered at the caller level. $^*p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$.

Table A9 reports the regressions of Table 5 in the natural units of each measure. The specifications and estimation samples are identical; only the scaling of the row variable differs. The estimated change in the probability of a high rating is statistically indistinguishable from zero for all four measures of recommendation quality spanning Decision 1 and Decision 2. Call length, which we read as a proxy for broker effort, raises the probability of a high rating by 0.43 percentage points per additional minute on the call.

Figure A.7 reports the adjusted margins behind the knowledge-proxy comparison of Section 5.5.

D.1 Calculating lost consumer surplus from an MA Recommendation

This appendix gives the compensating variation behind the annual and lifetime consumer surplus losses reported in Section 5.1 for Outcome 1.

We measure the consumer surplus cost of steering a Medigap-fit senior i into a \$0-premium

Table A9: Broker Rating by Recommendation Quality and Call Characteristics

	(1) % High Rating (4 or 5)	(2) % High Rating (4 or 5)	(3) % High Rating (4 or 5)	(4) % High Rating (4 or 5)	(5) % High Rating (4 or 5)
Recommend MA (%)	-0.05				
	(0.05)				
Medigap Markup ($100)		-0.27			
		(0.57)			
Medigap Broker Markup ($100)			-0.20		
			(0.56)		
Medigap Markup (%)				-0.03	
				(0.07)	
Call Length (mins)					0.43***
					(0.13)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055	637
R^2	0.05	0.08	0.08	0.08	0.06

Notes: Table A9 reports the regressions of Table 5 with each row variable in its natural units. The rating corresponds to the senior's answer to the question "On a scale from 1 (definitely would **not** recommend) to 5 (definitely **would** recommend), how likely are you to recommend this insurance broker to a friend or family member selecting insurance in this state?" A high rating is defined as a rating of a 4 (likely to recommend) or 5 (definitely would recommend). Each column reports a separate OLS regression of the high-rating indicator, scaled to 0–100, on the row variable and controls. Column (1) reports the coefficient on an indicator for a Medicare Advantage recommendation scaled to 0–100, so the effect of a recommendation is one hundred times the reported coefficient. Columns (2) and (3) report the effect of a $100 increase in the annual Medigap markup, column (4) the effect of a one percentage point increase in the percent markup, and column (5) the effect of one additional minute of call length. Columns (1) and (5) are estimated at the call level; columns (2)–(4) are estimated on the sample of Medigap price quotes (up to one quote per plan type and state per call), so the number of observations differs across columns. Baseline controls, as in Table 5. Standard errors are clustered at the caller level. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$.

Medicare Advantage (MA) plan as a compensating variation: the monthly dollar transfer that would leave senior i indifferent between the two plans. MA and Traditional Medicare + Medigap (TM+MG) plans differ on a set of well-defined features such as prior authorization requirements, network restrictions, and dental, vision, and hearing benefits. We assume that plan attributes are additively separable features and that utility is linear. This allows us to calculate the compensating variation as the sum of the individual plan attributes.[75]

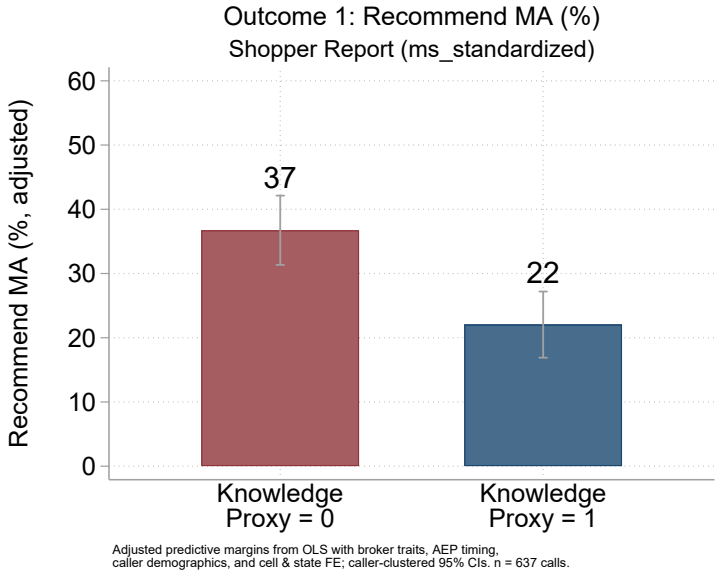
$$\mathrm{CV}_i = \underbrace{S + D + v_i^{\mathrm{dvh}}}_{\text{gains from MA}} - \underbrace{\left(v_i^{\mathrm{net}} + v_i^{\mathrm{auth}} + v_i^{\mathrm{claim}} + v_i^{\mathrm{oop}}\right)}_{\text{Medigap protections lost in MA}} - \underbrace{v_i^{\mathrm{switch}}}_{\text{forgone MA}\to\text{MG switching option}} . \tag{26}$$

Each v_i^{attrib} is i's willingness to pay ($/month) for each Medicare plan attribute, including dental/vision/hearing (DVH), network freedom, no prior authorization, freedom from claims-denial hassle costs, and predictable out-of-pocket costs; S is the avoided Medigap premium and D is the money value of MA's bundled drug coverage. To be conservative we set the MA premium and the MAPD drug cost to $0.[76] A negative CV_i implies the MA plan leaves consumer i worse off. Figure A.8 plots each component of Equation (26).

[75]This additive part-worth structure is methodologically related to conjoint analysis in marketing (Green and Srinivasan, 1990) and to discrete-choice experiments in health economics (De Bekker-Grob et al., 2012).

[76]The average MA-PD premium for Part D coverage was $7 in 2025 ($8 in 2026) (Cubanski and Damico, 2026); as with D above, we use the 2025 figure.

Figure A.7: Medicare Advantage Recommendations by Broker Knowledge Proxy



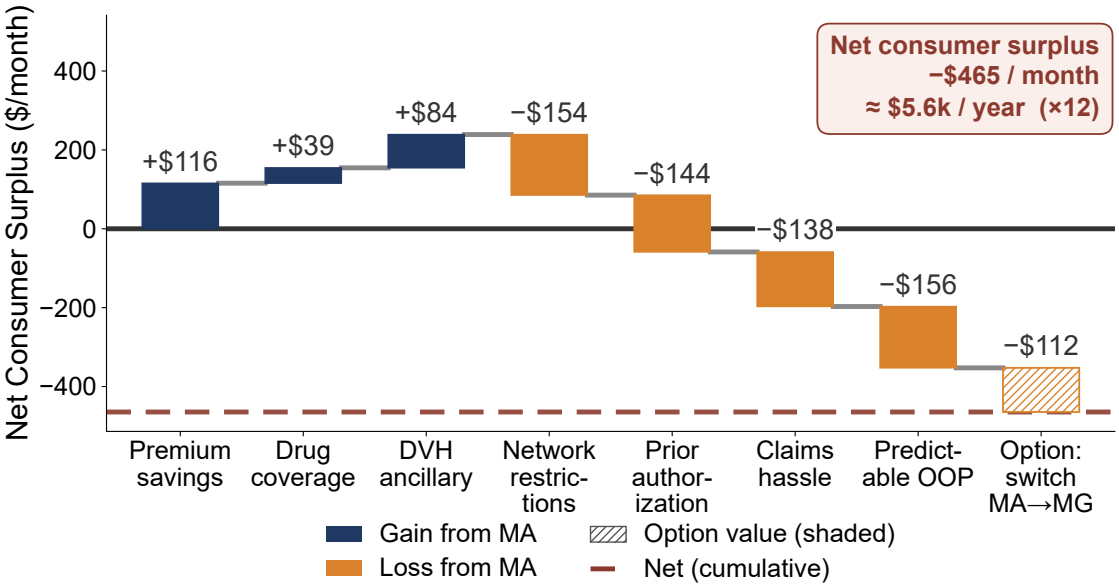
Notes: Figure A.7 shows adjusted predictive margins for the share of calls on which the broker recommends Medicare Advantage, by whether the shopper reported that the broker stated Medigap plans within a plan letter are standardized. Estimates come from OLS with broker traits, Annual Enrollment Period timing, caller demographics, and cell and state fixed effects; bars are 95% confidence intervals from standard errors clustered at the caller level, over 637 calls. The proxy is an affirmative-statement indicator: it equals one when the shopper reported the broker made the statement, so its zero pools brokers who said the opposite, brokers who never discussed Medigap, and calls the shopper could not classify. We use the shopper-coded proxy rather than the transcript-coded one because it is observed for every call, while the transcript measure conditions on a complete recording.

Given existing evidence of information frictions, switching costs, and consumer inertia resulting in consumer choice that is inconsistent with expected-utility, we elicit the valuations directly as a money-metric from the Medicare broker survey described in Appendix C.5. While it may seem counterintuitive to elicit the welfare measure directly from brokers given our evidence of unsuitable broker advice, the survey is designed to elicit the informed expert benchmark without the misaligned incentives present in our audit setting since they are not tied to real enrollments or commissions.[77] Any residual self-serving bias in the valuations would work against our results: a broker inclined to rationalize their own MA steering would understate the Medigap-distinguishing protections or overstate MA's ancillary benefits, shrinking our measured consumer surplus loss toward zero. Thus, our estimate of the consumer surplus loss from an MA recommendation is, if anything, a conservative lower bound on the true consumer surplus cost of an MA recommendation.

The second concern with the broker-elicited measure is broker competence. If brokers do not understand the different Medicare products, a concern we show in Section 5.5 to be somewhat founded, the elicited measures would be noisy and Δ may measure confusion rather than consumer welfare. In Appendix D.1 we show that restricting the sample to the brokers who score highest on the knowledge test would increase our MA consumer surplus loss estimates. We use the full-sample

[77]To incentivize effort brokers are told that the aggregate responses will be shown to real Medicare beneficiaries, but the responses are not tied to potential clients or enrollments, avoiding commission-driven conflicts of interest.

Figure A.8: MA Consumer Surplus for a Medigap-fit Senior



Notes: Figure A.8 shows the components of a net annualized consumer surplus loss of −$5,574 from MA restrictions by Medicare plan attribute for a Medigap-fit senior. Measures for each attribute are elicited as a money-metric from a survey of brokers. Premium savings calculated from the minimum average Medigap Plan N/G monthly premium available to 65-year-old non-smokers in our audit study regions. Drug coverage valued at the 2025 national average monthly stand-alone Part D (PDP) premium from Cubanski and Damico (2026). The full methodology is described in Appendix D.1. Figure A.12 reports the same decomposition for the average senior.

estimate to be conservative.

For each one of the key attributes that distinguish MA from TM+MG, we ask brokers to report the "maximum additional monthly premium ($/month)" that a given senior would pay to have that feature. To account for the lower MA premium cost we include two offsetting attributes: S, the average avoided minimum monthly Medigap premium ($116/mo from our audit data), and D ($39/mo), the 2025 national average monthly stand-alone Part D (PDP) premium (Cubanski and Damico, 2026).[78]

We assume that each plan attribute is additively separable. This assumption seems reasonable given the plan attributes. For example, it seems reasonable that how much a consumer values avoiding prior authorization would depend on their valuation of network restrictions.

We report an annual consumer surplus loss as well as a mortality-adjusted present-value consumer surplus loss, both in constant 2025 dollars, both for a Medigap-fit senior and for the average senior. Given the difficulty switching from an MA plan to Traditional Medicare plus Medigap due to the underwriting restrictions of Medigap plans, we hold the senior in the MA plan with annual persistence φ, and our headline sets $\varphi = 1$: the senior stays in the MA plan for the rest of their life. We express the consumer surplus loss in constant 2025 dollars; the annual loss Δ is flat in

[78]Concretely, the premium savings from Medigap are calculated from the minimum average Medigap Plan N/G monthly premium available to 65-year-old non-smokers in our audit study region.

real terms. We accumulate the loss from age 65 through the life table's terminal age of 100, weight each year by survival, and discount at the real rate $r = 3\%$, the U.S. federal benefit–cost default for consumption effects set by the OMB (United States Office of Management and Budget, 2025).[79] The mortality adjustment uses the survival rate of the 2022 NCHS United States Life Tables.[80] The 36-year estimates reported in Section 5.1 apply these weights; undiscounted, the cumulative 36-year loss is $200,682 for a Medigap-fit senior and $97,604 for the average senior.

MA consumer surplus loss by knowledge test score The broker survey includes three knowledge check questions on fundamental facts about the Medicare market that Medicare brokers should know. We give brokers one point for each correct answer and the total score is out of three points. The knowledge check questions and answers are shown in Figure A.9.

Figure A.11 plots Δ, the consumer surplus loss from an MA recommendation, against the knowledge score. Brokers who miss all three items value the lost protections at $4,807/yr; brokers who get all three right value them at $6,404/yr. The results suggest that the estimated consumer surplus loss from MA is *increasing* in broker knowledge, making the pooled estimate across all knowledge levels a lower bound. The brokers who are more knowledgeable about Medicare report larger consumer surplus loss, so restricting the sample to the most knowledgeable brokers would only increase our consumer surplus loss estimates. We continue to use the full-sample estimate to be conservative.

D.2 Calculating lost consumer surplus from the Medigap markup

This appendix gives the back-of-the-envelope calculation behind the lifetime Medigap markup loss reported in Section 5.1 for Outcome 2. We project the monetary loss forward over an expected enrollment horizon for our hypothetical 65-year-old senior. All values are in constant 2025 dollars. We make two simplifying assumptions in our calculation. The first concerns switching: the senior remains enrolled each year with persistence φ, and the headline calculation sets $\varphi = 1$, so the senior stays in the same Medicare plan for the remainder of their life. Given underwriting restrictions on switching outside the initial Medicare enrollment window, consumer inertia, and the low rates of switching behavior observed in other Medicare markets, we believe this is a reasonable benchmark.

The second simplifying assumption is that both the low-cost Medigap plan and the broker-

[79] The 3% figure is a real (inflation-adjusted) social discount rate, which the OMB defines as the rate at which society trades present consumption for future consumption. This is the standard rate used by the OMB for similar calculations. In particular, it is distinct from the Federal Reserve's 2% inflation objective, which targets the growth rate of the price level, and from nominal market interest rates, which embed expected inflation (at the Medicare Trustees' intermediate inflation assumption of 2.5%, the nominal counterpart to our rate is $(1+3\%)(1+2.5\%)-1 \approx 5.6\%$). We also report the estimates with $r = 5\%$ in the Appendix.

[80] Specifically, we use Table 1, the NCHS life table for the total population, reported in Arias et al. (2025).

A healthy 67-year-old in a Medicare Advantage plan asks: *“If I try MA now, can I switch to a Medigap plan in a few years if I don’t like it?”*

How would you answer this question?

- ○ “Yes — you can switch to any Medigap plan anytime, no health questions.”
- ○ “You can apply, but in most states you can be turned down or charged more based on your health.”
- ○ “No — switching from MA to Medigap isn’t allowed.”

A client asks: *"Does Original Medicare by itself — with no Medigap — protect me from very high bills in a catastrophic year?"*

How would you answer this question?

- ○ “Yes — Original Medicare has an annual out-of-pocket maximum.”
- ○ “No — Original Medicare alone has no annual out-of-pocket limit.”
- ○ “It depends on the state.”

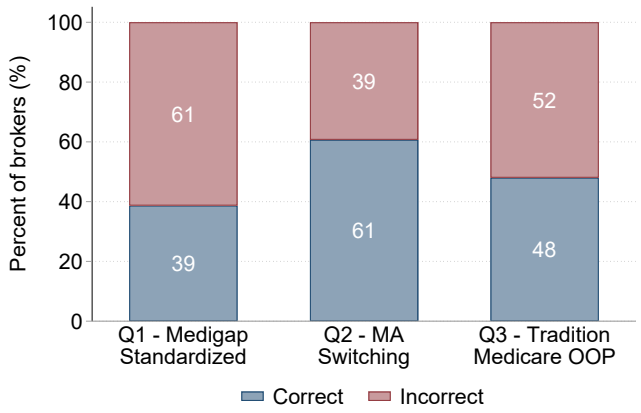
A client comparing Medigap quotes from two different insurance companies asks: *“If it’s the same standardized plan letter (like Plan G or Plan N), can one company’s plan have better coverage than another company?”*

How would you answer this question?

- ○ “Yes — each company designs its own version, so the benefits can differ quite a bit.”
- ○ “No — for a given plan letter the benefits are set by law and identical across companies; they mainly differ on price.”
- ○ “Most of it is the same, but each company adds or drops a few benefits.”

Figure A.9: Broker Survey Knowledge Check Questions

Figure A.10: Broker Knowledge Score by Question



Notes: Figure A.10 shows the share of brokers who answered each question in the multiple choice broker knowledge check correctly.

recommended plan follow the same premium growth rates over time. This is reasonable given the institutional setting; whether a Medigap policy is community-rated, issue-age-rated, or attained-age-rated, the annual rate revision is always reported as a percent increase relative to the existing premium. This percentage increase is then reviewed and approved by state insurance commissioners. In interviews with former actuaries, individuals who work at insurance companies, and employees at state department of insurance offices, individuals reported that most rate increases were approved as long as they were reasonably in line with the rest of the market.

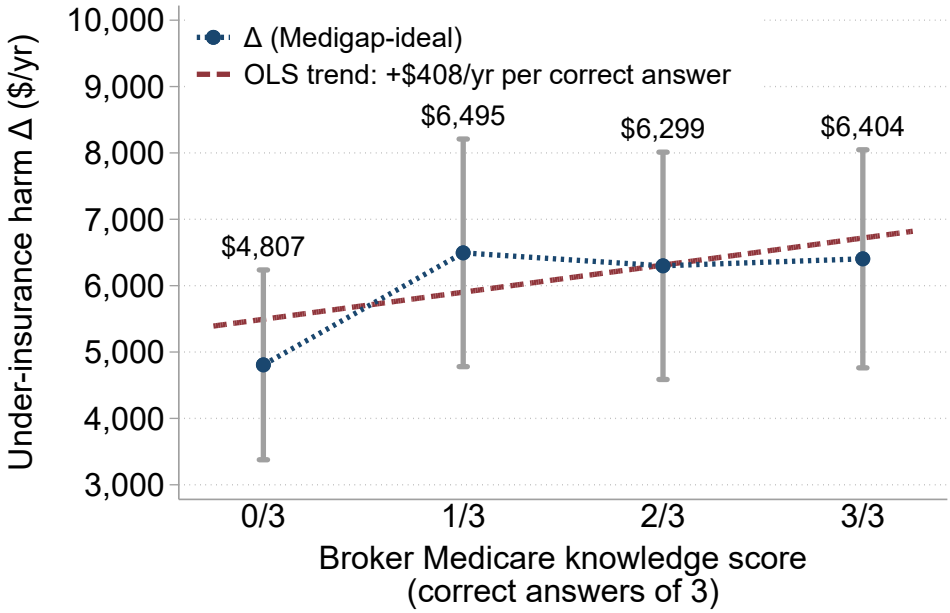
With these assumptions, we calculate the survival-weighted present value of the loss from the Medigap markup as:

$$L_{\mathrm{PV},S} \;=\; \sum_{t=1}^{T} S(t)\,\varphi^{\,t-1}\,\frac{\Delta p_0\,(1+g_{\mathrm{real}})^{t-1}}{(1+r)^{t-1}}, \qquad 1+g_{\mathrm{real}} \;=\; \frac{1+g}{1+\pi}, \tag{27}$$

where Δp_0 is the observed year-1 markup in 2025 dollars, g is the annual nominal Medigap premium growth rate, π is expected inflation, φ is the annual plan persistence, r is the real discount rate, and $S(t)$ is the conditional probability that an enrollee alive at age 65 is still alive in year t. Because carriers file rate revisions as nominal percent increases while r is a real rate, we deflate each growth scenario by expected inflation before compounding, keeping the numerator and denominator of (27) in the same units; we set $\pi = 2.5\%$, the intermediate inflation assumption of the Medicare Trustees.[81] The horizon T runs from age 65 to the life table's terminal age of 100, so no arbitrary cutoff truncates the projection and the survival weights make the contribution of late years small. We use a real discount rate of $r = 3\%$ and a survival rate constructed from the 2022 NCHS

[81]Equivalently, discounting the nominal wedge at the nominal counterpart rate $(1+r)(1+\pi)-1 \approx 5.6\%$ yields identical values; we present the calculation in real terms following the OMB and cost-effectiveness conventions.

Figure A.11: The under-insurance harm Δ by broker knowledge



Notes: Figure A.11 shows Δ, the mean consumer surplus loss or under-insurance harm when a Medigap-fit senior enrolls in MA, by broker knowledge score. The grey bars show the 95% confidence intervals. The knowledge score counts the number of Medicare knowledge check questions the broker answers correctly (out of three). The dashed line is the fitted OLS trend: Δ rises $408/yr per correct answer (s.e. $364, $p = 0.26$).

United States Life Tables.[82] We calculate the consumer surplus loss over a range of nominal premium growth rates $g \in \{5\%, 10\%, 15\%, 20\%\}$ and focus on the 10% and 15% scenarios, which better represent recent rate increases.[83] The methodology parallels the welfare-loss calculations for dominated financial products in Handel (2013) and Hortaçsu and Syverson (2004).

[82]We also report the estimates with $r = 5\%$ in Table A10.

[83]The 5% scenario reflects the lower end of the distribution and corresponds to the long-run historical average until roughly 2022. The 10% and 15% scenarios reflect the bulk of the rate-increase mass observed in 2023–2026 carrier filings. The 20% scenario reflects the higher end of carrier rate filings observed.

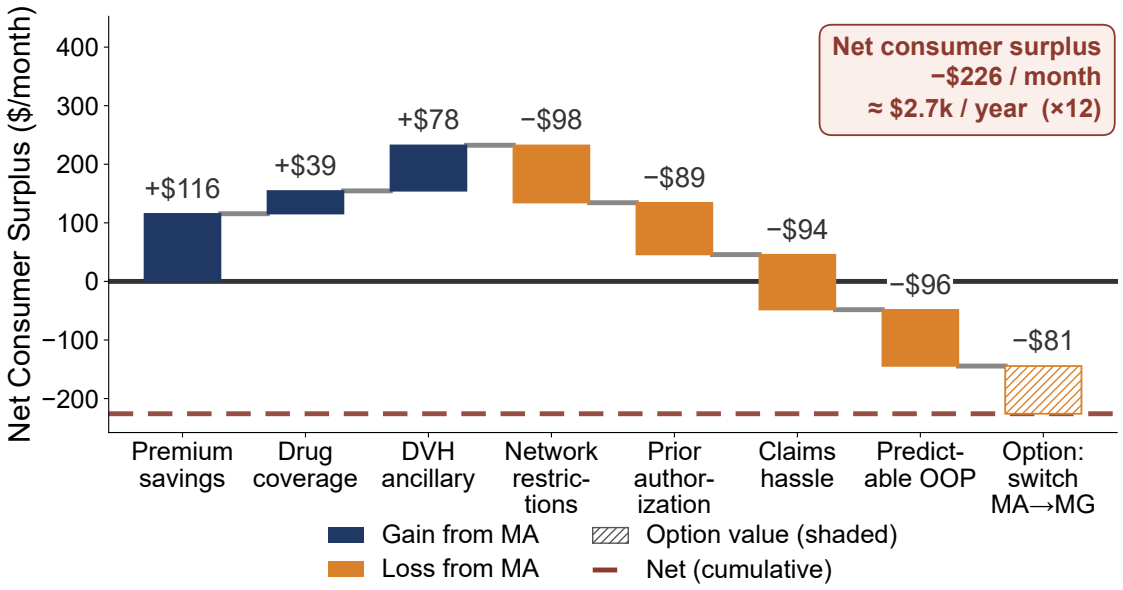
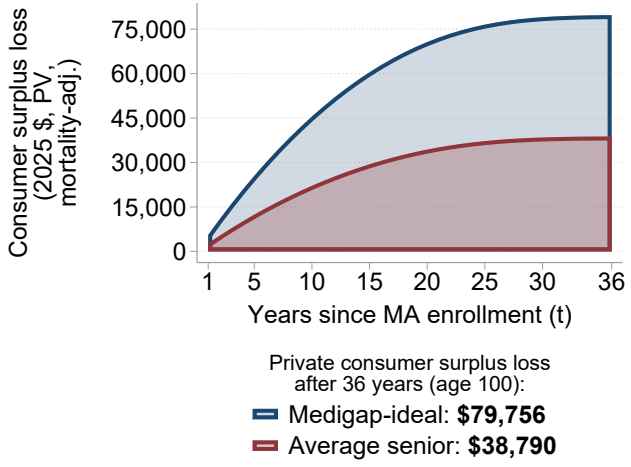
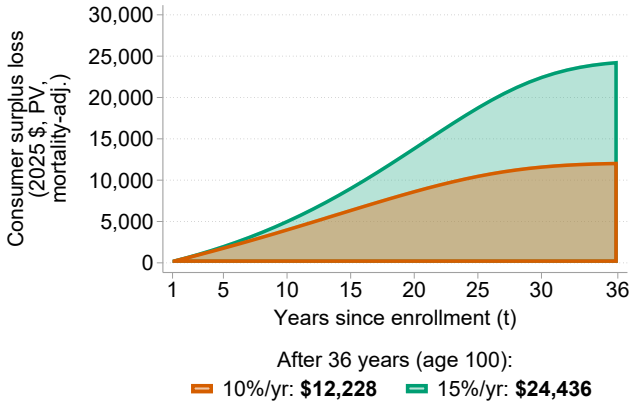


Figure A.12: MA consumer surplus waterfall: average senior (DVH wanted)

Notes: Figure A.12 shows the components of a net annualized consumer surplus loss of $-\$2,711$ from MA restrictions by Medicare plan attribute for the average senior (versus the Medigap-fit senior in Figure A.8). Measures for each attribute are elicited as a money-metric from a survey of brokers. Premium savings calculated from the minimum average Medigap Plan N/G monthly premium available to 65-year-old non-smokers in our audit study regions. Drug coverage valued at the 2025 national average monthly stand-alone Part D (PDP) premium from Cubanski and Damico (2026).



(a) Outcome 1: Medicare Advantage



(b) Outcome 2: Medigap Markup

Figure A.13: Lifetime Consumer Surplus Loss

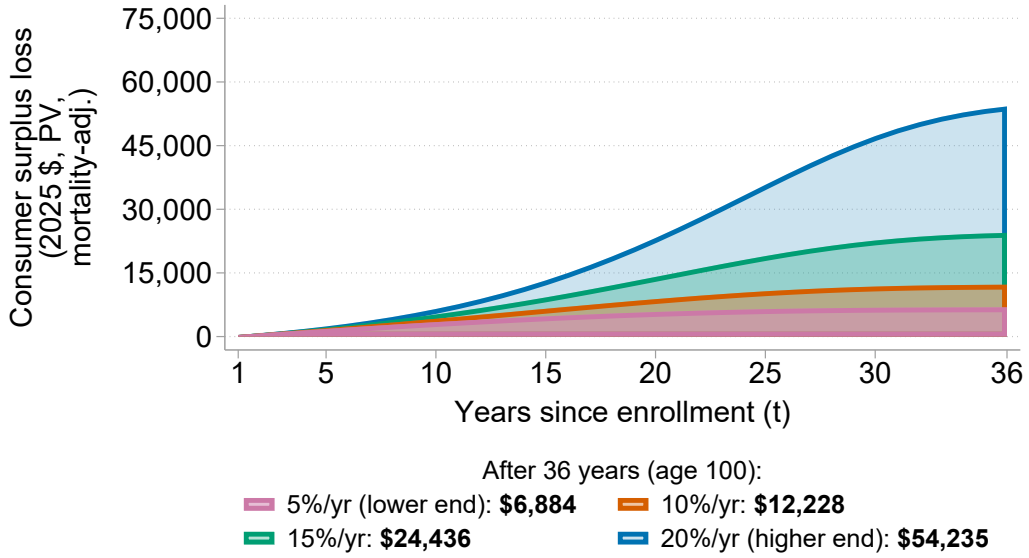
Notes: Figure A.13(a) and A.13(b) show the cumulative mortality-adjusted present value of the annual consumer surplus loss from an incorrect MA recommendation and from the Medigap markup, respectively, in constant 2025 dollars. Figure A.13(a) shows the consumer surplus loss for both a Medigap-fit senior and for the average senior. Figure A.13(b) shows the consumer surplus loss from a high-markup Medigap plan with a growth rate of 10%/year and 15%/year. The annual loss is held flat in real terms and discounted at $r = 3\%$ real; survival weights use the 2022 NCHS life tables.

Annual premium growth (g)	Cumulative welfare loss ($, PV, mortality-adj.)	
	$r = 3\%$ (primary) $L_{\mathrm{PV},S}(36;\, g,\, 0.03)$	$r = 5\%$ (robustness) $L_{\mathrm{PV},S}(36;\, g,\, 0.05)$
5% (lower end)	$6,884	$5,652
10%	$12,228	$9,473
15%	$24,436	$17,791
20% (higher end)	$54,235	$37,235

Table A10: Cumulative lifetime monetary loss by premium-growth rate and discount rate.

Notes: Table A10 reports the survival-weighted present value of the Medigap markup monetary loss, Equation (27), from age 65 through the life-table terminal age of 100, in constant 2025 dollars. Each row is a nominal premium-growth scenario from carrier rate filings, deflated by expected inflation ($\pi = 2.5\%$) before compounding; columns show the primary ($r = 3\%$) and robustness ($r = 5\%$) real discount rates.

Figure A.14: Medigap Markup Monetary Loss – All Growth Rates

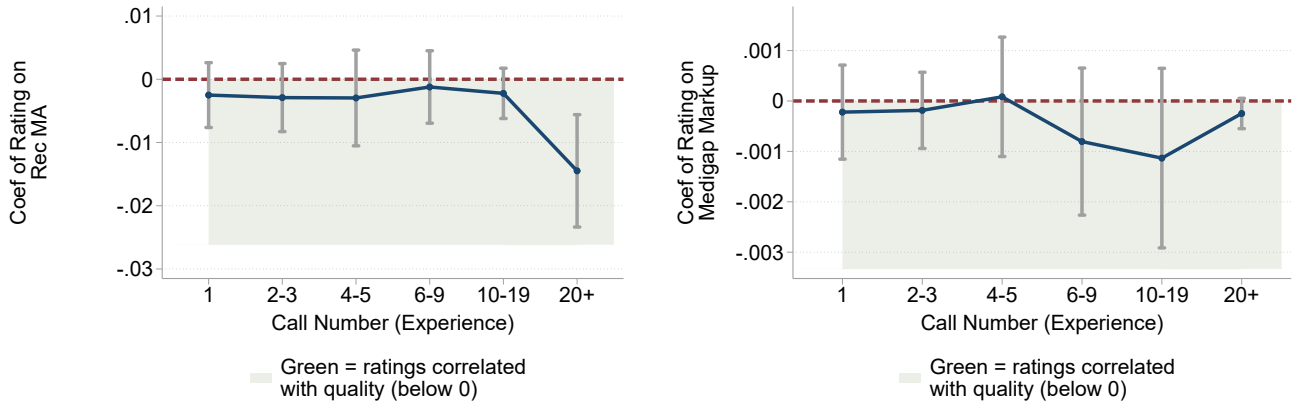


D.3 Caller Learning

Information frictions could also ease over time if callers learn to identify recommendation quality as they gain experience with the market, even if they are initially unable to identify high-quality advice. As an indirect test for caller learning, we examine how the correlation between a caller's rating and our objective measures of broker quality evolves as the caller gains greater exposure to the market. We split calls into six caller experience buckets indexed by n, from the caller's first call to the twentieth or later call. The lower-experience call buckets are kept narrow to show the regions where caller learning is most likely. Higher-experience call buckets are pooled to maintain adequate sample size. Within each bucket we estimate:

$$\text{Rating}_i \;=\; \alpha_n \;+\; \beta_n\, Q_i \;+\; \mathbf{X}_i'\boldsymbol{\gamma}_n \;+\; \varepsilon_i, \tag{28}$$

Figure A.15: Caller Learning



(a) Outcome 1: Recommend MA

(b) Outcome 2: Medigap Markup

Notes: Each point is the OLS coefficient $\hat{\beta}$ from a separate regression, one per caller-experience bin (their 1st call, 2nd–3rd, . . . , 20th-or-later), of the senior's recommendation rating ("on a scale from 1 to 5, how likely are you to recommend this broker?") on a measure of broker quality. In Panel (a) broker quality is an indicator for steering the consumer toward Medicare Advantage; in Panel (b) it is the Medigap markup (dollars per year). Both quality measures are coded so that a higher value is worse, so a coefficient below zero means the rating falls as broker quality worsens, that is, ratings track quality. The green shading marks this region. Both specifications include the baseline controls. Vertical bars are 95% confidence intervals from standard errors clustered at the caller level.

where i indexes the audit call, Rating_i is the rating (1 to 5) the caller assigns the broker at the end of call i, Q_i is either Outcome 1 (whether the broker recommended MA) or Outcome 2 (the Medigap markup), and $\mathbf{X}_i$ is the vector of baseline controls. Estimating a separate β_n in each bucket leaves the shape of the learning profile to the data rather than imposing it by functional form.

If caller ratings better reflect recommendation quality over time we would expect β_n to become increasingly negative as n increases (that is, an increase in recommending MA and recommending a higher markup Medigap plan should be correlated with lower ratings). Panel (a) of Figure A.15 shows $\hat{\beta}_n$ for the broker's MA recommendation rate; Panel (b) shows the results for the Medigap Markup.

Through the nineteenth call $\hat{\beta}_n$ is statistically indistinguishable from zero in both panels and exhibits no systematic trend. Only at twenty or more calls does the Medicare Advantage estimate become meaningfully negative in the direction predicted by learning, but with a wide confidence interval. The corresponding Medigap estimate is slightly more negative, but is not significant at the 5% level. Since this call bucket contains only a handful of callers and the confidence intervals remain wide, we read the pattern as suggestive of modest learning, but only at a number of calls far higher than the numbers seniors actually make.

Taken together, the panels suggest that our senior callers' stated willingness to recommend a broker is not correlated with either one of our two main measures of recommendation quality, even after repeated exposure to the market. If consumer ratings are thought of as a proxy for con-

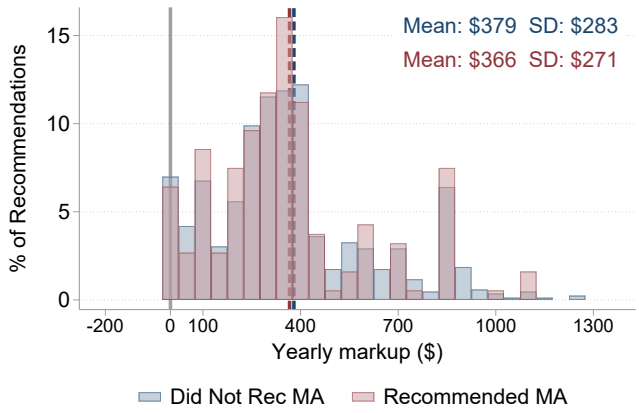
sumer feedback and the primary mechanism through which consumers might discipline low-quality brokers, the results raise concerns about how the mechanism may work in practice. The results highlight that demand-side learning is an unlikely source of market discipline in this setting.

D.4 Medigap Markup by MA Recommendation

To test whether the Medigap markup is intentionally exaggerated when brokers aim to steer the senior into Medicare Advantage, we plot the Medigap markup outcome in Decision 2 by broker recommendation in Decision 1.

Figure A.16 plots the distribution of the Medigap markup separately for calls in which the broker did and did not recommend MA; the two distributions are nearly identical, and the average markup is, if anything, marginally lower among brokers who recommend MA. The Medigap markup therefore reflects broker pricing in the standardized Medigap market itself, not a rationalization of the initial MA recommendation.

Figure A.16: Medigap Markup by Whether the Broker Recommended Medicare Advantage



Notes: Figure A.16 plots the distribution of the annual Medigap markup separately for baseline (normal-incentive) audit calls in which the broker recommended Medicare Advantage (MA) and calls in which the broker did not. The markup is the difference between the premium of the broker-recommended Medigap plan and the low-cost identical plan available to the senior. The overlaid histograms report the percent of recommendations in each markup bin, and the dashed vertical lines mark the group means. The two distributions are nearly identical, and the mean markup is marginally lower among brokers who recommended MA than among those who did not. This indicates that the Medigap markup is not inflated to rationalize an MA recommendation, so Outcome 2 is not mechanically driven by Outcome 1.

D.5 Caller Experience & Time Trends

Given concern around non-compliance when data collection for the treatment and control arms was conducted at the same time, for the majority of the study we collect data in waves. Control call waves were generally implemented before the treatment call waves; however, the waves did partially overlap in calendar time since the two secret shopper firms we used had different implementation

timelines.[84] In this section, we include several additional time and caller experience controls to ensure we are fully isolating the impact of the competition treatment.

Table A11: Robustness Check: Controlling for Caller Experience & Time Trends

	Recommend MA (%)				Medigap Markup ($)			
	(1) Caller Experience (Linear)	(2) Caller Experience FE	(3) + Linear Experience × Perceived Competition	(4) + Linear Time Trend	(5) Caller Experience (Linear)	(6) Caller Experience FE	(7) + Linear Experience × Perceived Competition	(8) + Linear Time Trend
Perceived Competition Assignment	11.33**	15.87***	18.39***	17.99***	-84.98***	-96.72***	-134.93***	-148.59***
	(4.92)	(5.68)	(6.73)	(6.50)	(19.84)	(24.51)	(34.52)	(37.73)
Call-Center Brokers	16.81***	18.55***	18.32***	18.71***	-8.98	-12.75	-7.16	-18.75
	(4.85)	(5.74)	(5.70)	(5.85)	(18.48)	(25.38)	(25.58)	(26.40)
Caller Experience	-0.12				0.88*			
	(0.08)				(0.49)			
Linear Experience x Perceived Competition			-0.42	-0.42			5.89	7.02*
			(0.64)	(0.64)			(3.95)	(4.17)
Linear Time Trend				0.02				0.37
				(0.06)				(0.26)
Control Mean	28.13	31.82	31.82	31.82	407.25	407.01	407.01	407.01
Caller Experience FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	637	532	532	532	1,055	1,043	1,043	1,043
R^2	0.16	0.18	0.18	0.18	0.15	0.26	0.26	0.27

Notes: Table A11 supplements our main specification by adding additional controls to account for caller experience and time trends. Columns (1) and (5) are the same as our baseline specification Equation (14) and control linearly for caller experience. Columns (2) and (6) replace the linear trend with indicators for each level of caller experience. Columns (3) and (7) keep the caller experience fixed effects from Columns (2) and (6) and add an interaction between Competition Assignment and Linear Experience. Finally, Columns (4) and (8) add both the interaction between competition assignment and linear experience and a linear time trend for the date of the call. Baseline controls, with the caller-experience specification varying by column as described above; the Medigap columns (5)–(8) omit the Annual Enrollment Period interaction. Standard errors are clustered at the caller level. $^{*}p < 0.10,^{**} p < 0.05,^{***} p < 0.01$.

Table A11 summarizes the following regression specifications. Columns (1) and (5) are the same as our baseline specification Equation (14), and include linear controls for caller experience (i.e., the linear trend of the number of calls the shopper has previously completed). Columns (2) and (6) replace the linear trend with indicators for each level of caller experience as in Equation (29)

$$Y_{is} = \alpha + \beta_1(\text{Perceived Competition Treatment}_{is}) + \beta_2 \mathbb{1}(\text{Caller Experience}_{is}) + \gamma' \mathbf{X}_{is} + \epsilon_{is} \quad (29)$$

where Y_{is} represents a recommendation outcome from broker i to shopper s and $\text{Caller Experience}_{is}$ is the number of calls the senior has completed (i.e. call number minus 1). Columns (3) and (7) keep the caller experience fixed effects from Columns (2) and (6) and add an interaction between Competition Assignment and Linear Experience. This interaction allows us to isolate the impact of competition assignment from shopper learning over time.

$$\begin{aligned} Y_{is} = \alpha + \beta_1(\text{Perceived Comp Treatment}_{is}) + \beta_2 \mathbb{1}(\text{Caller Experience}_{is}) + \\ \beta_3(\text{Perceived Comp Treatment}_{is} \times \text{Caller Experience}_{is}) + \gamma' \mathbf{X}_{is} + \epsilon_{is} \end{aligned} \quad (30)$$

Finally, Columns (4) and (8) add both the interaction between competition assignment and linear experience and a linear time trend for the date of the call. All regressions cluster standard errors

[84] We choose to implement the control arm first given concerns that once shoppers have delivered the competition line, they may be tempted to repeat the competition sentence if they feel it makes them seem more sophisticated.

at the caller level.

$$Y_{is} = \alpha + \beta_1(\text{Perceived Comp Treatment}_{is}) + \beta_2 \mathbb{1}(\text{Caller Experience}_{is}) + \beta_3(\text{Perceived Comp Treatment}_{is} \times \text{Caller Experience}_{is}) + \beta_4 \text{Date} + \gamma' \mathbf{X}_{is} + \epsilon_{is} \quad (31)$$

By controlling for experience (or number of previous calls completed), we interpret the coefficient on the competition treatment as the impact of competition for the *first phone call* the shopper makes. In all specifications, the competition coefficient remains significant and if anything increases in magnitude and significance as additional controls are added. The results suggest that the competition treatment has a larger impact on the first call seniors make. This makes the impact of the competition treatment especially important given our survey results that suggest many seniors only call one insurance agent.

D.6 Perceived Competition IV Analysis

Table A12 reports the effect of the Perceived Competition Treatment rather than Perceived Competition Assignment. This variable is endogenous because shoppers may deviate from their randomized perceived competition treatment assignment, that is, from whether the shopper is randomly assigned to deliver or to not deliver the treatment line of the script.[85]

Columns (1)-(4) report as-treated OLS estimates; Columns (5)-(8) instrument Perceived Competition Treatment with randomly assigned Perceived Competition Assignment, which identifies the effect of delivering the script for compliers when delivery is reported accurately. The instrument is strong (first-stage coefficients of 0.74–0.85; F-statistics of 170–407), and the OLS and IV results are similar in magnitude: for every outcome the two estimates have the same sign and statistical significance, and the differences between them are small relative to the standard errors. For the Medigap outcomes the IV estimates are modestly larger than the OLS estimates, as expected if misreporting of delivery attenuates the as-treated OLS; the overall similarity suggests that selection into delivering the script introduces little bias, though the OLS–IV comparison cannot separate selection from misreporting.

[85]Noncompliance is two-sided: a shopper assigned to control may deliver the competition script, and a shopper assigned to treatment may forget to. Random assignment therefore identifies the average effect of delivering the script for compliers (a local average treatment effect) rather than the effect of treatment on the treated. The complier interpretation also requires the standard monotonicity (no-defiers) assumption: no shopper would deliver the script only when assigned to control. This is plausible because delivery follows a scripted protocol and shoppers have no reason to invert their instructions, and the strong first stage leaves little room for defiers.

Table A12: OLS and IV Comparison

	OLS				Instrumental Variable (IV)			
	(1) Recommend MA (%)	(2) Medigap Markup ($)	(3) Medigap Broker Markup ($)	(4) Medigap % Markup	(5) Recommend MA (%)	(6) Medigap Markup ($)	(7) Medigap Broker Markup ($)	(8) Medigap % Markup
Panel A: OLS & IV Second Stage								
Perceived Competition Treatment	16.03***	-84.08***	-85.14***	-6.81***	15.26**	-99.05***	-99.64***	-8.32***
	(4.68)	(19.08)	(16.72)	(1.56)	(6.95)	(24.62)	(21.86)	(2.06)
Control Mean	28.13	407.25	217.69	31.47				
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	637	1,055	1,002	1,055	637	1,055	1,002	1,055
R^2	0.17	0.16	0.16	0.23	0.17	0.16	0.16	0.23
Panel B: First Stage								
First Stage Coefficient					0.74***	0.85***	0.84***	0.85***
					(0.06)	(0.04)	(0.04)	(0.04)
First-Stage F-stat					*169.81*	*407.43*	*358.21*	*407.43*

Notes: Table A12 reports the effect of the Perceived Competition Treatment, an indicator for whether the shopper reports actually delivering the competition script during the call. This variable is endogenous because shoppers may deviate from their randomized Perceived Competition Assignment, that is, from whether the shopper is randomly assigned to deliver or not deliver the treatment line of the script. Columns (1) - (4) report as-treated OLS estimates. Columns (5) - (8) report IV estimates that instrument Perceived Competition Treatment with Perceived Competition Assignment. Because noncompliance is two-sided, the IV identifies the average effect of delivering the competition script for compliers, assuming shoppers report delivery accurately and no shopper delivers the script when assigned to control (the monotonicity assumption). Panel B reports the first stage; first-stage F-statistics are shown in italics. The identifying assumption is that random assignment affects broker recommendations only through delivery of the competition script. Baseline controls. Standard errors are clustered at the caller level. $^{*}p < 0.10$,** $p < 0.05$,*** $p < 0.01$.

D.7 LLM analysis and additional results

We use a large language model (LLM) to conduct text analysis on the call transcripts from the audit study. Where a call was recorded, we transcribe it and pass the transcript to GPT-4.1-mini (`gpt-4.1-mini-2025-04-14`) through the OpenAI Batch API, which returns a fixed set of structured fields. The model receives one context line naming the caller's and the broker's gender, followed by the full transcript. It receives nothing about the treatment arm, the recommendation the shopper recorded after the call, or any other outcome, so the coding is blind to treatment except through the conversation itself. Transcripts are available for 497 of the 637 calls.

The measure of caller price sensitivity used in Section 5.4 comes from the following prompt, quoted in full:

> "Rate the caller's overall price sensitivity on a scale of 1 to 5 based on the entire conversation. 1=Not at all price sensitive (cost never mentioned, budget not a concern); 2=Slightly price sensitive; 3=Moderately price sensitive (mentioned cost a few times); 4=Quite price sensitive; 5=Very price sensitive (cost was a dominant concern, repeatedly mentioned budget constraints or inability to afford higher premiums)."

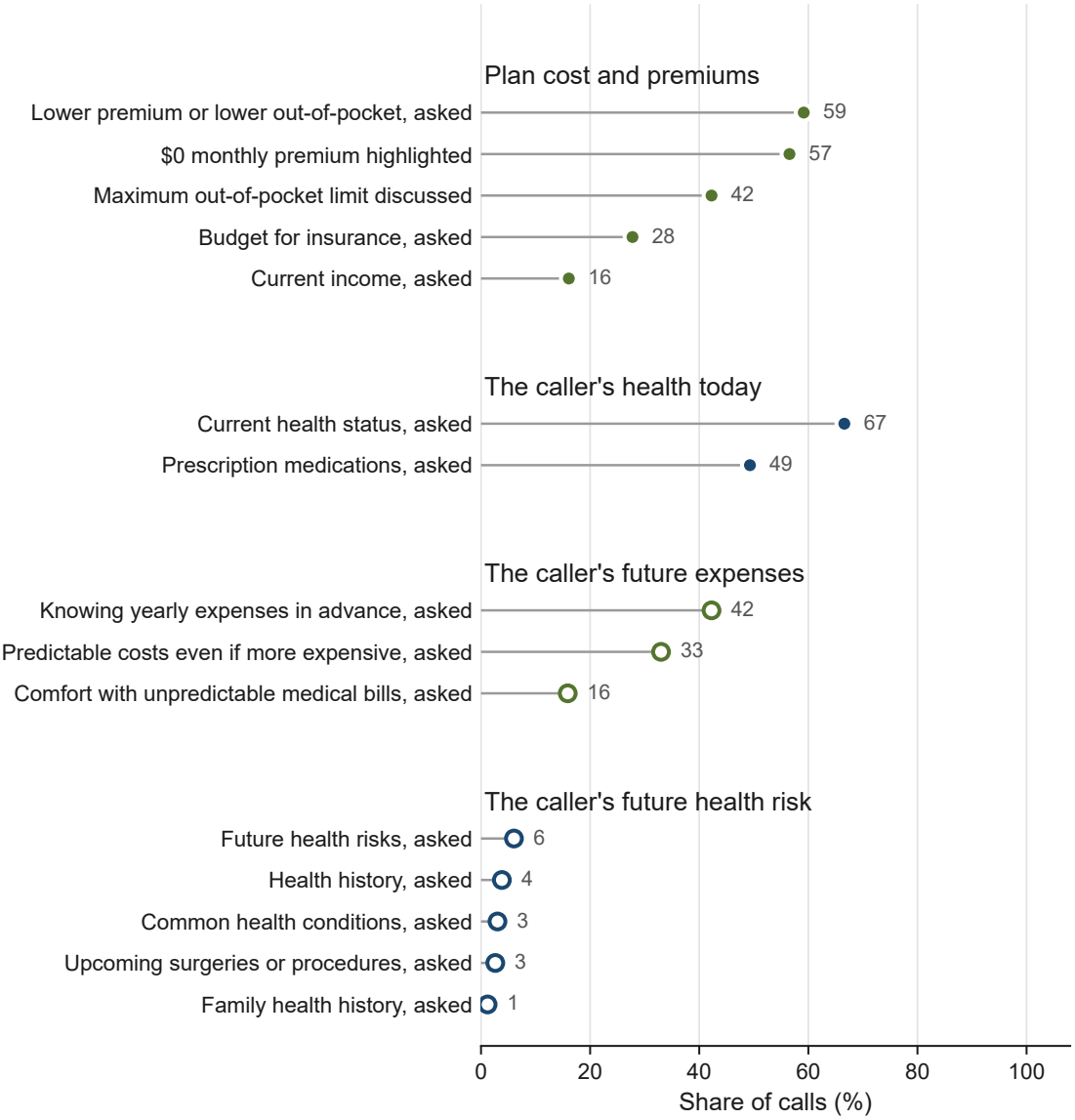
Table 8 tests whether the perceived competition treatment shifts perceived price sensitivity and finds no significant difference between the treatment and control calls.

Table A13: Coverage of Key Recommendation Topics by Broker Tenure

	All Mean (SE)	High Tenure Mean (SE)	Low Tenure Mean (SE)	Difference Δ	OLS p-value (robust)
9 Core Topics Covered (Z-Score)	0.00	6.44	-14.07	20.50	0.03
	(4.49)	(5.57)	(7.39)		
Network Restrict	90.14	89.15	92.31	-3.16	0.25
	(1.34)	(1.69)	(2.14)		
Prior Auth	24.95	27.57	19.23	8.34	0.04
	(1.94)	(2.42)	(3.17)		
Switching Plans	69.42	73.02	61.54	11.48	0.01
	(2.07)	(2.41)	(3.91)		
Cost Predictability	92.96	92.67	93.59	-0.92	0.70
	(1.15)	(1.41)	(1.97)		
Drug Plan	73.84	73.31	75.00	-1.69	0.69
	(1.97)	(2.40)	(3.48)		
Travel	93.56	92.96	94.87	-1.91	0.40
	(1.10)	(1.39)	(1.77)		
Medigap Standardized	57.34	62.17	46.79	15.38	0.00
	(2.22)	(2.63)	(4.01)		
OOP Costs	42.25	44.57	37.18	7.40	0.12
	(2.22)	(2.70)	(3.88)		
Plan Tradeoffs	98.19	97.95	98.72	-0.77	0.52
	(0.60)	(0.77)	(0.90)		
N (Calls)	497	341	156		

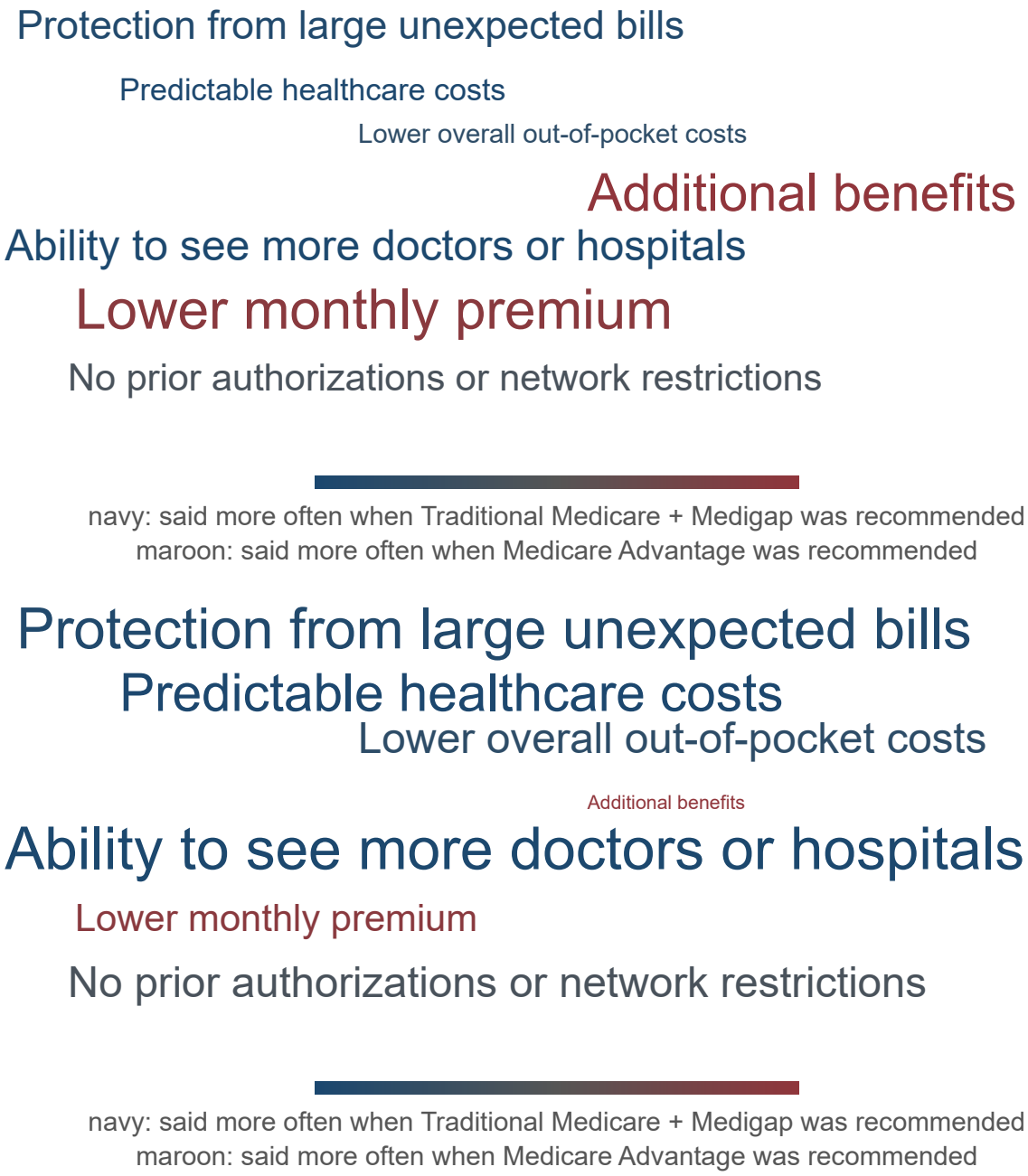
Notes: Table A13 reports means (robust standard errors in parentheses) of the standardized nine-topic coverage index and of each individual coverage topic, split by the high-tenure indicator, together with the raw difference and the p-value from a bivariate OLS test. Entries are percentages except the first row, which is the standardized index. A positive difference indicates greater coverage by high-tenure brokers.

Figure A.17: Topics Covered on the Call, by Theme



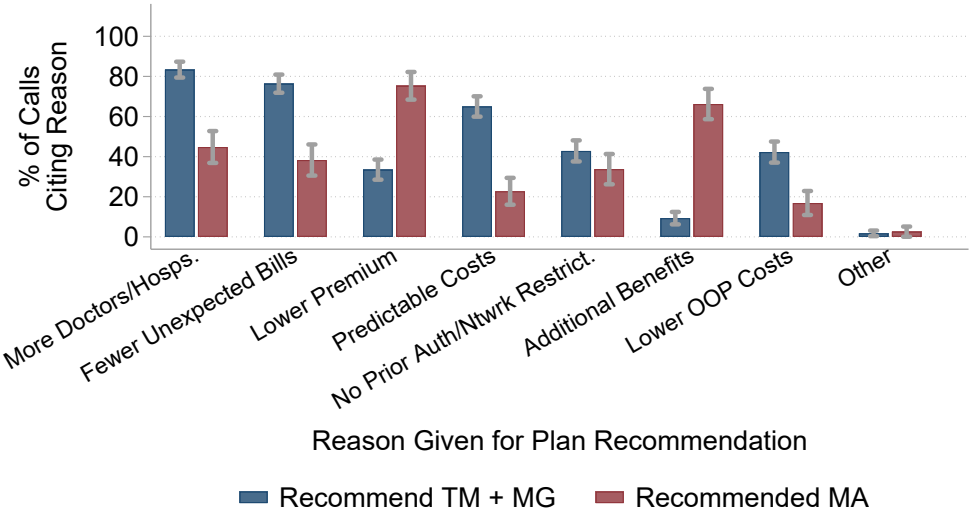
Notes: Figure A.17 reports the share of calls where each topic was discussed, grouped into four themes. The analysis is done on the 497 calls for which we have transcripts. Topics are ordered by frequency within each theme. Green markers are used to denote financial topics and blue markers are used for health topics. Solid markers are used for "present-focused" topics and hollow markers are used for "future-focused" topics.

Figure A.18: Reasons Given for the Decision 1 Recommendation, by Recommended Plan



Notes: Figure A.18 reports the reasons brokers gave for the Decision 1 recommendation, split by the plan recommended: 154 calls recommending Medicare Advantage (upper panel) and 313 recommending Traditional Medicare plus Medigap (lower panel), of the 497 LLM-coded transcripts; 30 recommended neither and are dropped. The split is on the broker's first recommendation, so the Medicare Advantage group includes 66 calls that also recommended Medigap later in the call. Type size is proportional to the square root of the share of calls in that panel's group citing the reason, on one scale and one layout common to both panels, so a reason sits in the same place in each and only its size changes. A reason is drawn only if cited on at least 5% of calls in one group.

Figure A.19: Recommendation Reasons, by Broker Recommendation in Decision 1



Notes: Figure A.19 reports the share of calls citing each reason for the Decision 1 recommendation, split by the plan the broker recommended. The figure reports the shares behind the word clouds in Figure A.18 with 95% confidence intervals.

E Enrollment Effects of the MA Commission Removal: Evidence from CMS Administrative Data

We supplement the natural experiment within the audit study with an analysis using the monthly MA enrollment and contract data collected by CMS to test whether the MA commission removals affect actual MA enrollments at the county level. Recall from Section 4 that we intentionally choose rural areas with few MA plans for the audit study. As a result, there are few MA insurance companies in the zip codes we choose, potentially leaving the actions of the few national carriers visible in enrollments at the county level.

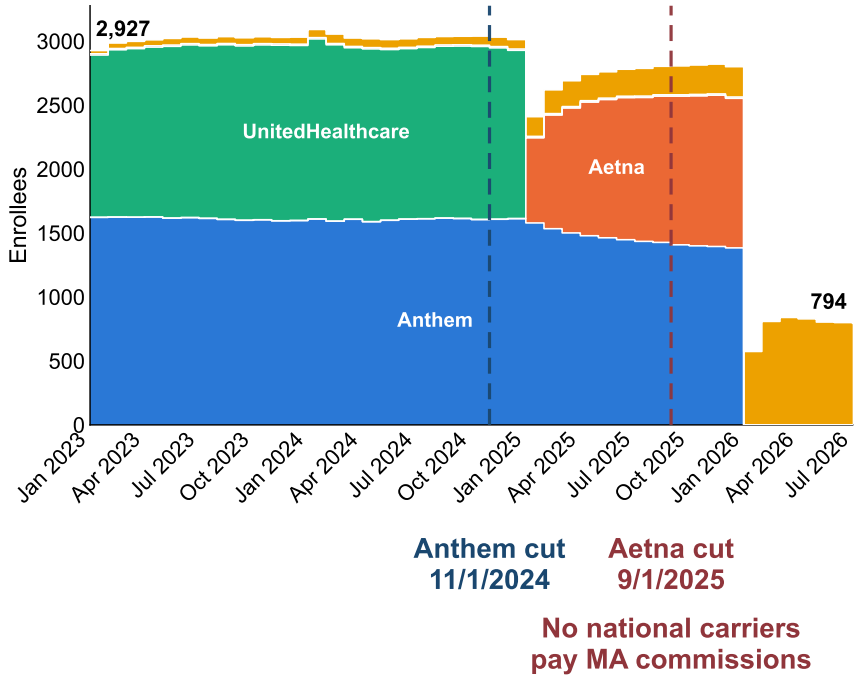
Data and sample We use the publicly available CMS Monthly Enrollment by Contract, Plan, State, and County (CPSC) data files for January 2021 through September 2026. The data report enrollment for every MA plan in every county in the US. We combine this dataset with the State and County Penetration (SCP) files, which give us the number of Medicare-eligible individuals in the county and total MA enrollment.[86] Our sample is the county's individual MA market or standard, general-enrollment MA plans only, excluding special needs plans and employer or union group plans. These are the MA plans sold by the brokers in our audit study to our audit study senior, and they are the object of focus in Outcome 1. The excluded MA plans, which include the employer and special needs plans described below, create a comparison group that we use below.

4 carriers offer individual MA plans in the county. Anthem's MediBlue Access PPO (H8552-029) is the incumbent, offered every year from 2021 and holding just over half the market (53% in October 2024). UnitedHealthcare offers 3 HMOs over the window, 2 of which are still in the county in 2024. Aetna enters for plan year 2025 with 2 PPOs. Imperial Health Plan, a California-only carrier that no national call center is appointed to sell, offers 4 HMOs throughout. Anthem stopped paying commissions on H8552-029 effective November 1, 2024 and Aetna on its plans effective September 1, 2025. Imperial offered MA plans that paid commissions during the entire period. UnitedHealthcare's plans were also made non-commissionable, but on a date we cannot document, so we leave that date missing rather than impute it.

MA enrollment response after the AEP enrollment window Figure A.20 shows the county's individual MA market month by month. Anthem's commission ended on November 1, 2024, but Medicare beneficiaries can move only in the Annual Enrollment Period, which runs from October 15 to December 7 for coverage starting January 1, or through more limited windows. November

[86] CPSC suppresses enrollment count cells with ten or fewer observations, but continues to report the plan-county row. For these suppressed cells we use the midpoint of $[0, 10]$. We assume missing plan-month observations have zero enrollees. Because CPSC reports by county of residence rather than service area, we verify every plan entry and exit against the CMS landscape files rather than inferring it from a missing row.

Figure A.20: Individual MA enrollment in the audit county, by carrier

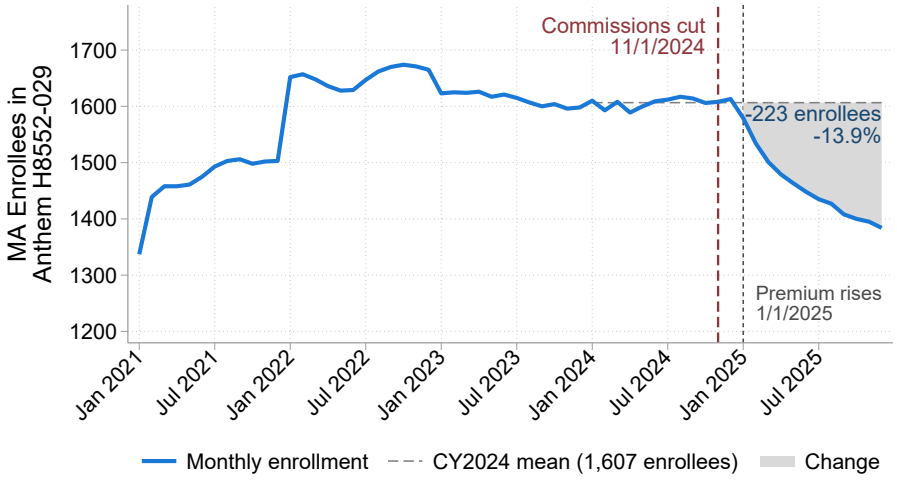


Notes: Figure A.20 stacks monthly CPSC enrollment in the county's individual MA plans by carrier, January 2023 through September 2026. Band heights are enrollee counts, so the height of the stack is the size of the county's individual MA market: it falls from 2,927 in January 2023 to 786 in September 2026. Special needs plans and employer or union group plans are excluded throughout. Carriers are labelled inside their own bands; the gold band, unlabelled because it is too thin to carry text before 2026, is Imperial Health Plan, the only carrier that never stopped paying commissions and the only one left in the county in 2026. The two dashed vertical rules mark the commission removals: Anthem effective November 1, 2024 and Aetna effective September 1, 2025, after which no carrier available to a national call center paid an MA commission in this county. Plan non-renewals, which are distinct from the commission cuts, take effect January 1, 2025 (UnitedHealthcare) and January 1, 2026 (Anthem and Aetna). Anthem's monthly premium was $30.00 in each of plan years 2021 through 2024 and $44.00 in 2025.

and December 2024 are therefore the only two months in which Anthem's commission had gone and nothing else had changed—the premium was still fixed at its plan-year 2024 value, and the plan was unchanged. Anthem's enrollment does not move in those two months: net changes of +2 and +5, against a mean net change of −0.8 per month over January 2023 to October 2024. The impact of the commission change becomes visible in January 2025: net enrollment flows into the plan are −34, −45, and −33 in the first three months of 2025, or −19.1 per month on average across 2025. This is consistent with the institutional setting: since consumers cannot switch plans outside of enrollment windows (October through December), a commission change cannot move *existing* enrollees out of the plan during periods outside these enrollment windows. Thus the commission changes outside an enrollment window can only influence the flow of *new* enrollees who are choosing an MA plan.

Enrollments shift to MA plans that pay commissions The clearest plan-level pattern comes from the 2024 enrollment period, when UnitedHealthcare's plans were non-renewed and their 1,316 enrollees had to choose again. Two national plans were available for 2025: Anthem, the

Figure A.21: Anthem MA Enrollment



four-year incumbent with 1,613 enrollees and no commission, and Aetna, brand new to the county with no enrollees and a commission. In January 2025, Aetna had 667 enrollees and Anthem had lost 34.[87] By September 2025 Aetna held 1,163.

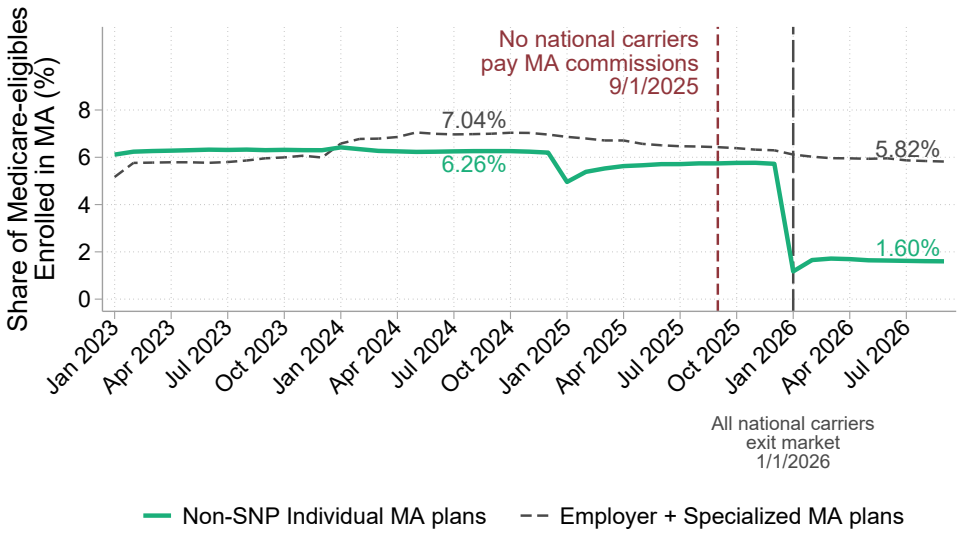
Natural Experiment: MA enrollments change when no national carrier pays commissions In January 2026 Anthem and Aetna both left the county. Of the 2,554 enrollees in plans that disappeared, Imperial picked up 327, or 13%; the rest left individual MA. County MA enrollment fell by 2,317 in a single month, from 5,879 to 3,562.[88] We flag two reasons not to read this as a commission effect. The plans were gone, so the enrollment decline is mechanical rather than behavioral; and a non-renewal gives beneficiaries a special enrollment period together with guaranteed-issue rights to a Medigap policy, so the move out of MA is partly a regulatory artifact. The fact worth keeping is narrower and still relevant to the audit: when the only individual MA plan left in the county was one that no national call center is appointed to sell, the overwhelming majority of displaced enrollees ended up in Traditional Medicare rather than in the MA plan that remained available to them.

Two features of this setting make the administrative data results comparable to our audit study. First, the size of the MA market is small enough that the entire individual MA market is easily identified. Because we targeted rural counties with few plans, the entire MA market is 10 plans from 4 carriers over the two-year window. This allows us to manually verify the date and scope of each market change against carrier documents and the CMS landscape files. Second, the enrollment data cover the Anthem cut of November 2024 as well as the September 2025 cut that identifies Experiment A. This allows us to observe two commission removal changes in the same market and isolate the potential confounds specific to each change.

[87]Imperial gained 80 over the same month; total individual MA fell from 3,013 to 2,410.
[88]Individual MA fell from 2,801 to 574 and Imperial rose from 247 to 574.

Individual versus Employer and Special Needs Plans This analysis compares the share of Medicare-eligible individuals in non-Special Needs Plan (SNP) individual MA plans to a placebo group of employer and specialized MA plans.[89] The individual non-SNP plan population is the population our audit study seniors fall into; the plans these beneficiaries are enrolled in are the exact Anthem, Aetna, and Imperial Health MA plans described above. To create a placebo group of Medicare enrollees that was not affected by MA commission changes in the same county, we use the share of Medicare-eligible individuals in employer and specialized plans. This group includes all other MA plans in the county not available for general enrollment. These plans face different commission schedules and were not subject to the same MA commission change natural experiment.[90]

Figure A.22: Broker-sold and non-broker-sold MA in the audit county



Notes: Figure A.22 plots the share of MA enrollees as a percent of Medicare-eligibles in Shasta County over time. The maroon and grey vertical dashed lines identify the date of the key commission changes. The solid green line shows the share of eligible enrollees in the individual MA market (excluding special needs (SNP) plans, employer and union group plans, and a very small group of cost contracts and PACE contracts). This represents plans sold on the open-market commission schedule. The grey dashed line represents the excluded plans sold through a different distribution channel and commission schedule (special needs (SNP) plans, employer and union group plans, and cost contracts and PACE contracts). These plans are very different from the individual plans and the commission schedules set by insurance companies for these plans are different, though most of these plans are also sold through brokers. These plans are also closed to seniors on the individual open market. Most importantly, none of the plans in this group were impacted by the commission change studied in Experiment A.

Figure A.22 plots the share of Medicare-eligible individuals enrolled in MA in the individual MA market against the share of individuals enrolled in employer and specialized MA plans that open-market brokers do not sell. Both are for Shasta County, our California audit study county of interest.

[89]We exclude Special Needs Plans (SNP) MA plans since these are different plans that are only available to Medicare beneficiaries with special needs and not available to the general population. These plans often face different broker commission schedules.

[90]One SNP MA plan did become non-commissionable, but there were several other SNP MA plans, including ones offered by national carriers, available for enrollment in the region. In contrast, all of the national insurance companies that offer MA plans in Shasta County stopped paying MA commissions.

The two shares are at similar levels before the national carrier change: 6% and 7% of Medicare-eligibles in October 2024 (a ratio of 0.89); however, the two lines begin to diverge after the 2024 AEP enrollments take effect in January 2025 and continue to diverge through 2025, the months in which the commissions were removed. Both national carriers (Anthem and Aetna) exit the MA market in January 2026; thus, the gap after that date reflects both the removal of national MA carrier plans that pay commissions and the plan exits themselves. However, Imperial, a local carrier, remained in the market after January 2026 and throughout 2026. Imperial continued to pay MA commissions and had very similar (in some cases identical) provider networks and higher star ratings. Beneficiaries who are strong fits for MA would have substituted into these MA plans. Limited substitution into the remaining MA plans, as shown in Figure A.20, suggests a portion of MA enrollments may have been driven by commissions.

F Structural Model: Estimation Details

Why the baseline sale-probability ratio is calibrated Since the audit study variation is insufficient to identify R_0, we choose to calibrate R_0 using the broker survey. We calibrate $R_0 = 0.61$ using the ratio of the mean sale close probabilities reported by brokers in our survey for a Medigap sale and for a Medicare Advantage sale. We then use the experimental variation in the perceived competition treatment to estimate R_1 against R_0. This makes the direction test ($\hat{R}_1 < R_0$) a genuine test and not a normalization.

Perceived harms The two types believe they are causing different damage: under-insurance for the informed broker's misfit MA sale and, for the misinformed broker's misfit Medigap sale, a harm the survey does not elicit. Only the product of each harm with the Decision 1 suitability cost enters the likelihood, so a freely estimated misinformed harm and the misinformed share are not separately identified from the no-MA-commission floor. The main specification therefore sets the misinformed type's perceived harm equal to the survey's under-insurance harm, a normalization that adds no parameter and preserves the identification of λ.

As a robustness check, we re-estimate the model setting the net harm from over-insurance equal to the cost of overpaying for a Medigap plan (the premium gap plus the baseline mean markup). The survey's under-insurance harm is 2.5 times this over-insurance harm. This calibration has the same parameter count and changes the log likelihood by -1.3 points (Vuong $z = 0.78$, which does not rank the two decisively), and a profile of the likelihood over the ratio of the two harms reaches its maximum at equality, with the asymmetric ratio at a deviance of 2.70. The calibration changes one measure of model fit. Under the asymmetric calibration the sale-probability mechanism reproduces 71% of the perceived competition treatment's effect on the share of brokers recommending Medicare Advantage, against 101% under the common harm, because the smaller misinformed penalty lowers

the estimated commission sensitivity. Appendix F.2 notes the one counterfactual the calibration changes.

Appointment portfolios Each broker's portfolio is measured from state insurance carrier appointment records. Both Texas and California, the two states our audit study targets, require appointments before brokers can sell a plan from an insurance company. However, both allow just-in-time appointments. We match the broker in our audit study to the appointment data using national producer number (NPN), carrier (NAIC number), and call date. We match 95% of observations.

Commission multipliers and the Part D stream Selling product r pays a first-year commission and a renewal in each later year the enrollee persists up to year 6, after which CMS no longer requires commissions to be paid (Section 2). A forward-looking broker values the discounted stream $W_r = w_r^1 \mathcal{M}_r$, where w_r^1 is the first-year commission and the multiplier $\mathcal{M}_r$ collects the discount factor, the enrollee's persistence, and the horizon. Medigap pays a percentage z of the premium, so the Medigap stream $W_{\mathrm{MG},i} = z\,\rho_{\min,i}\mathcal{M}_{\mathrm{MG}}$ is anchored on $\rho_{\min,i}$, the cheapest coverage-identical premium in the consumer's choice set. The commission multipliers come from the published CMS term structure, valued over a horizon of 6 years. The year-1 payment plus renewals through year 6, all discounted at 0.94 per year, make each commission stream worth 3.08 times its year-1 payment. That horizon is a conservative truncation since renewal commissions are no longer guaranteed after year 6, but could continue to be paid.

Since every product's renewal is half its year-1 payment, the multiplier rescales all products' present values by the same factor and does not shift the relative stakes between them. A Medigap sale is accompanied by a stand-alone Part D plan, which pays the broker a fixed first-year commission and renewals under the same CMS term structure. Both Medigap commission terms of the model, $W_{\mathrm{MG},i}$ in Decision 1 and the commission in equation (18), add the present value of that Part D stream to the premium-linked Medigap commission. In Decision 2 the Part D stream is the same for every plan in the choice set, so it cancels from the plan choice on control calls; on treated calls the deflator multiplies it together with the premium-linked commission.

The Decision 2 choice utility The Decision 2 suitability cost κ_2 is a distinct parameter from the Decision 1 suitability cost κ_1, being a common per-dollar cost borne at the recommendation and not a cost dispersed across calls and borne only at close. The appointment cost a_{appt} is what it costs the broker to place business with a carrier the broker is not yet appointed to sell. Since brokers can become appointed to an insurance company after a sale using "just-in-time appointments," this is a cost the broker can avoid rather than a binding constraint. In equation (18) the cheapest plan ($m_j = 0$) plays the role of L and every marked-up plan the role of H, so the equation generalizes the pairwise comparison of Section 3.4 to the consumer's full choice set.

Units of the commission sensitivity Section 6.1 states Decision 1 in dollars, equation (17). We maximize the likelihood in the equivalent index form, which multiplies out the bracket in (17) and replaces the dollar location with the index location, the logit intercept $\beta\bar{K}_1$. Thus, the dollar spread of the suitability cost is $\mu_K = \$1{,}000/\beta$. Equations (17) and (18) price a PV \$1,000 of commission at the same rate β by assumption; the plan-choice data identify only the net markup slope α and the appointment penalty relative to their own unobservable, so the equality of the two scales is maintained and not estimated (Section 6.2). The calibrated harm enters Decision 1 only through the dollar suitability cost $K_1 = \kappa_1 N$, and equation (17) is written in that dollar cost, so under the main specification's equal harms the calibrated harm does not enter the Decision 1 likelihood at all: a different calibrated harm changes the implied weight $\kappa_1 = K_1/N$ and nothing else, leaving β, $\bar{K}_1$, and every predicted recommendation unchanged. Only the dollar value of the counterfactuals in Section 7, where the net harm values each averted MA recommendation, moves with it.

Plan rows in the likelihood A call can contribute more than one quoted plan, because the audit records a quote for each plan letter (G or N) and for each of the two audited states the broker priced. Each quote is matched to the administrative choice set for its state, plan letter, date, and consumer gender, and the quotes from one call are weighted so that the call counts once in the likelihood.

Tenure assumptions Tenure enters as $\log(1 + \text{tenure in years})$, standardized over the brokers the licensure database matches. Three brokers, all in call centers, have no licensure record and are assigned the median tenure of call-center brokers.

Estimation The likelihood is maximized over a vector of unrestricted real parameters. Three parameters in Θ are positive by definition and enter as logs: the suitability-cost location, through the log of the logit intercept, $\log(\beta\bar{K}_{1,i}) = \log(\beta\bar{K}_{1,ncc}) + \log(\bar{K}_{1,cc}/\bar{K}_{1,ncc})\, cc_i$, the treated sale-probability ratio, $\log R_1$, and the appointment cost, $\log a_{\text{appt}}$. The unrestricted parameters are the log intercept for non-call-center brokers, $\log(\beta\bar{K}_{1,ncc})$, and the call-center log ratio, $\log(\bar{K}_{1,cc}/\bar{K}_{1,ncc})$, which multiplies the call-center indicator cc_i; the dollar location $\bar{K}_{1,i}$ is the intercept divided by β. The commission sensitivity β and the treatment deflator η enter unrestricted, so a negative estimate of either would represent a rejection of the mechanism as opposed to a constraint binding.

The Decision 2 suitability cost κ_2 is not estimated directly. On control calls the plan recommendation data identify only the net coefficient on the markup m_j in (18), the net markup slope $\alpha = \beta(z\mathcal{M}_{\text{MG}} - \kappa_2)$ of Section 6.1. The likelihood is maximized over α, which combines the commission per dollar of markup, $z\mathcal{M}_{\text{MG}}$, and the suitability cost per dollar of markup, κ_2. The suitability cost per dollar of markup, $\kappa_2 = z\mathcal{M}_{\text{MG}} - \alpha/\beta$, is then computed from the estimates of α and β.

Testing for a one-type broker model Testing whether the mixture has a single component places the null on the boundary of the parameter space and leaves the type-probability coefficients unidentified under it, so the reference distribution is not χ^2 with the nominal degrees of freedom. At a likelihood ratio of 29.7 we reject the one-type broker model in which the suitability cost is drawn independently on each call. The rejection comes from the agreement of recommendations across the calls of brokers called more than once, which that model cannot reproduce. A one-type model in which part of the suitability cost persists across a broker's calls fits the recommendations as well as the two-type model. The likelihood therefore identifies persistent differences between brokers, and the interpretation of those differences as a knowledge type depends on the evidence in Section 5.5.

A common suitability-cost location We also estimate the model with one suitability-cost location for call-center and non-call-center brokers, $\bar{K}_{1,cc} = \bar{K}_{1,ncc}$. The data do not reject this restriction (likelihood ratio 2.4, $p = 0.12$), consistent with the imprecision of the call-center shift in the main specification (call-center log ratio $\log(\bar{K}_{1,cc}/\bar{K}_{1,ncc}) = -0.24$, SE 0.17). The common location, \$3,533, is close to the main specification's location for non-call-center brokers, \$3,558, but the restriction changes how the model explains why call-center brokers recommend MA more often. With one location, the difference between the call-center and non-call-center firm types not explained by commission terms must run through the broker type logit: the estimated misinformed share rises to 39% for call-center brokers and falls to 3% for non-call-center brokers, against 37% and 9% in the main specification. The commission sensitivity also falls, to $\beta = 0.78$ per PV \$1,000.

The commission counterfactuals are largely unchanged: pay parity changes the share of brokers recommending MA by -9.7 percentage points and removing the MA commission by -15.7, against -9.8 and -15.3 in the main specification. The counterfactual that acts on information is smaller: training every broker to the informed type changes the share by -13.0 percentage points, against -14.7, because fewer brokers are estimated to be misinformed. By comparison, the opposite restriction, which keeps the call-center shift in the suitability cost but removes the call-center term from the type logit, is rejected (likelihood ratio 9.5, $p = 0.002$).

This exercise shows that the data require the difference between the firm types to run partly through the type logit and cannot tell whether part of the difference also runs through the suitability cost. We keep the call-center shift in the main specification because call-center brokers work under conditions that can lower the cost of recommending a plan that does not fit, such as sales targets and liability coverage held by the firm and not the broker, and the estimated shift has that sign. Without the shift, the model would attribute the entire difference between the two firm types that the commission terms do not explain to broker information by assumption.

Robustness check: re-estimating the model using the calibrated non-call-center misinformed share The misinformed share among non-call-center brokers is the least precisely estimated model parameter. The non-call-center brokers are never observed at a zero MA commission; therefore, the share of misinformed non-call-center brokers is inferred from the one-third-stream calls through the commission sensitivity and the logistic shape. In the bootstrap refits, the estimate reaches zero in 8% of resamples.

As a robustness check, we re-estimate the model with the misinformed non-call-center share calibrated using a value from the broker survey. We use the share of non-call-center brokers who self-report that they only sell MA, or 13.3% of the 90 non-call-center respondents. A broker who sells both products can still default to MA or face a high effort cost of recommending Medigap due to asymmetric training, so we assume this estimate is a lower bound on the misinformed share.

An apples-to-apples comparison requires us to re-specify the model without broker tenure in the type logit (a fixed share for one firm type does not nest in a specification with a common-tenure slope). The log-likelihood of the tenure-free model is -1583.13 versus -1579.03 in our headline specification.

Table A14 reports the estimates. The data do not reject the calibrated share (likelihood ratio 0.56 on one degree of freedom, $p = 0.454$). The commission sensitivity rises from 1.04 to 1.20 per PV \$1,000, and the dollar suitability-cost locations remain within one standard error of the main specification. The call-center share, which the zero-commission calls identify directly, is unchanged at 36.7%.

Table A15 reports the counterfactuals. In the re-estimated model, pay parity recovers \$466 per senior per year against \$510 in the main specification. The change is caused by a larger misinformed share, which means fewer brokers would respond to a commission change. The other policies in Figure 6 remain fairly similar.

The treated sale-probability ratio at the boundary The sale-probability ratio on perceived competition-treated calls, R_1, lies at the lower limit of its parameter space, zero. At that value the usual standard error is not informative, because it relies on the curvature of the likelihood around an interior maximum, and the bootstrap refits all land on the same boundary (below). Table 10 therefore reports a profile-likelihood bound for R_1 instead of a point estimate and standard error.

To construct the bound, we fix R_1 at each value on a grid, re-estimate all other parameters, and record the maximized log likelihood. The 95% profile interval, $R_1 \leq 0.50$, contains the values of R_1 whose log likelihood is close enough to the maximum that a likelihood-ratio test does not reject them at the 5% level. The data therefore rule out values of R_1 above this bound, but cannot

Table A14: Parameter Estimates with the Non-Call-Center Misinformed Share Calibrated

	Headline	Tenure-free parent	Independents at 13.3%
Misinformed share			
Independents (%)	8.9	9.5	13.3†
	(5.9)	(4.8)	
Call centers (%)	37.3	36.7	36.7
	(4.9)	(5.2)	(5.1)
All brokers (%)	18.4	18.7	21.2
	(4.2)	(3.6)	(1.7)
Estimated parameters			
β (per PV-$1,000)	1.01	1.04	1.20
	(0.32)	(0.30)	(0.26)
$\bar{K}_1$, independents ($)	3,558	3,552	3,603
	(521)	(513)	(544)
$\bar{K}_1$, call centers ($)	2,803	2,737	2,603
	(473)	(389)	(278)
R_1	0.00	0.00	0.00
η	0.60	0.57	0.48
	(0.41)	(0.40)	(0.29)
κ_2	2.37	2.39	2.49
	(0.28)	(0.26)	(0.25)
a_{appt} (PV-$1,000)	1.78	1.77	1.77
Fit			
Log-likelihood	-1579.03	-1583.13	-1583.41
Parameters	10	9	8
LR vs. parent (df)	–	–	0.56 (1)
p-value	–	–	0.454

Notes: †: fixed at the broker survey's own MA-only licensure share among non-call-center respondents (13.3%, $n = 90$). Parentheses are caller-clustered sandwich standard errors in every column, the convention for the alternative specifications in this appendix; the main specification's standard errors in Table 10 are bootstrap standard deviations. The type prior carries no tenure term in the second and third columns, and the third column is a one-parameter restriction of the second (the likelihood-ratio test). Suitability-cost locations $\bar{K}_1$ are the logit intercept divided by β, in present-value dollars. R_1 lies at its boundary in every column.

reject any value below it.

The interval excludes the calibrated control-call ratio $R_0 = 0.61$. Equivalently, we test the hypothesis that the perceived competition treatment has no effect on Decision 1, $R_1 = R_0$, with a likelihood-ratio test. The test rejects this hypothesis with a statistic of 5.2 ($p = 0.022$). The economic question is one-sided, whether the treatment lowers the ratio ($R_1 < R_0$), so the corresponding one-sided p-value is half this value.

One concern is that standard tests are unreliable when an estimate lies on a boundary. That problem arises when the hypothesis being tested itself lies on the boundary. Here, this is not the case: the hypothesis $R_1 = R_0$ sets R_1 to 0.61, an interior value, so in large samples the usual χ^2_1 critical values apply. If anything, the boundary works against rejecting the hypothesis. The test statistic measures how much better the model fits at the best value of R_1 than at $R_1 = R_0$. Because R_1 cannot fall below zero, the best value is zero, and allowing negative values could only

Table A15: Counterfactuals with the Non-Call-Center Misinformed Share Calibrated

	Headline	Tenure-free parent	Independents at 13.3%
Share of commission-paying calls recommending MA (%)			
Baseline	34.1	34.1	34.4
Pay parity	24.2	24.3	25.4
MA commission removed	18.8	19.0	20.8
Every broker informed	19.3	19.2	17.1
Pay parity and every broker informed	7.0	6.8	5.1
All control ($s = 0$)	31.3	31.3	31.6
Universal shopping ($s = 1$)	39.4	39.6	39.8
Annual consumer-surplus gain per Medigap-fit senior ($; % of the baseline loss)			
Baseline loss per senior ($)	2,154	2,158	2,169
Universal shopping	-276 (-13%)	-282 (-13%)	-286 (-13%)
High consumer ratings only	35 (2%)	31 (1%)	40 (2%)
MA/Medigap pay parity	510 (24%)	509 (24%)	466 (21%)
25% Medigap markup cap	127 (6%)	127 (6%)	127 (6%)

Notes: Top panel: model-implied share of commission-paying calls on which the broker recommends MA, re-solved under each policy; universal shopping holds R_1 at its estimate. Bottom panel: the policy menu of Figure 6 valued per Medigap-fit senior per year with the same accounting (steering valued at the net harm less the markup the senior would otherwise pay; markup gains on the share recommended a Medigap plan), in dollars and as a share of the baseline loss. The Headline column equals Figure 6.

improve the fit and produce a larger statistic. The rejection is therefore conservative.

Inference The standard errors and intervals reported for the structural model are estimated using a caller-cluster block bootstrap that re-estimates the model on 1000 resamples of the 101 caller clusters. Caller clusters match the reduced-form analysis. A caller cluster drawn more than once enters the resample once per draw, and the brokers in each copy are treated as distinct brokers, so the likelihood keeps the product over a broker's calls within each copy, as in equation (19). A parameter's standard error is its standard deviation across the refits and its interval is the 2.5th to 97.5th percentile; the 7 refits in which an index parameter ran to a numerical limit are excluded from the standard deviations and kept in the intervals.

We use the refits for inference instead of the normal approximation around the maximum because the misinformed share among non-call-center brokers is identified only indirectly, from the one-third-stream calls (Section 6.2), and it reaches zero in 8% of resamples; a percentile interval represents that boundary and a normal approximation cannot. The sale-probability ratio on perceived competition-treated calls, R_1, is the exception. Its estimate reaches the boundary in every refit, so the bootstrap distribution of R_1 collapses to a single point and provides no interval. The bootstrap is also not a reliable measure of uncertainty when the true value of a parameter may lie on the boundary of its parameter space (Andrews, 2000). We therefore report the profile-likelihood bound in its place, and the counterfactual draws of Appendix F.2 hold $\log R_1$ at its estimate for the same reason.

F.1 Model Fit

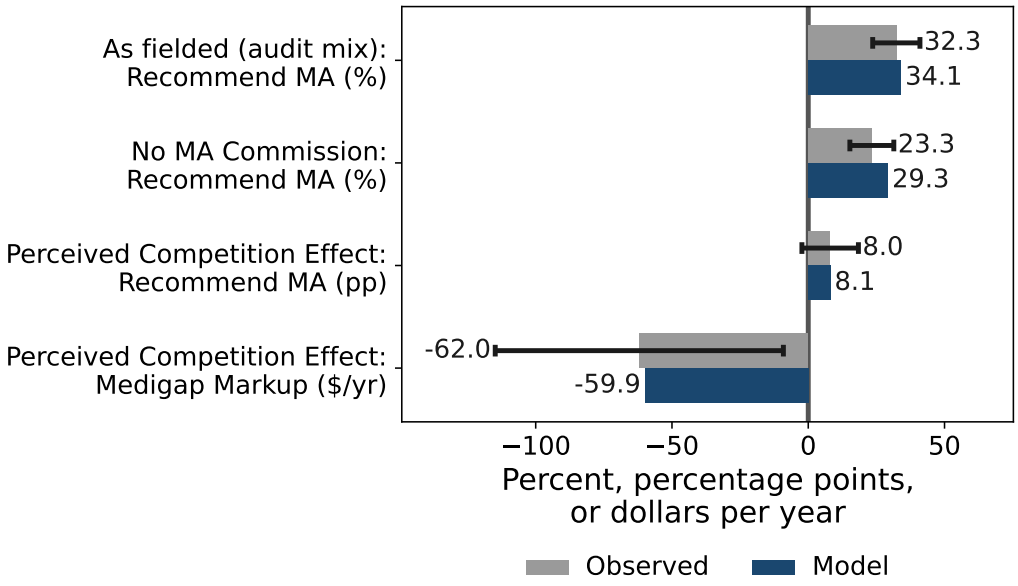


Figure A.23: Observed versus predicted treatment effects

Notes: Each pair compares a moment in the audit data with the model's prediction at the estimated parameters. The moments are cell means, so they differ from the reduced-form regression coefficients, which include controls. The share of brokers recommending MA on commission-paying calls, the share on no-MA-commission calls, and Experiment B's effects on Decisions 1 and 2 are the main sources of variation used to estimate $\bar{K}_1$, λ, R_1 and η, so we do not present them as tests of the model.

Figure A.23 plots each moment observed in the audit data against the model's prediction at the estimated parameters. These moments are the main sources of variation used to estimate $\bar{K}_1$, λ, η and R_1, so a close fit to them is expected and we do not present them as tests of the model: the baseline share of brokers recommending MA (34% fitted), the no-MA-commission floor (29% fitted against 23% observed), Experiment B's effect on Decision 2 (a \$60 annual reduction in the mean recommended markup against \$62 observed, 97% of the effect), and Experiment B's effect on Decision 1 (8.1 percentage points fitted against 8.0 observed, 101% of the effect). The rest of this subsection compares the model with moments and reduced-form estimates that the estimation does not target.

Since the model mixes over latent types the likelihood's first-order conditions do not equate fitted and observed shares exactly. Thus, the fitted floor differs from the observed floor; the observed floor lies inside the range of floors the model generates across 1000 audits simulated at the estimated parameters (simulated $p = 0.370$, twice the smaller of the shares of simulated audits with a value above or below the observed one).

Fit of the perceived competition treatment's effect on Decision 1 In the model, the perceived competition treatment changes Decision 1 through a single channel. It lowers the sale-probability ratio R, how likely a Medigap recommendation is to end in a sale relative to a Medicare

Advantage recommendation, from $R_0 = 0.61$ on control calls to R_1 on treated calls. Because R is a ratio of two probabilities, it cannot fall below zero, and this caps how far the treatment can move a broker of the informed type toward Medicare Advantage. At $R_1 = 0$ the broker expects a Medigap recommendation never to close, so the Medigap commission no longer counts against the Medicare Advantage commission in (16).[91] The estimate reaches this cap: $\hat{R}_1$ is zero, and at that value the model fits a 8.1 percentage-point effect against the observed 8.0 percentage points.

We perform two checks to make sure the cap does not distort the fit of the model. First, we re-estimate the model without the requirement $R_1 \geq 0$. In this model the estimate of R_1 is -0.17, a value with no economic meaning, but the log likelihood improves by only 0.63 points and the fit to treated MA-commission-paying calls does not improve. The data therefore do not push meaningfully against the cap. Second, because R_1 is fitted to this effect, we simulate 1000 audits from the estimated model and re-estimate the paper's reduced-form regression on each. The perceived competition coefficient on the share of brokers recommending Medicare Advantage in the actual audit lies well inside the simulated distribution (simulated $p = 0.388$; $p = 0.897$ for the within-cell contrast), and it remains inside under the asymmetric harm calibration of Appendix F ($p = 0.168$).

We do not read $\hat{R}_1 = 0$ literally, as brokers believing that a senior who is shopping around never buys Medigap: the data bound R_1 only from above ($R_1 \leq 0.50$, Table 10), and Appendix F.2 reports the universal shopping counterfactual across that range.

The survey bound on the misinformed share One statistic outside the likelihood was specified in advance, and the estimated share exceeds it. Believing Medicare Advantage suits the caller is necessary but not sufficient for recommending it at zero commission, so the share of surveyed brokers whose elicited perceived-fit probability falls below the broker's indifference threshold bounds the misinformed share from above. The threshold is 0.38 under the main specification's common harm and 0.20 under the asymmetric calibration, and the survey's 313 elicitations put 11% and 2% of brokers below them. The estimate, 0.18 (SE 0.04), is above both: by less than one standard error under the main specification and by several under the asymmetric calibration.

Because the threshold is a function of the same calibrated harms, the bound and the estimate are not independent measurements. That is, lowering the over-insurance harm tightens the bound and raises the estimated share simultaneously. Another caveat is that the broker survey and the audit measure different populations: a broker who sells only Medicare Advantage is in the information-friction group by construction and has no Medigap belief to elicit. Since the estimated share lies above the bound the model overstates the share of brokers of the misinformed type.

[91] The cap limits only the informed type. A broker of the misinformed type already regards Medicare Advantage as the suitable plan, so as R_1 approaches zero the model predicts that this broker recommends Medicare Advantage on every treated call that pays an MA commission.

F.2 Additional Counterfactuals and Welfare Details

Medigap markup benchmark Table A16 compares the $386 annual recommended markup with four benchmarks and splits it into the markup of the cheapest plan the broker is appointed to sell and the markup the broker adds above that plan. The split is an accounting identity over measured objects, not a causal attribution; Table A18 reports what each force in the model contributes to the markup.

Table A16: Benchmarks for the Medigap markup

	$/yr	95% CI
Benchmarks		
Random draw from the menu	584	[575, 592]
Random draw from the broker's portfolio	631	[622, 642]
Recommended plan	386	[355, 407]
Cheapest plan in the broker's portfolio	207	[186, 222]
Cheapest plan on the menu	0	
Recommended plan above the cheapest plan in the broker's portfolio		
All recommendations	179	[154, 200]
recommended plan in the portfolio	154	[137, 174]
recommended plan outside the portfolio	25	[10, 36]

Notes: Annual markup over the cheapest coverage-identical plan in the consumer's choice set, per Medigap purchaser, with every benchmark computed over the same choice sets with the same weights. The second panel reports how much of the recommended markup lies above the cheapest plan in the broker's portfolio at the time of the call, and how much of that comes from calls on which the recommended plan is in the portfolio and from calls on which it is not. Brackets are 95% caller-clustered block-bootstrap intervals.

We compare the recommended markup to four benchmarks. Choosing at random from the senior's choice set would cost the senior $584 a year. Choosing at random from the plans the broker is appointed to sell would cost the senior $631, because the plans a broker is appointed to sell are more expensive on average than the full choice set available to the senior. The broker in fact recommends $386, and the cheapest plan already in the broker's portfolio would have cost $207. Broker discretion divides further by where the recommended plan lies: $154 comes from calls on which the broker recommended a plan with a higher premium than the cheapest plan the broker was appointed to sell at the time of the call, and $25 from calls on which the broker went outside the portfolio altogether. Only 5% of recommendations are the cheapest plan in the senior's choice set and 18% are the cheapest in the broker's own portfolio.

Accounting Commissions are transfers from insurance companies to brokers and net out when calculating total surplus. Markups are transfers from consumers to brokers and insurance companies, so a markup reduction is a dollar-for-dollar consumer gain and exactly zero in total surplus. The steering component that does impact total surplus is the misallocation of a Medigap-fit senior being steered into Medicare Advantage and losing both the value of the coverage and the option to switch plans later, net of the premium savings from enrolling in MA. We value the unsuitable MA recommendation at the broker's perceived consumer harm elicited from the broker survey and

not at the consumers' revealed or stated willingness to pay for the plan attributes. The choice is in line with existing literature that relies on expert valuations when well-documented consumer information frictions distort choice (Abaluck and Gruber, 2011; Handel and Kolstad, 2015; Handel and Schwartzstein, 2018). We use the broker-elicited harm range in every estimate.

Partial equilibrium All counterfactuals hold premiums, commission schedules, plan choice sets, and the population of practicing brokers fixed. This assumption is restrictive. In health insurance markets, policies that improve individual choices can reverse sign once prices re-equilibrate (Handel, 2013), and the natural margin of adjustment here is the supply of advice. The audit contains direct evidence on that margin: when insurance companies removed Medicare Advantage commissions, the call centers removed the non-commissionable plans from their internal search tools (Section 4.2). That evidence weighs against removing the Medicare Advantage commission outright, and in favor of pay parity.

Inference and Calibration The confidence intervals on the policy counterfactuals are calculated using the same caller-cluster block-bootstrap refits that supply every standard error in the paper (Appendix F). Calibrated parameters, including the two perceived harms, the baseline sale-probability ratio, the commission PV multipliers, and the mortality-weighted annuity, are held at their calibrated values in every draw. Sensitivity analysis for these items is reported separately (e.g. the harm range and the alternative calibrations in Appendix F). Similarly, there is no bootstrap confidence interval estimated for the treated sale-probability ratio since the treated sale-probability ratio lies at a boundary. Counterfactual calculations using this ratio hold $\log R_1$ constant at its estimate in each refit.

The universal shopping counterfactual is also outside the support of Experiment B. In the audit study, the perceived competition treatment is a statement the broker cannot verify and the treatment is delivered on calls to randomly assigned brokers. The brokers' response to it, R_1, was estimated against brokers who did not hear it. This counterfactual applies the statement to every caller and holds the broker response at R_1. It therefore assumes that brokers do not update their beliefs when every senior makes the same statement. If seniors who make the statement do not in fact search more, a statement every senior makes carries no information and the response would disappear, so the counterfactual is a benchmark for the largest response the treatment can generate, not an equilibrium prediction.

Universal shopping: alternative baselines, the reduced-form valuation, and the R_1 range Section 7.1 values universal shopping as the contrast between the all-control and the all-treated state, which is the estimand of Experiment B. Figure 6 values it against the audit's observed mix of treated and control calls, the baseline every other counterfactual in the policy menu uses; on that basis the policy lowers consumer surplus by $276 per senior per year. The

reduced-form treatment effects give the same sign and a larger magnitude. The model's figure is smaller because it prices the treatment's fitted effects, a $56 reduction in the mean recommended markup and a rise in the share of brokers recommending Medicare Advantage from 31% to 39% of calls, and not the reduced-form coefficients. Because the treated sale-probability ratio lies on the boundary of its parameter space, we report the R_1 range instead of a standard error: at the profile-likelihood upper bound $R_1 = 0.50$ the change in consumer surplus is $-28 per senior per year, and with no Decision 1 response, $R_1 = R_0$, it is $+39, the Decision 2 gain alone.

The ban-versus-parity break-even Removing the Medicare Advantage commission entirely dominates pay parity only if the Medigap-fit share of the advised population exceeds a break-even we compute at 73% under the main specification's common harm and 57% under the asymmetric calibration. This is the one result of the main specification that the harm calibration moves, and we report it as the harm range. The sensitivity comes from one convention. The calculation values the broker's Decision 1 suitability cost for a misdirected Medigap sale at the over-insurance harm under every specification, including the main specification, whose likelihood carries a common harm, because the relevant broker faces a genuinely Medicare-Advantage-fit consumer and not the audited caller. An outright Medicare Advantage commission ban is not a realistic instrument, so the exercise indicates the direction of the ranking and is not a policy recommendation.

Robustness Every counterfactual stays within its interval under the alternative harm calibration of Appendix F, and re-running the estimation from many starting values returns the reported likelihood optimum.

The full policy roster Figure A.24 shows the main counterfactuals in Figure 6, plus the additional counterfactuals described below.

Broker composition counterfactuals The broker composition counterfactuals restrict the pool of brokers to those with a given characteristic. These counterfactuals can be thought of as redirecting consumers to brokers with certain characteristics, assuming the alternative brokers have capacity to absorb the additional demand. Screens on broker characteristics re-weight the broker pool and hold the call environment at its population distribution.

Non-call-center brokers only This counterfactual removes every broker who works in a call center. In the type probability, working in a call center raises a broker's probability of being misinformed (Section 6.2), so the brokers who remain are more often informed and steer less. The screen leaves commissions unchanged, so an informed broker who remains still recommends Medicare Advantage whenever the commission gap outweighs the suitability cost. Restricting the pool to non-call-center brokers recovers $346 per senior per year, 16% of the loss.

Experience of more than 4 years This counterfactual removes every broker licensed for 4 years or less. In the type probability, a broker's probability of being misinformed falls with years licensed (Section 6.2), so the experienced brokers who remain steer less, again with commissions unchanged. Restricting the pool to brokers with experience of > 4 years recovers \$283 per senior per year, 13% of the loss.

Call-center onboarding This counterfactual changes what call-center brokers know, not which brokers advise. It sets the call-center term in the type probability to zero, so a call-center broker is as likely to be misinformed as a non-call-center broker with the same experience. Every call-center broker stays in the market and keeps the estimated commission response and suitability cost. The policy represents an onboarding program that trains call-center hires to the non-call-center standard. It recovers \$193 per senior per year (9% of the loss). That is less than restricting the pool to non-call-center brokers, because the screen also removes everything else that distinguishes call-center brokers in the model, while onboarding changes only their information.

Removing appointment frictions This counterfactual acts on Decision 2 only. It sets the cost of recommending a plan the broker is not appointed to sell to zero, so plans outside the broker's portfolio compete with plans inside it on equal terms. The commission pull toward higher premiums and the broker's discretion are unchanged. The expected markup falls by \$46 a year per Medigap purchaser, worth \$30 per senior per year (1% of the loss). The gain is small even though the cheapest plan in the broker's portfolio carries a markup of \$207 a year, because a broker free to recommend any plan still chooses among them with the same commission incentive. As with the markup cap, the gain is a transfer from brokers and insurance companies to seniors, so it is zero in total surplus.

What drives broker advice Tables A17 and A18 report the counterfactuals summarized in Section 6.2. Each row switches off one force, or two in the third panel of Table A17, from the same baseline, so the rows are separate counterfactuals and do not sum.

F.3 Scaling to the Medicare Population

Section 7.2 scales the loss per advised senior to a population as a back-of-the-envelope exercise: it multiplies a loss estimated from the audit by population counts, a broker-use share and a fit split that the audit does not identify, to give the estimates a scale rather than to produce an estimate with a sampling distribution. The annual loss of population u is

$$L_u \;=\; N_u \times b \times \sum_g \pi_g \, \ell_g, \tag{32}$$

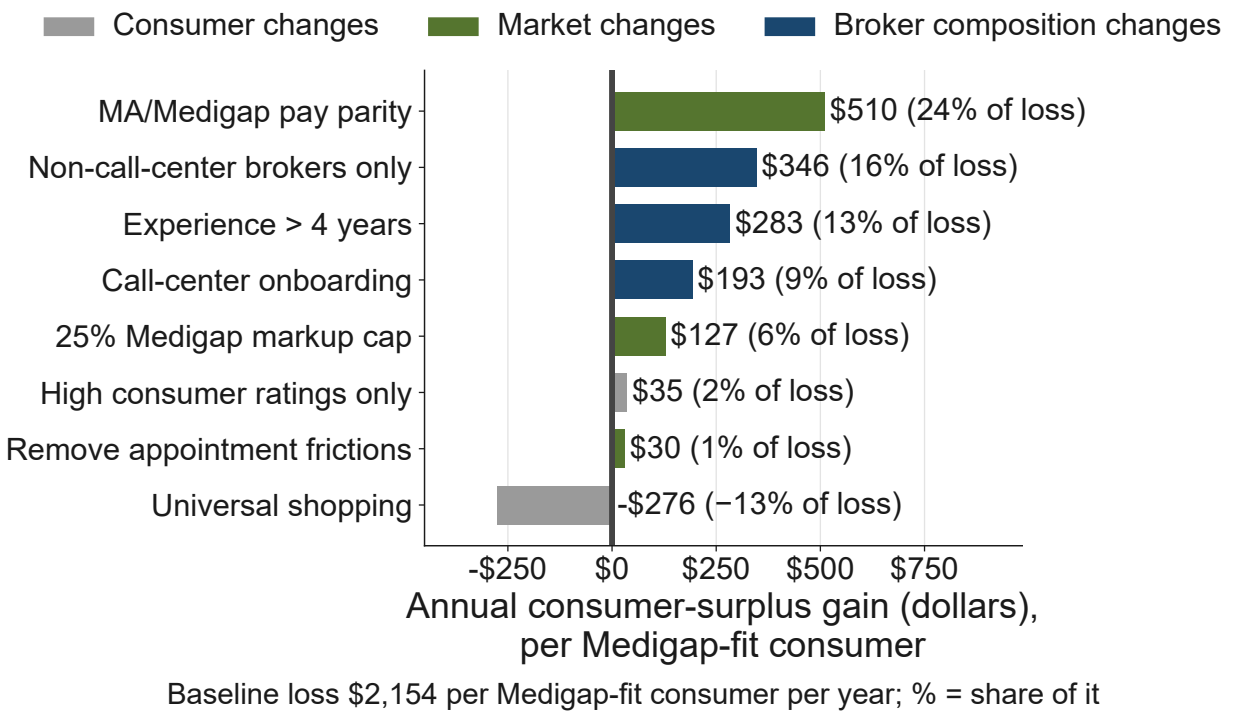


Figure A.24: The full policy roster on one axis

Notes: As in Figure 6, adding the two policies the main-text figure omits: call-center onboarding to the non-call-center standard, and removing appointment frictions.

in which N_u is the number of seniors in the population, b is the share who enroll through a broker, π_g is the share of seniors of fit type g, and ℓ_g is the annual consumer surplus loss per advised senior of that type. The gain from a policy is the change in L_u when ℓ_g is recomputed under the policy.

Population The headline population is the set of beneficiaries who choose between Medicare Advantage and Medigap. From the 60.7 million Medicare beneficiaries aged 65 and over in 2024 (CMS, 2025a) we subtract the 6.5 million with full Medicaid benefits, whose cost sharing Medicaid pays and to whom federal law bars the sale of a Medigap policy, and 12.6 million with employer or union coverage, whose plan the employer selects. The first count is the 13.6 million beneficiaries who were dually eligible in some month of 2022, times the share of them aged 65 and over (65%), times the share of that group with full benefits (73%), both from the December 2025 MedPAC and MACPAC data book on dually eligible beneficiaries. The second is the 14.1 million beneficiaries of all ages with employer or union coverage in 2023 (8.2 million with an employer supplement to Traditional Medicare and 5.9 million in employer-group Medicare Advantage plans, from KFF's analysis of the Medicare Current Beneficiary Survey), times the share of all Medicare beneficiaries who are aged 65 and over (60.7 million of 67.8 million in 2024). Partial-benefit dual eligibles, whose Medicaid coverage is limited to Medicare premiums and cost sharing, remain in the population.

Table A17: What Drives Medicare Advantage Recommendations to a Medigap-Fit Senior

Force switched off	Change (pp)	Change relative to baseline	95% interval (pp)
Commissions			
Pay parity	−9.8	−29%	−15.1 to −5.7
No pay for either product	−13.0	−38%	−19.5 to −7.6
No MA commission	−15.3	−45%	−22.6 to −9.0
Information			
Every broker of the informed type	−14.7	−43%	−21.3 to −8.6
Commissions and information together			
Pay parity, every broker informed	−27.0	−79%	−37.0 to −19.3
No pay, every broker informed	−30.3	−89%	−40.4 to −22.9
No MA commission, every broker informed	−31.5	−92%	−41.1 to −24.1
Perceived competition			
Treatment on every call rather than none	+8.1†	+24%	+4.5 to +14.9
Suitability cost			
No suitability cost	+53.5	+157%	+41.9 to +63.8
Infinite suitability cost	−14.9	−44%	−23.8 to −8.3

Notes: Each row switches off one force in the estimated model, or two in the third panel, holds every other parameter at its estimate, and re-solves the broker's recommendation. "Change" is the change in the model-implied share of commission-paying calls on which the broker recommends Medicare Advantage (MA) to a Medigap-fit senior, from the baseline of 34% averaged over the audit's mix of treated and control calls; "change relative to baseline" divides it by that baseline. Pay parity sets the MA year-1 commission equal to the Medigap year-1 commission. With no pay for either product, every broker's commission gap is zero. No MA commission sets the MA commission to zero and leaves the Medigap commission in place. Every broker informed sets the probability of the misinformed type to zero. The perceived competition row moves every call from the control arm to the treated arm, the contrast of Experiment B; it is measured from the share with no call treated, 31%, and divided by the same baseline as the other rows. No suitability cost sets its location to zero and keeps its dispersion across calls; with an infinite suitability cost only misinformed brokers recommend MA. Rows are separate counterfactuals and do not sum. The interval column reports 95% caller-cluster block-bootstrap percentile intervals (Appendix F). †The treated sale-probability ratio R_1 lies on the boundary of its parameter space and is held at its estimate in the refits; at its profile-likelihood upper bound, $R_1 = 0.50$, the change is +1.3 percentage points.

That leaves $60.7 - 6.5 - 12.6 = 41.6$ million beneficiaries who choose between Medicare Advantage and Medigap, or 69% of beneficiaries aged 65 and over. No count of coverage at the time of enrollment exists, so we apply the same ratio to the 4.1 million people who turn 65 each year: $4.1 \times 41.6/60.7 = 2.81$ million. In the UAS survey, 73% of respondents report neither Medicaid eligibility nor access to an employer plan (Table A7), and 56% of them used a broker (61% at ages 65 to 69), above the 50% we hold throughout, so both the population restriction and the broker-use share lean toward a smaller loss. Without the restriction, the baseline loss is $1.8 billion a year for the 2.05 million seniors who turn 65 in one year and use a broker.

Loss per advised senior A Medigap-fit senior loses the net harm on the share of calls on which the broker recommends Medicare Advantage and the mean Medigap markup on the remaining calls, which is $2,154 a year at the estimates of Section 6.1. An MA-fit senior loses the over-insurance harm plus the markup on the share of calls on which the broker recommends Medigap. Under our main assumption that share is 14.9%, and the loss is $243 a year. A senior who fits neither product loses the markup when placed in Medigap, or $225 a year. Weighted by the fit shares, the loss is $894 per advised senior per year.

Table A18: What Drives the Medigap Markup

Force switched off	Change ($/yr)	Change relative to baseline	95% interval ($/yr)
Commissions			
Flat Medigap pay (equivalent to no pay)	−35	−9%	−66 to −19
Perceived competition			
Treatment on every call rather than none	−56	−15%	−105 to −24
Appointment cost			
No appointment cost	−46	−12%	−52 to −41
Suitability cost			
No suitability cost	+305	+79%	+255 to +448
Infinite suitability cost	−386	−100%	–

Notes: Each row switches off one force in the estimated model, holds every other parameter at its estimate, and re-solves the plan-choice probabilities. "Change" is the change in the model-implied mean markup over the cheapest coverage-identical plan in the senior's choice set, in dollars per year per Medigap purchaser, from the baseline of $386 averaged over the audit's mix of treated and control calls; "change relative to baseline" divides it by that baseline. A flat Medigap commission pays the broker the same amount on every plan in the choice set; because only the commission's premium gradient enters the plan choice, this is also the markup with no pay for either product. The perceived competition row moves every call from the control arm to the treated arm and is measured from the markup with no call treated, $408. No appointment cost sets a_{appt} to zero. No suitability cost sets κ_2 to zero; with an infinite suitability cost every broker recommends the cheapest plan. The model has no broker type in the plan choice, so information frictions have no row. The split between the commission and the suitability cost rests on the assumption that the commission sensitivity is the same in both decisions (Section 6.1); the competition and appointment rows do not. Rows are separate counterfactuals and do not sum. The interval column reports 95% caller-cluster block-bootstrap percentile intervals.

MA-fit seniors No audited call presents an MA-fit senior, so the model supplies the broker's recommendation to that senior. An informed broker knows that Medicare Advantage fits and recommends Medigap only when the commission gap outweighs the suitability cost, which at the estimated parameters almost never occurs. We assume that a misinformed broker, who believes that Medicare Advantage fits the audited Medigap-fit senior, also believes that Medigap fits an MA-fit senior. The assumption is not tested by the audit. Without it, the loss of an MA-fit senior is $3 a year, the loss per advised senior is $773, and the baseline loss of seniors who turn 65 in one year is $1.09 billion instead of $1.26 billion. In the other direction, the broker on 21.0% of transcribed calls said that the broker sells only Medigap. Adding that share of brokers, who recommend Medigap to every senior, raises the loss per advised senior to $1,067 and the baseline loss to $1.50 billion. One piece of evidence outside the audit supports the higher value: in the broker survey, 41% of brokers recommend Medigap to a vignette senior whom Medicare Advantage fits, against 14.9% under our main assumption.

The order of the two policies Figure 7 adds the markup cap after pay parity. The cap applies to seniors who buy Medigap, and pay parity raises their number, so the cap is worth $141 million after pay parity and $106 million alone. The gain from the two policies together does not depend on the order.

Taxpayer savings The Medicare Payment Advisory Commission estimates that Medicare spends $84 billion more on Medicare Advantage enrollees than it would spend on the same beneficiaries in

Traditional Medicare (Medicare Payment Advisory Commission, 2025), or $2,407 for each of the 34.9 million enrollees. That comparison uses the spending of Traditional Medicare beneficiaries of whom 42% hold Medigap (Freed et al., 2024). A senior whom these policies move out of Medicare Advantage holds Medigap with certainty, and Medigap coverage raises Medicare spending by 22.2% (Cabral and Mahoney, 2019). Subtracting the part of that additional spending not already in the comparison, $1,561 a year, leaves $846 per enrollee per year. We apply that value to every senior whom a policy moves out of Medicare Advantage, of either fit type. Part of the taxpayer savings in bar (4) therefore comes from MA-fit seniors whom pay parity moves into Medigap, and those seniors lose consumer surplus. Bar (0) contains no taxpayer cost, so bar (4) is a change in federal spending and not a remaining loss.

Inference We report no interval on the baseline loss, which is a back-of-the-envelope product of estimated and assumed quantities. The population counts, the fit shares, the broker-use share and the two harms are held at their reported values, as in every counterfactual (Appendix F.2), and the range across the assumptions for MA-fit seniors above is wider than the sampling interval of the estimated parameters. For the policy steps, the bootstrap refits give a 95% interval of $108 million to $327 million for the $204 million gain from pay parity among seniors who turn 65 in one year.